Smt. Velagapudi Durgamba Siddhartha School of Law

LL.B. (3 Year) Degree Programme – Year I, Semester I

CYBER LAWS

Course Code: 26LLB105

Complete Academic Notes

Cyberspace and Jurisdiction, the Information Technology Act 2000, Cyber Crimes, Intermediaries and Emerging Technologies, Privacy, Data Protection and Cyber Security

Units 1 – 5 • Every syllabus topic explained in detail

With definitions, explanations, examples, flowcharts, tables and the ten prescribed landmark decisions

Updated to the DPDP Rules, 2025, the IT Rules amendments of 2025 and 2026, and the BNS, BNSS and BSA, 2023 (old IPC, CrPC and Evidence Act sections in brackets)

Syllabus Checklist (Unit-wise Topic List)

The table below lists every topic of the prescribed syllabus (Course Code 26LLB105) in the exact syllabus order (1.1 to 5.6). All ten landmark decisions prescribed for study are explained as full case laws under the relevant topics. Each topic begins on a fresh page.

Unit	No.	Topic (as in the syllabus)	Status
UNIT 1	1.1	Origin and meaning of cyberspace; cyberspace and physical space distinguished	Covered
	1.2	Legal issues in cyberspace and the need for regulation	Covered
	1.3	Models of cyberspace regulation	Covered
	1.4	Development of cyber law: national and international perspective	Covered
	1.5	Concept of jurisdiction and jurisdiction in cyberspace	Covered
	1.6	Tests of cyber jurisdiction; the position in India; judicial interpretation	Covered
UNIT 2	2.1	Scheme, objects, definitions and excluded transactions; the Amendment Act, 2008	Covered
	2.2	Legal recognition of electronic records and electronic signatures	Covered
	2.3	Digital signature: cryptography, public key infrastructure, certificates	Covered
	2.4	Controller and Certifying Authorities; duties of subscribers	Covered
	2.5	E-governance and e-commerce; UNCITRAL Model Laws	Covered
	2.6	E-contracts: click-wrap, shrink-wrap and browse-wrap; adjudication and appeal	Covered
UNIT 3	3.1	Cyber contravention and cybercrime distinguished	Covered
	3.2	Compensation and penalties under the Information Technology Act, 2000	Covered
	3.3	Cyber offences under the Information Technology Act, 2000	Covered
	3.4	Cyber offences under the Bharatiya Nyaya Sanhita, 2023 and other laws	Covered
	3.5	Reporting and investigation of cybercrimes	Covered
UNIT 4	4.1	Intermediary; safe harbour and due diligence	Covered
	4.2	Intermediary Guidelines and Digital Media Ethics Code Rules, 2021	Covered
	4.3	Freedom of speech in cyberspace; defamation, hate speech and obscenity	Covered
	4.4	Deepfakes, misinformation and internet shutdowns	Covered
	4.5	Artificial intelligence, algorithmic accountability and platform governance	Covered
	4.6	Cloud computing, Internet of Things, blockchain and virtual digital assets	Covered
UNIT 5	5.1	Concept of privacy; constitutional framework; judicial interpretation in India	Covered
	5.2	Digital Personal Data Protection Act, 2023: scope, definitions and consent	Covered
	5.3	Obligations of Data Fiduciaries; rights and duties of Data Principals	Covered
	5.4	Cyber security: confidentiality, integrity and availability; threat landscape and risk management; Critical Information Infrastructure; CERT-In and the Directions of 2022	Covered
	5.5	Data Protection Board of India; penalties, exemptions and cross-border transfer	Covered

Unit	No.	Topic (as in the syllabus)	Status
	5.6	Comparative regimes: GDPR and CCPA; international cyber security cooperation	Covered

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UNIT 1
Foundations of Cyberspace and Cyber Law

Topics in this Unit

1.1 Origin and meaning of cyberspace; cyberspace and physical space distinguished
1.2 Legal issues in cyberspace and the need for regulation
1.3 Models of cyberspace regulation
1.4 Development of cyber law: national and international perspective
1.5 Concept of jurisdiction and jurisdiction in cyberspace
1.6 Tests of cyber jurisdiction; the position in India; judicial interpretation

Topic 1.1: Origin and meaning of cyberspace; cyberspace and physical space distinguished

Introduction

Today, a large part of human life takes place online: we talk, shop, bank, study, work, vote, find entertainment and even commit crimes through computers and the internet. The "place" where all this happens is called **cyberspace**. It is not a physical place, but a virtual environment created by interconnected computers and networks. Because people act, transact and cause harm there, the law must regulate it. **Cyber law** is the branch of law that deals with the legal issues arising from the use of computers, the internet and digital technologies.

Definition

Cyberspace: the virtual environment created by interconnected computer systems and communication networks (especially the internet), in which people communicate, store and exchange information, and carry out transactions.

Cyber law: the body of law that governs cyberspace, i.e. the legal aspects of computers, the internet, electronic communications, electronic commerce, data and digital technologies.

Section 2(1)(nb), IT Act, 2000 defines "cyber security" as "protecting information, equipment, devices, computer, computer resource, communication device and information stored therein from unauthorised access, use, disclosure, disruption, modification or destruction". (The Act does not define "cyberspace" itself.)

Detailed Explanation

1. Origin of the Term "Cyberspace"

1. The word "cyber" comes from **cybernetics**, a term used by Norbert Wiener (1948) for the science of control and communication in animals and machines, from the Greek *kybernetes* ("steersman").
2. The word **"cyberspace"** was coined by the science-fiction writer **William Gibson** in his short story *Burning Chrome* (1982) and popularised in his novel *Neuromancer* (1984), where he described it as a "consensual hallucination" experienced by millions of users.
3. In the 1990s, the term came to be used for the internet and the online world generally. In 1996, John Perry Barlow's *A Declaration of the Independence of Cyberspace* described cyberspace as a new world beyond the reach of governments.

2. Development of the Internet

Year	Development
1969	ARPANET, a network of the US Department of Defense, connects four university computers

Year	Development
1983	ARPANET adopts the TCP/IP protocol, the technical basis of the modern internet
1989–1991	Tim Berners-Lee at CERN develops the World Wide Web (HTTP, HTML, URLs)
1995	Public internet access in India launched by VSNL (15 August 1995)
1998	ICANN set up to manage domain names and IP addresses
2000s	Broadband, e-commerce, search engines, social media
2010s–2020s	Smartphones, cloud computing, UPI and digital payments, Internet of Things, artificial intelligence

3. Components (Layers) of Cyberspace

1. **Physical layer:** computers, servers, cables, routers, satellites, mobile towers, data centres.
2. **Logical (code) layer:** software, protocols (TCP/IP, HTTP), the domain name system and operating systems.
3. **Content layer:** information, data, websites, messages, images and videos.
4. **Social (persona) layer:** users, their online identities (accounts, email IDs, usernames) and online communities.

4. Characteristics of Cyberspace

1. **Borderless:** information travels across national boundaries instantly.
2. **Virtual (non-physical):** it has no physical location, although it depends on physical infrastructure.
3. **Anonymity and pseudonymity:** users can hide or fake their identities.
4. **Speed and scale:** information can reach millions of people in seconds.
5. **Interactivity:** users are both consumers and creators of content.
6. **Decentralisation:** no single authority controls the internet.
7. **Permanence and replicability:** digital information can be copied perfectly and may remain online indefinitely.
8. **Convergence:** telephony, broadcasting, publishing and computing merge on one platform.

5. Cyberspace and Physical Space Distinguished

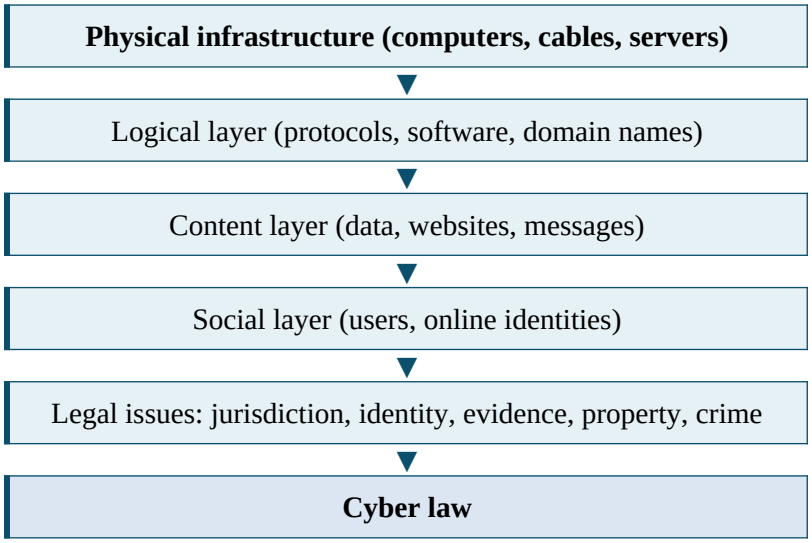
Basis	Physical space	Cyberspace
Nature	Tangible, material world	Intangible, virtual environment
Boundaries	Defined national and territorial boundaries	Borderless; a website can be accessed anywhere
Identity	Persons can usually be identified physically	Users may be anonymous or use fake identities
Location of acts	Acts have a clear place of commission	An act may originate in one country, pass through servers in others and have effects everywhere

Basis	Physical space	Cyberspace
Speed	Physical movement takes time	Information moves almost instantly
Property	Mainly tangible property	Mainly intangible property: data, domain names, software, digital assets
Evidence	Physical evidence (documents, objects)	Electronic evidence that is easily altered or deleted
Jurisdiction	Generally based on territory	Difficult to apply territorial jurisdiction
Copying	Copies are imperfect and costly	Copies are perfect and cost almost nothing
Regulation	By the State through laws enforced in its territory	By law, technical code, private platforms and international cooperation

6. Significance for Law

Because cyberspace differs from physical space, traditional legal concepts such as "document", "signature", "place of contract", "jurisdiction", "evidence" and "property" had to be adapted. For example, the IT Act, 2000 gives legal recognition to electronic records and signatures, and courts have developed new tests of jurisdiction for websites.

Flowchart: From physical world to cyberspace



Examples

- A person in Hyderabad buys a product from a website hosted on a server in the USA, owned by a company registered in Singapore. The "place" of the transaction is unclear, which shows the borderless nature of cyberspace.
- A false post on social media is shared by thousands of users within minutes and becomes impossible to remove completely, showing the speed and permanence of cyberspace.

Case Laws

There is no judicial decision that defines "cyberspace" as such; the term is descriptive. However, Indian courts have repeatedly recognised the special features of cyberspace. In *Shreya Singhal v Union of India* (2015), the Supreme Court accepted that the internet is a medium that reaches a very large number of people instantly, but held that this "intelligible differentia" could justify separate laws only if they satisfied Article 19(2) (see Topics 4.1 and 4.3). In *Anuradha Bhasin v Union of India* (2020), it held that freedom of speech and the freedom to carry on trade through the internet are constitutionally protected (see Topic 4.4).

Important Points

- "Cyberspace" was coined by William Gibson (*Burning Chrome*, 1982; *Neuromancer*, 1984).
- Cyberspace is a virtual environment created by interconnected computers and networks.
- Layers: physical, logical, content and social.
- Characteristics: borderless, virtual, anonymous, fast, decentralised, permanent.
- Cyberspace differs from physical space in boundaries, identity, location, evidence, property and jurisdiction.
- The IT Act defines "cyber security" (s.2(1)(nb)) but not "cyberspace".

Remember: Traditional legal concepts based on territory and tangible things had to be adapted for cyberspace.

Topic 1.2: Legal issues in cyberspace and the need for regulation

Introduction

The internet brings great benefits, but it also creates new legal problems and makes old problems harder to solve. Fraud, defamation, theft, harassment and obscenity existed before the internet, but cyberspace allows them to be committed on a larger scale, anonymously and across borders. New problems such as hacking, data breaches, identity theft, ransomware, deepfakes and the liability of platforms have also arisen. This topic explains the main legal issues and why cyberspace needs regulation.

Definition

> **Regulation of cyberspace** means the use of laws, rules, technical standards, institutions and other mechanisms to control conduct in cyberspace, protect rights and prevent harm.

Detailed Explanation

1. Main Legal Issues in Cyberspace

Issue	Explanation	Indian law (examples)
Jurisdiction	Which country's courts and laws apply to online acts with cross-border effects?	IT Act ss.1(2), 75; CPC s.20; *Banyan Tree* (2009)
Recognition of electronic transactions	Are electronic records, signatures and contracts valid?	IT Act ss.4, 5, 10A
Electronic evidence	How to prove emails, CCTV footage, WhatsApp messages?	BSA, 2023 ss.61–63 (IEA ss.65A, 65B); *Anvar*, *Arjun Panditrao*
Cybercrime	Hacking, identity theft, phishing, cyber stalking, child sexual abuse material, cyber terrorism	IT Act ss.65–67B; BNS
Privacy and data protection	Collection and misuse of personal data by companies and the State	Art 21 (*Puttaswamy*); DPDP Act, 2023
Intellectual property	Piracy, domain name disputes, software protection, AI-generated works	Copyright Act, 1957; Trade Marks Act, 1999; *Yahoo v Akash Arora*
Freedom of speech and content regulation	Hate speech, fake news, obscenity, defamation, blocking of content	Art 19; IT Act s.69A; *Shreya Singhal*
Liability of intermediaries	Should platforms be liable for content posted by users?	IT Act s.79; IT Rules, 2021
E-commerce and consumer protection	Online fraud, unfair contracts, dark patterns, defective products	Consumer Protection Act, 2019; E-Commerce Rules, 2020

Issue	Explanation	Indian law (examples)
Cyber security	Protecting networks and critical infrastructure from attacks	IT Act ss.70, 70A, 70B; CERT-In Directions, 2022
Taxation	Taxing digital services and virtual digital assets	Income Tax Act (VDA provisions); equalisation levy
Emerging technologies	AI, deepfakes, blockchain, IoT, cloud	IT Rules (2026 amendment on synthetic content); advisories

2. Why Traditional Law Was Inadequate

1. **Paper-based concepts:** laws required "writing", "signature" and "original document", which did not cover electronic forms.
2. **Territorial limits:** national laws apply within territory, but online acts cross borders.
3. **Anonymity:** it is difficult to identify offenders.
4. **New kinds of wrongs:** hacking, denial of service attacks, malware and data theft were not covered by traditional definitions of theft or mischief.
5. **Evidence:** electronic evidence is easy to alter and difficult to authenticate.
6. **Speed and scale:** harm spreads too quickly for traditional remedies.

3. Need for Regulation

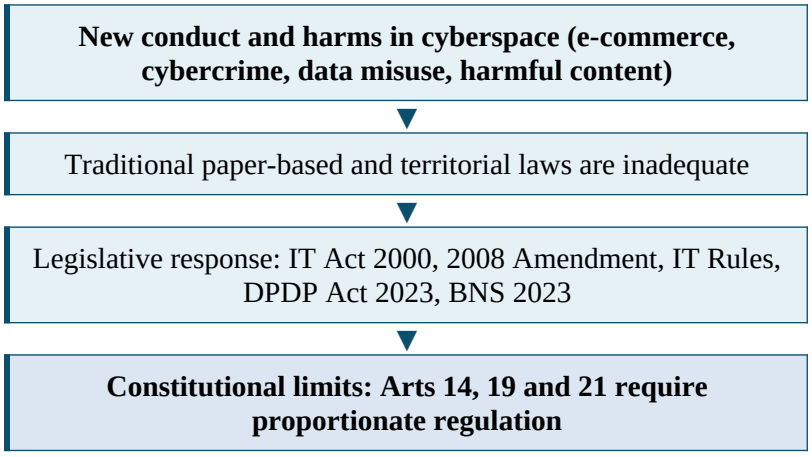
1. **To give legal certainty to e-commerce and e-governance:** businesses and governments need assurance that electronic contracts, records and signatures are legally valid.
2. **To prevent and punish cybercrime:** to protect individuals, businesses and the nation.
3. **To protect privacy and personal data.**
4. **To protect freedom of speech while preventing harmful content**, balancing Article 19(1)(a) with Article 19(2).
5. **To protect consumers** in online transactions.
6. **To protect intellectual property.**
7. **To protect national security and critical infrastructure** (power grids, banking, telecom, defence).
8. **To clarify the responsibilities of intermediaries** such as internet service providers, social media platforms and e-commerce marketplaces.
9. **To facilitate international cooperation**, since cybercrime is often transnational.
10. **To protect children and vulnerable groups** from online abuse and exploitation.

4. Arguments against Excessive Regulation

1. The internet developed through openness and freedom; excessive regulation may stifle innovation.
2. Censorship and surveillance may threaten free speech and privacy.
3. National laws may fragment the global internet ("splinternet").
4. Regulation may be ineffective because of technology and borders.

5. Hence the need for **balanced, proportionate and rights-respecting regulation** (*Shreya Singhal*, *Puttaswamy*, *Anuradha Bhasin*).

Flowchart: Why cyberspace needs regulation



Examples

- A fraudster in another country sends phishing emails to Indian bank customers and steals their money. Issues of jurisdiction, evidence, identification and international cooperation arise.
- A company collects users' location data without their knowledge and sells it to advertisers. This raises issues of privacy and data protection.

Case Laws

Case Law: Shreya Singhal v Union of India

Year: 2015 **Court:** Supreme Court of India

Facts / Background: Section 66A of the IT Act, inserted in 2008, punished sending, by means of a computer resource or communication device, information that was "grossly offensive" or had a "menacing character", or false information sent to cause "annoyance, inconvenience, danger, obstruction, insult, injury, criminal intimidation, enmity, hatred or ill will". After arrests for social media posts (including two young women in Palghar, Maharashtra, for a Facebook post questioning a bandh after a political leader's death), several petitions challenged s.66A, s.69A and s.79 of the IT Act.

Legal Issue: Is s.66A constitutional, and how should the regulation of online speech be balanced with Article 19(1)(a)?

Court's Decision / What the Court Held: The Court struck down s.66A in its entirety as violating Article 19(1)(a) and not saved by Article 19(2). The terms used were vague, overbroad and undefined, and the provision had a chilling effect on free speech. The Court distinguished between discussion, advocacy and incitement, and held that only incitement can be restricted. It upheld s.69A and the Blocking Rules, 2009, because of their procedural safeguards, and read down s.79(3)(b) (see Topic 4.1). It also held that the internet is a distinct medium and could be treated differently for some purposes, but that any restriction must fall within Article 19(2).

Relevance to the Topic: The leading case showing that the **need to regulate cyberspace** must be balanced against fundamental rights. (This is one of the landmark decisions prescribed for study; its intermediary aspect is explained in Topic 4.1 and its free speech aspect in Topic 4.3.)

Important Points

- Legal issues: jurisdiction, electronic transactions, evidence, cybercrime, privacy, IP, speech, intermediaries, e-commerce, cyber security, taxation, emerging technologies.
- Traditional laws were paper-based and territorial, and so inadequate.
- Regulation is needed for legal certainty, crime prevention, privacy, consumer protection, security and international cooperation.
- Regulation must be balanced and proportionate (*Shreya Singhal*, 2015).

Remember: Section 66A of the IT Act was struck down in *Shreya Singhal* (2015); no one can be prosecuted under it.

Topic 1.3: Models of cyberspace regulation

Introduction

Can cyberspace be regulated at all, and if so, by whom and how? Scholars have given different answers. Early internet enthusiasts argued that cyberspace was beyond the reach of national governments and should govern itself. Others argued that national laws can and do regulate the internet. Still others pointed out that technology itself (software code) regulates behaviour. These different views are called **models (theories) of cyberspace regulation**.

Definition

> A **model of cyberspace regulation** is a theory that explains who regulates conduct in cyberspace and by what means, e.g. self-regulation by users, regulation by States, regulation by technical code, or a combination.

Detailed Explanation

1. Cyber-Libertarian (Self-Regulation) Model

1. **John Perry Barlow,* A Declaration of the Independence of Cyberspace *(1996):** "Governments of the Industrial World ... You have no sovereignty where we gather." He argued that cyberspace is a new world, and governments have no moral right or practical ability to rule it.
2. **David Johnson and David Post,* Law and Borders: The Rise of Law in Cyberspace *(1996):** since cyberspace has no territorial borders, territorial laws are illegitimate and ineffective there. Cyberspace should be treated as a separate "place" with its own rules developed by its users (net-citizens, system operators).
3. **Criticism:** the internet depends on physical infrastructure located in States; users and companies are subject to national laws; self-regulation failed to stop crime, fraud and abuse.

2. Cyber-Paternalist (State Regulation / Traditional Law) Model

1. **Jack Goldsmith and Tim Wu (*Who Controls the Internet?*, 2006): national governments can and do regulate the internet effectively, by acting on internet service providers, payment systems, platforms and local assets. Example:* Yahoo! v LICRA* (2000), where a French court ordered Yahoo! to block French users' access to Nazi memorabilia auctions.
2. Online transactions are not fundamentally different from offline transactions with cross-border elements; ordinary rules of conflict of laws can be adapted.
3. Examples in India: the IT Act, 2000, blocking orders under s.69A, intermediary rules, data localisation requirements.

3. "Code is Law": Lessig's Four Modalities of Regulation

1. **Lawrence Lessig (*Code and Other Laws of Cyberspace, *1999):** behaviour in cyberspace is regulated by four forces:

Modality	How it regulates	Example
Law	Commands backed by sanctions	IT Act penalties for hacking
Social norms	Community expectations and disapproval	Online etiquette; community standards on platforms
Market	Price and availability	Cost of data; subscription models
Architecture (code)	Software and hardware design that permits or prevents conduct	Passwords, encryption, filters, DRM, content moderation algorithms

2. **"Code is law":** in cyberspace, software code (architecture) is a powerful regulator, since it determines what users can and cannot do.
3. The State can regulate **indirectly** by requiring changes in code (e.g. requiring platforms to label AI-generated content, or to enable traceability).
4. **Concern:** code is written by private companies and may embed values without democratic accountability.

4. Network Communitarianism

1. **Andrew Murray (*The Regulation of Cyberspace, *2007):** regulation in cyberspace results from the interaction of many actors (States, companies, users, technical bodies) in a network. Users are not passive subjects; they shape regulation through their choices and communities.

5. Co-Regulation and Multistakeholder Model

1. **Co-regulation:** the State sets the framework and the industry develops detailed codes, subject to State oversight. Example: the three-tier structure for digital news and OTT platforms under Part III of the IT Rules, 2021 (self-regulation by publishers, self-regulatory bodies and oversight by the Ministry of Information and Broadcasting).
2. **Multistakeholder governance:** internet governance involving governments, private sector, civil society and the technical community, e.g. ICANN (domain names), the Internet Engineering Task Force (technical standards) and the UN Internet Governance Forum (dialogue).

6. International (Treaty-Based) Model

1. Harmonisation through international instruments: UNCITRAL Model Laws, the Budapest Convention on Cybercrime (2001), and the UN Convention against Cybercrime (adopted by the UN General Assembly in December 2024).
2. Helps address cross-border issues, but depends on State cooperation.

7. Comparison

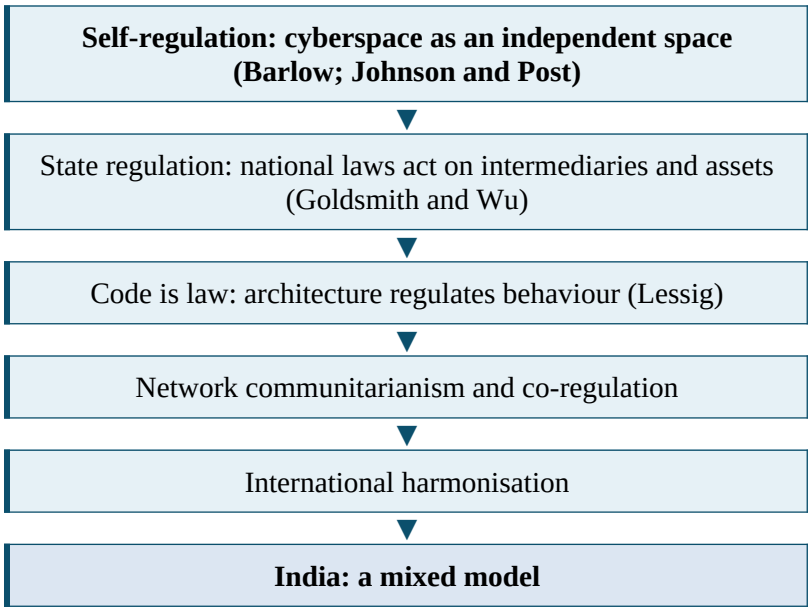
Model	Who regulates	Main thinkers	Strength	Weakness
Self-regulation (libertarian)	Users and online communities	Barlow; Johnson and Post	Respects freedom and innovation	Cannot control crime and abuse

Model	Who regulates	Main thinkers	Strength	Weakness
State regulation (paternalist)	National governments	Goldsmith and Wu	Enforceable; democratic legitimacy	Territorial limits; risk of censorship
Code (architecture)	Software designers, platforms, and States through them	Lessig	Very effective	Private power; lack of transparency
Network communitarianism	Interacting networks of actors	Murray	Realistic view of complexity	Diffuse accountability
Co-regulation and multistakeholder	State plus industry plus civil society	ICANN, IGF practice	Flexible and expert	Risk of capture by industry
International	States acting together	UNCITRAL; Council of Europe; UN	Addresses cross-border issues	Slow; not all States join

8. Indian Approach

India follows a **mixed model**: State regulation through the IT Act and DPDP Act, co-regulation for digital media (IT Rules, 2021), regulation through code (e.g. labelling and provenance duties for synthetic content under the 2026 amendment, traceability requirements), and international cooperation through MLATs and model laws.

Flowchart: Models of cyberspace regulation



Examples

- A platform's algorithm automatically blocks the upload of copyrighted songs. This is regulation by code.
- The Government orders an intermediary to block a website under s.69A. This is State regulation.

Case Laws

Case Law: Yahoo! Inc v La Ligue Contre Le Racisme et L'Antisemitisme (LICRA)

Year: 2000 **Court:** Tribunal de Grande Instance de Paris (France); later proceedings in US courts (2001, 2006)

Facts / Background: Yahoo!'s US-based auction website allowed users to sell Nazi memorabilia. French organisations sued Yahoo! in France, since the sale and display of Nazi objects is a crime under French law, and the website was accessible in France.

Legal Issue: Can a national court order a foreign website to prevent access by its nationals to content that is illegal in that country?

Court's Decision / What the Court Held: The Paris court held that it had jurisdiction because the content was accessible in France and caused harm there, and ordered Yahoo! to take all measures to prevent French users from accessing the auctions, after expert evidence that geolocation technology could identify most French users. Yahoo! sought a declaration in US courts that the French order was unenforceable under the First Amendment; the US Court of Appeals for the Ninth Circuit ultimately dismissed Yahoo!'s suit in 2006, without deciding the First Amendment question.

Relevance to the Topic: An important example supporting the **State regulation** model, showing that national courts can regulate global websites through technical measures (code).

Important Points

- Self-regulation (libertarian): Barlow (1996); Johnson and Post (1996).
- State regulation: Goldsmith and Wu; *Yahoo! v LICRA* (2000).
- Lessig: law, norms, market and architecture; "code is law".
- Murray: network communitarianism.
- Co-regulation and multistakeholder governance: IT Rules Part III; ICANN; IGF.
- International model: UNCITRAL, Budapest Convention, UN Convention against Cybercrime (2024).
- India follows a mixed model.

Remember: Lessig's insight is that software code can regulate behaviour as effectively as law.

Topic 1.4: Development of cyber law: national and international perspective

Introduction

Cyber law developed in response to the growth of computers and the internet. At the international level, organisations such as the United Nations Commission on International Trade Law (UNCITRAL), the Council of Europe, the OECD and the European Union developed model laws, conventions and guidelines. Individual countries enacted laws on computer crime, electronic commerce, data protection and online content. India enacted the Information Technology Act, 2000, which has since been supplemented by amendments, rules and new laws.

Definition

> The **development of cyber law** refers to the historical evolution of legal rules, at national and international levels, dealing with computers, electronic commerce, cybercrime, data protection and the internet.

Detailed Explanation

1. International Perspective

A. Electronic Commerce and Signatures (UNCITRAL)

1. **UNCITRAL Model Law on Electronic Commerce, 1996:** adopted by the UN General Assembly by Resolution 51/162 of 30 January 1997. It established the principles of non-discrimination against electronic records, functional equivalence and technology neutrality. The IT Act, 2000 is based on it.
2. **UNCITRAL Model Law on Electronic Signatures, 2001.**
3. **UN Convention on the Use of Electronic Communications in International Contracts, 2005.**
4. **UNCITRAL Model Law on Electronic Transferable Records, 2017.**
5. **UNCITRAL Model Law on the Use and Cross-border Recognition of Identity Management and Trust Services, 2022.**

B. Cybercrime

1. **Council of Europe Convention on Cybercrime (Budapest Convention), 2001:** the first international treaty on cybercrime. It requires States to criminalise offences such as illegal access, illegal interception, data interference, system interference, computer-related forgery and fraud, child pornography and copyright offences, and provides for procedural powers and international cooperation. India is not a party.
2. **UN Convention against Cybercrime:** adopted by the UN General Assembly in December 2024 and opened for signature in Hanoi in October 2025. It is the first global (UN) treaty on cybercrime.

C. Data Protection

1. **OECD Guidelines on the Protection of Privacy and Transborder Flows of Personal Data, 1980** (revised 2013).
2. **Council of Europe Convention 108, 1981** (modernised as Convention 108+ in 2018).
3. **EU Data Protection Directive, 1995**, replaced by the **General Data Protection Regulation (GDPR), 2016**, which applies from 25 May 2018 (see Topic 5.6).

D. Content, Platforms and Digital Markets

1. **United States:** Computer Fraud and Abuse Act, 1986; Electronic Communications Privacy Act, 1986; Communications Decency Act, 1996 (s.230 immunity for platforms); Digital Millennium Copyright Act, 1998 (notice and takedown).
2. **European Union:** E-Commerce Directive, 2000 (safe harbour); Digital Services Act, 2022; Digital Markets Act, 2022; Artificial Intelligence Act, 2024.
3. **WIPO Internet Treaties, 1996:** WIPO Copyright Treaty and WIPO Performances and Phonograms Treaty.

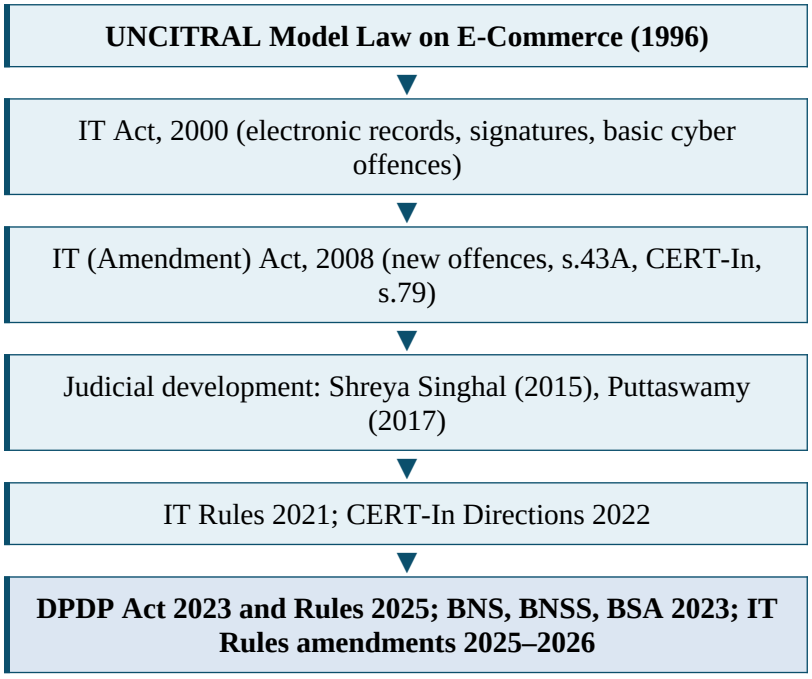
2. National (Indian) Perspective

Year	Development
1998	Draft E-Commerce Act prepared by the Department of Electronics
2000	**Information Technology Act, 2000** enacted (assent 9 June 2000; in force 17 October 2000), based on the UNCITRAL Model Law
2000	Amendments to the IPC, Evidence Act, Bankers' Books Evidence Act and RBI Act through schedules to the IT Act (e.g. electronic records as documents; s.65B of the Evidence Act)
2008	**Information Technology (Amendment) Act, 2008** (in force 27 October 2009): electronic signatures, data protection (s.43A, s.72A), new offences (ss.66A–66F, 67A–67C), CERT-In (s.70B), revised s.79 safe harbour
2009	Blocking Rules, Interception Rules and other rules under the IT Act
2011	Reasonable Security Practices (SPDI) Rules; Intermediary Guidelines Rules, 2011
2013	National Cyber Security Policy, 2013
2015	*Shreya Singhal*: s.66A struck down
2017	*K.S. Puttaswamy*: privacy recognised as a fundamental right
2021	IT (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021
2022	CERT-In Directions (28 April 2022)
2023	**Digital Personal Data Protection Act, 2023**; **Bharatiya Nyaya Sanhita, Bharatiya Nagarik Suraksha Sanhita and Bharatiya Sakshya Adhiniyam, 2023** (in force 1 July 2024); **Telecommunications Act, 2023**
2025	**DPDP Rules, 2025** (notified November 2025, phased commencement); amendment to Rule 3(1)(d) of the IT Rules (takedown procedure)
2026	IT Rules amendment on synthetically generated information (deepfakes), in force 20 February 2026

3. Other Indian Laws Relevant to Cyberspace

1. **Copyright Act, 1957** (amended 2012: ss.65A and 65B on technological protection measures and rights management information; s.52(1)(b)–(c) safe harbour for transient storage).
2. **Trade Marks Act, 1999** (domain names protected through passing off and infringement).
3. **Consumer Protection Act, 2019** and **Consumer Protection (E-Commerce) Rules, 2020**.
4. **Payment and Settlement Systems Act, 2007** and RBI directions on digital payments.
5. **Protection of Children from Sexual Offences Act, 2012** (online child sexual abuse material).
6. **Indecent Representation of Women (Prohibition) Act, 1986.**
7. The proposed **Digital India Act**, intended to replace the IT Act, has been discussed by the Government but not enacted as of the date of these notes.

Flowchart: Development of Indian cyber law



Examples

- Before 2000, a contract by email had no clear legal recognition in India; the IT Act gave it recognition.
- Before 2008, there was no specific offence of identity theft in India; s.66C was added by the 2008 Amendment.

Case Laws

No single judgment describes the development of cyber law as such. The development is mainly legislative, but key judicial decisions have shaped it: *Shreya Singhal v Union of India* (2015) (Topics 1.2, 4.1, 4.3), *K.S. Puttaswamy v Union of India* (2017) (Topic 5.1), *Anvar P.V.*

v P.K. Basheer (2014) and *Arjun Panditrao Khotkar* (2020) on electronic evidence (Topic 3.5), and *Anuradha Bhasin v Union of India* (2020) on internet shutdowns (Topic 4.4).

Important Points

- International instruments: UNCITRAL Model Laws (1996, 2001), Budapest Convention (2001), UN Convention against Cybercrime (2024), OECD Guidelines (1980), GDPR (2016).
- The IT Act, 2000 is based on the UNCITRAL Model Law on Electronic Commerce (UNGA Res. 51/162).
- The IT (Amendment) Act, 2008 added major offences and data protection provisions.
- DPDP Act, 2023 and DPDP Rules, 2025 created India's data protection law.
- BNS, BNSS and BSA, 2023 replaced the IPC, CrPC and Evidence Act from 1 July 2024.
- India is not a party to the Budapest Convention.

Remember: The IT Act, 2000 came into force on 17 October 2000, and the 2008 Amendment on 27 October 2009.

Topic 1.5: Concept of jurisdiction and jurisdiction in cyberspace

Introduction

Jurisdiction is the power of a State or a court to make, apply and enforce law. In the physical world, jurisdiction is mainly based on territory: a State's laws apply to acts within its borders. Cyberspace challenges this, because a single online act may involve persons, servers and effects in many countries. The question "which court can hear the case, and which law applies?" is therefore one of the most difficult issues in cyber law.

Definition

> **Jurisdiction:** the authority of a State to prescribe laws, of its courts to adjudicate disputes, and of its agencies to enforce laws, over persons, property and events.

> **Cyber jurisdiction:** the application of the principles of jurisdiction to persons, transactions and events in cyberspace.

Detailed Explanation

1. Kinds of Jurisdiction

Kind	Meaning
Prescriptive (legislative) jurisdiction	The power of a State to make laws applicable to persons and conduct
Adjudicative (judicial) jurisdiction	The power of courts to hear and decide cases
Enforcement (executive) jurisdiction	The power to enforce laws, e.g. by arrest, search or execution of judgments; generally limited to the State's territory

Other classifications:

1. **Territorial jurisdiction:** based on where the event occurred or the defendant is located.
2. **Pecuniary jurisdiction:** based on the value of the subject matter.
3. **Subject-matter jurisdiction:** based on the type of case (e.g. consumer, family, criminal).
4. **Personal jurisdiction:** power over the defendant (important in US law).

2. Bases of Jurisdiction in International Law

Principle	Basis	Example
Territorial principle	Act committed in the State's territory (subjective) or its effects felt there (objective)	An Indian court tries a hacker who attacked an Indian bank from abroad, because the effect was in India

Principle	Basis	Example
Nationality (active personality) principle	The offender is a national of the State	India prosecutes an Indian citizen for an offence committed abroad
Passive personality principle	The victim is a national of the State	India claims jurisdiction over an offence against an Indian abroad
Protective principle	The act threatens the State's security or vital interests	Cyber terrorism against Indian critical infrastructure
Universality principle	Crimes of universal concern	Piracy, genocide; arguably some serious cybercrimes

3. Problems of Jurisdiction in Cyberspace

1. **Location problem:** it is difficult to locate the place of an online act: where the user is, where the server is, or where the content is accessed.
2. **Multiple jurisdictions:** a website can be accessed in every country, so every country may claim jurisdiction, exposing persons to many laws.
3. **Anonymity:** the identity and location of the wrongdoer may be unknown.
4. **Conflict of laws:** an act lawful in one country may be unlawful in another (e.g. *Yahoo! v LICRA*).
5. **Enforcement:** even if a court gives judgment, it may not be enforceable in the country where the defendant or assets are located.
6. **Data stored abroad:** evidence may be on servers in other countries, requiring MLATs or cooperation from service providers.

4. Jurisdiction under Indian Law

A. Civil Jurisdiction

1. **Section 20, Code of Civil Procedure, 1908:** a suit may be filed where (a) the defendant resides or carries on business or personally works for gain, or (b) the cause of action, wholly or in part, arises.
2. **Section 19, CPC:** suits for compensation for wrongs to the person or movable property may be filed where the wrong was done or where the defendant resides.
3. **Special provisions:**
   - **Section 134(2), Trade Marks Act, 1999** and **Section 62(2), Copyright Act, 1957**: the plaintiff may sue where he resides or carries on business.
   - **Section 34, Consumer Protection Act, 2019:** a consumer may file a complaint where he resides or personally works for gain.

B. Criminal Jurisdiction

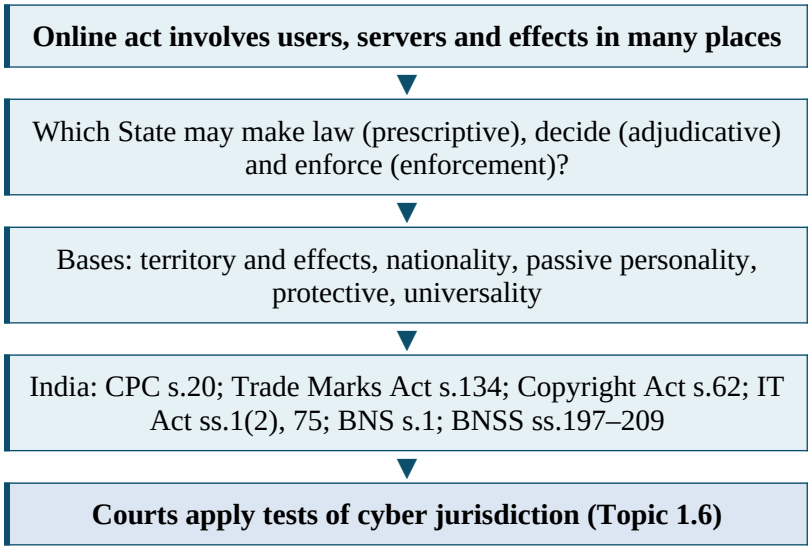
1. **Section 1(2), IT Act:** the Act extends to the whole of India and, save as otherwise provided, applies to any offence or contravention committed outside India by any person.

2. **Section 75, IT Act:** the Act applies to an offence or contravention committed outside India by any person, irrespective of nationality, if the act or conduct involves a computer, computer system or computer network located in India.
3. **Section 1(4) and (5), Bharatiya Nyaya Sanhita, 2023:** extra-territorial application to offences committed by Indian citizens outside India, on Indian ships or aircraft, and by any person outside India targeting a computer resource located in India.
4. **BNSS, 2023 (ss.197–209; formerly ss.177–189 CrPC):** an offence may be tried where it was committed, or where the consequence ensued, or where any part of it occurred; offences committed outside India may be tried in India subject to the sanction of the Central Government (s.208 BNSS; s.188 CrPC).

C. Choice of Law and Forum Clauses

1. Online contracts often contain clauses choosing the law and courts of a particular country.
2. Indian courts generally enforce such clauses if they choose a court that has jurisdiction, but consumer protection law may override them.

Flowchart: Jurisdiction in cyberspace



Examples

- A person in Germany hacks the server of an Indian company located in Bengaluru. Under s.75 of the IT Act, Indian law applies, because the computer system is located in India.
- A company in the USA sells counterfeit goods through a website to customers in Delhi. The trade mark owner may sue in Delhi if the defendant's website targets and makes sales in Delhi (*World Wrestling Entertainment v Reshma Collection*, Delhi High Court, 2014; see Topic 1.6).

Case Laws

Case Law: Dow Jones & Co Inc v Gutnick

Year: 2002 **Court:** High Court of Australia

Facts / Background: Dow Jones published an article on its subscription website (hosted on

servers in New Jersey, USA) alleging that Joseph Gutnick, a Melbourne businessman, was involved in money laundering. Gutnick sued for defamation in Victoria, where he lived and had his reputation, and where the article had been downloaded by subscribers.

Legal Issue: Where is online material "published" for the purpose of defamation jurisdiction: where it is uploaded or where it is downloaded and read?

Court's Decision / What the Court Held: The High Court of Australia held that defamation occurs where the material is downloaded and comprehended by the reader. Therefore, the article was published in Victoria, and the Victorian courts had jurisdiction, applying Victorian law. The Court rejected the argument that a single law of the place of uploading should apply to internet publications.

Relevance to the Topic: An important illustration of the **effects-based** (place of harm) approach to jurisdiction in cyberspace.

Important Points

- Jurisdiction: prescriptive, adjudicative and enforcement.
- Bases: territorial (subjective and objective), nationality, passive personality, protective, universality.
- Cyber jurisdiction problems: location, multiple jurisdictions, anonymity, conflict of laws, enforcement, data abroad.
- India: CPC ss.19–20; Trade Marks Act s.134(2); Copyright Act s.62(2); IT Act ss.1(2), 75; BNS s.1(4)–(5); BNSS ss.197–209.
- *Dow Jones v Gutnick* (2002): online defamation is published where it is downloaded.

Remember: Section 75 of the IT Act applies the Act to anyone, anywhere, if a computer system in India is involved.

Topic 1.6: Tests of cyber jurisdiction; the position in India; judicial interpretation

Introduction

Since a website can be accessed from anywhere, courts have had to decide when a court in a particular place can exercise jurisdiction over a website owner located elsewhere. If mere accessibility were enough, every website owner could be sued in every court in the world. Courts in the United States developed several tests, such as the **minimum contacts** test, the **Zippo sliding scale** test, the **effects** test and the **targeting** test. Indian courts, especially the Delhi High Court in *Banyan Tree Holding v A. Murali Krishna Reddy* (2009), have adopted and adapted these tests.

Definition

> **Tests of cyber jurisdiction** are the judicial standards used to determine whether a court can exercise jurisdiction over a defendant on the basis of his online activities, such as a website accessible in the forum State.

Detailed Explanation

1. The Minimum Contacts Test

1. **International Shoe Co v Washington (US Supreme Court, 1945):** a court may exercise personal jurisdiction over a non-resident defendant if he has "certain minimum contacts" with the forum State, such that the suit does not offend "traditional notions of fair play and substantial justice".
2. **Purposeful availment:** the defendant must have purposefully availed himself of the privilege of conducting activities in the forum State (*Hanson v Denckla*, 1958).

2. The Zippo Sliding Scale Test

1. **Zippo Manufacturing Co v Zippo Dot Com Inc (US District Court, W.D. Pennsylvania, 1997):** the likelihood that personal jurisdiction can be exercised is directly proportionate to the nature and quality of commercial activity that an entity conducts over the internet.
2. **Three categories:**

Type of website	Description	Jurisdiction
Active	The defendant clearly does business over the internet, entering into contracts and repeatedly transmitting files to residents of the forum	Jurisdiction exists
Interactive (middle ground)	The user can exchange information with the host computer	Depends on the level of interactivity and commercial nature of the exchange
Passive	The website merely posts information accessible to users	No jurisdiction merely because it is accessible

3. **Facts of Zippo:** Zippo Dot Com, a news service company in California, had about 3,000 subscribers in Pennsylvania and contracts with seven ISPs there. The court held that this was "doing business" in Pennsylvania, and exercised jurisdiction.

3. The Effects Test

1. **Calder v Jones (US Supreme Court, 1984):** jurisdiction can be exercised over a non-resident defendant whose intentional conduct outside the forum was **expressly aimed** at the forum and caused harm there that the defendant knew was likely to be suffered there.
2. Applied to cyberspace in cases of intentional torts (e.g. defamation, trade mark infringement) aimed at a particular State.

4. The Targeting Test

1. Jurisdiction exists where the website **specifically targets** users in the forum State, e.g. by using the local language or currency, accepting orders from that State, giving local contact details, or advertising there.
2. Courts look at factors such as the website's content, terms of use, payment and delivery options, and actual transactions with residents of the forum.

5. Comparison of Tests

Test	Focus	Origin
Minimum contacts	Purposeful contacts with the forum; fair play	*International Shoe* (1945)
Sliding scale	Level of interactivity and commercial activity of the website	*Zippo* (1997)
Effects	Intentional conduct aimed at the forum, causing harm there	*Calder v Jones* (1984)
Targeting	Whether the website specifically targets the forum	Later US and Indian cases

6. The Position in India: Judicial Interpretation

A. Early Approach: Mere Accessibility

1. **Casio India Co Ltd v Ashita Tele Systems Pvt Ltd (Delhi High Court, single judge, 2003):** in a trade mark case concerning a domain name, the Court held that since the defendant's website could be accessed in Delhi, the Delhi court had jurisdiction. This "mere accessibility" approach was later overruled.
2. **India TV Independent News Service Pvt Ltd v India Broadcast Live LLC (Delhi High Court, 2007):** the Court held that the website "indiatvlive.com", which was interactive and targeted Indian users, was subject to the Delhi court's jurisdiction, and discussed the US tests, rejecting mere accessibility as sufficient.

B. Banyan Tree (2009): The Leading Case

1. A Division Bench of the Delhi High Court overruled *Casio* and laid down the tests (explained below).

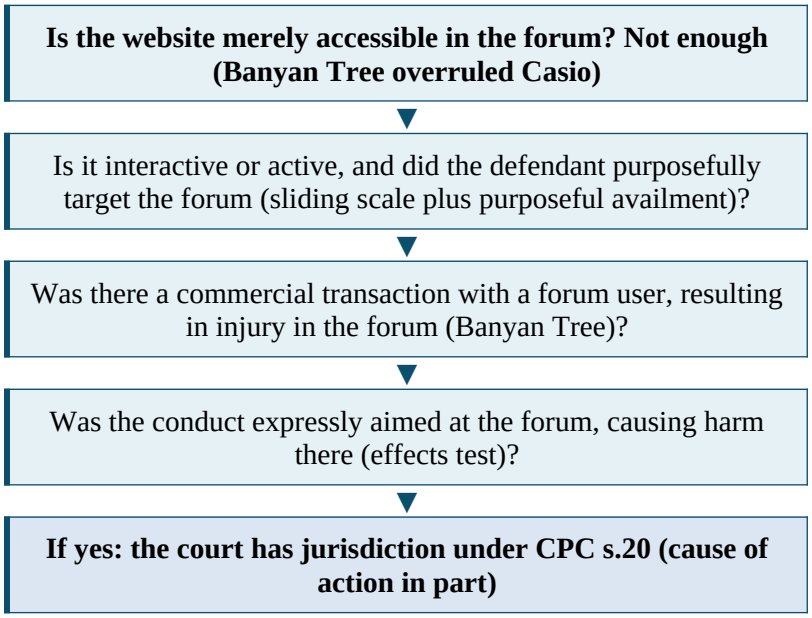
C. Later Cases

1. **World Wrestling Entertainment Inc v Reshma Collection (Delhi High Court, Division Bench, 2014):** WWE sold its products through its website to customers in Delhi. The Court held that when a customer in Delhi accepts the website's offer and makes payment, the contract is concluded in Delhi, and the plaintiff can be said to "carry on business" in Delhi for the purposes of s.134(2) of the Trade Marks Act and s.62(2) of the Copyright Act.
2. **Courts in later decisions have applied the "purposeful targeting" and "specific harm" requirements of* Banyan Tree*** in trade mark, copyright and defamation cases involving websites and social media.

D. Domain Names and Jurisdiction

1. Domain names are protected as trade marks in India through the action of passing off (*Yahoo! Inc v Akash Arora*, 1999; *Satyam Infoway Ltd v Sifynet Solutions Pvt Ltd*, Supreme Court, 2004).
2. Disputes about ".in" domain names are resolved through the **.IN Domain Name Dispute Resolution Policy (INDRP)**, and generic domain names through ICANN's **Uniform Domain Name Dispute Resolution Policy (UDRP)**.

Flowchart: Can an Indian court exercise jurisdiction over a foreign website?



Examples

- A blog in Canada with general information about Indian cooking, not selling anything, is a passive website; a Delhi court will not have jurisdiction merely because it can be read in Delhi.
- An online store in Dubai that sells counterfeit branded goods, quotes prices in rupees and ships to Delhi addresses targets Delhi customers; a Delhi court can exercise jurisdiction.

Case Laws

Case Law: Banyan Tree Holding (P) Ltd v A. Murali Krishna Reddy

Year: 2009 **Court:** Delhi High Court (Division Bench)

Facts / Background: The plaintiff, a Singapore company with no office in Delhi, ran hotels and spas under the mark "Banyan Tree". The defendants, based in Hyderabad, were developing a residential project called "Banyan Tree Retreat" and had a website advertising it. The plaintiff sued in the Delhi High Court for passing off, claiming jurisdiction because the defendants' website was accessible in Delhi. The single judge referred the question of jurisdiction to a Division Bench.

Legal Issue: When does a court have jurisdiction in a passing off or infringement action merely because the defendant's website is accessible within its territory?

Court's Decision / What the Court Held: The Division Bench held that mere accessibility of a website in the forum is not sufficient to confer jurisdiction (overruling *Casio India* on this point). For a court to have jurisdiction in a passing off or infringement action where neither party is located in the forum, the plaintiff must show that the defendant **purposefully availed** himself of the jurisdiction of the forum court, by showing that the website was not merely passive but was interactive with a commercial nature, and that the defendant **specifically targeted** the forum State for commercial transactions resulting in injury to the plaintiff there. A "trap transaction" by the plaintiff alone may not be sufficient; there must be real commercial transactions. The Court discussed and adopted elements of the *Zippo* sliding scale and the *Calder* effects test. The matter was sent back for decision on the facts.

Relevance to the Topic: The **leading Indian case on cyber jurisdiction**, and one of the landmark decisions prescribed for study.

Case Law: Yahoo! Inc v Akash Arora

Year: 1999 **Court:** Delhi High Court

Facts / Background: The plaintiff, Yahoo! Inc., owned the well-known trade mark and domain name "yahoo.com". The defendants started an internet service under the domain name "yahooindia.com", offering similar services, and claimed that the plaintiff's mark was not registered in India in respect of services, and that they had put a disclaimer on their website.

Legal Issue: Can a domain name be protected by the action of passing off, and does the defendants' use of "yahooindia.com" amount to passing off?

Court's Decision / What the Court Held: The Court held that a domain name is not merely an internet address but serves the function of a trade mark, identifying the business and services, and is entitled to protection against passing off. The defendants' domain name was almost identical and deceptively similar to the plaintiff's, and was likely to deceive and confuse internet users, who might be led to believe that both domain names belonged to the same source. A disclaimer was not sufficient, because internet users might not read it. An interim injunction was granted restraining the defendants from using "yahooindia.com" or any similar name.

Relevance to the Topic: The first major Indian case recognising **domain names as protectable under trade mark (passing off) law**, and one of the landmark decisions prescribed for study.

Important Points

- Tests: minimum contacts (*International Shoe*, 1945), sliding scale (*Zippo*, 1997), effects (*Calder v Jones*, 1984), targeting.
- *Casio* (2003): mere accessibility, overruled by *Banyan Tree* (2009).
- *Banyan Tree* (2009): purposeful availment, interactive commercial website, specific targeting and injury in the forum.
- *WWE v Reshma Collection* (2014): online sales mean carrying on business in the forum.
- *Yahoo! v Akash Arora* (1999): domain names protected by passing off; disclaimers insufficient.
- *Satyam Infoway* (2004): Supreme Court recognised domain names as having all the characteristics of trade marks.

Remember: In India, mere accessibility of a website is not enough to give a court jurisdiction (*Banyan Tree*).

UNIT 2
The Information Technology Act, 2000

Topics in this Unit

2.1 Scheme, objects, definitions and excluded transactions; the Amendment Act, 2008
2.2 Legal recognition of electronic records and electronic signatures
2.3 Digital signature: cryptography, public key infrastructure, certificates
2.4 Controller and Certifying Authorities; duties of subscribers
2.5 E-governance and e-commerce; UNCITRAL Model Laws
2.6 E-contracts: click-wrap, shrink-wrap and browse-wrap; adjudication and appeal

Topic 2.1: Scheme, objects, definitions and excluded transactions; the Amendment Act, 2008

Introduction

The **Information Technology Act, 2000** (IT Act) is India's primary law on electronic commerce, electronic governance and cybercrime. It was enacted on the basis of the **UNCITRAL Model Law on Electronic Commerce, 1996**, which the UN General Assembly recommended to all States by its Resolution 51/162 of 30 January 1997. The Act received the President's assent on 9 June 2000 and came into force on **17 October 2000**. It was substantially amended by the **Information Technology (Amendment) Act, 2008**, which came into force on **27 October 2009**.

Definition

> **Long title of the IT Act:** "An Act to provide legal recognition for transactions carried out by means of electronic data interchange and other means of electronic communication, commonly referred to as 'electronic commerce', which involve the use of alternatives to paper-based methods of communication and storage of information, to facilitate electronic filing of documents with the Government agencies and further to amend the Indian Penal Code, the Indian Evidence Act, 1872, the Bankers' Books Evidence Act, 1891 and the Reserve Bank of India Act, 1934 and for matters connected therewith or incidental thereto."

Detailed Explanation

1. Objects of the Act

1. To give **legal recognition to electronic records** and electronic transactions (e-commerce).
2. To give **legal recognition to digital and electronic signatures** for authentication.
3. To facilitate **electronic filing** of documents with Government agencies (e-governance).
4. To facilitate **electronic storage** of data.
5. To establish a **regulatory framework** for Certifying Authorities.
6. To define **cyber contraventions and cyber offences** and provide penalties.
7. To provide for **adjudication** and appeals in cyber matters.
8. To amend other laws (IPC, Evidence Act, Bankers' Books Evidence Act and RBI Act) to suit the electronic environment.
9. After 2008: to provide for **data protection**, **cyber security** (CERT-In and protection of critical information infrastructure) and the **liability of intermediaries**.

2. Scheme of the Act

Chapter	Sections	Subject
I	1–2	Preliminary: extent, application, excluded documents, definitions
II	3–3A	Digital signature and electronic signature

Chapter	Sections	Subject
III	4–10A	Electronic governance (legal recognition of electronic records and signatures; validity of e-contracts, s.10A)
IV	11–13	Attribution, acknowledgment and despatch of electronic records
V	14–16	Secure electronic records and secure electronic signatures
VI	17–34	Regulation of Certifying Authorities
VII	35–39	Electronic signature certificates
VIII	40–42	Duties of subscribers
IX	43–47	Penalties, compensation and adjudication
X	48–64	The Appellate Tribunal
XI	65–78	Offences
XII	79	Intermediaries not to be liable in certain cases
XIIA	79A	Examiner of electronic evidence
XIII	80–90	Miscellaneous (power of police to enter and search, overriding effect, rules)
Schedules	—	First Schedule: documents to which the Act does not apply; Second Schedule: electronic signature techniques

3. Extent and Application (Section 1)

1. The Act extends to the **whole of India**.
2. It applies also to any offence or contravention committed **outside India** by any person (s.1(2), read with s.75).

4. Excluded Transactions (Section 1(4) and the First Schedule)

The Act does **not** apply to the following documents or transactions:

No.	Excluded document or transaction
1	A **negotiable instrument** (as defined in s.13 of the Negotiable Instruments Act, 1881) other than a cheque
2	A **power of attorney** (s.1A, Powers of Attorney Act, 1882)
3	A **trust** (s.3, Indian Trusts Act, 1882)
4	A **will** (s.2(h), Indian Succession Act, 1925), including any other testamentary disposition
5	Any **contract for the sale or conveyance of immovable property** or any interest in such property

Reasons for exclusion: these documents are of great legal importance, often require registration, stamping, witnesses or special formalities, and involve risks of fraud; so they were kept outside the electronic regime. The Central Government may amend the First Schedule by

notification. (Cheques were excluded originally but brought in by the 2008 Amendment, allowing "electronic cheques" and "truncated cheques" under the Negotiable Instruments Act.)

5. Important Definitions (Section 2(1))

Clause	Term	Meaning (in brief)
(a)	Access	Gaining entry into, instructing or communicating with the logical, arithmetical or memory function resources of a computer, computer system or computer network
(ha)	Communication device	Cell phones, personal digital assistants, or combination of both, or any other device used to communicate, send or transmit any text, video, audio or image
(i)	Computer	Any electronic, magnetic, optical or other high-speed data processing device or system which performs logical, arithmetic and memory functions by manipulations of electronic, magnetic or optical impulses
(k)	Computer resource	Computer, computer system, computer network, data, computer database or software
(l)	Computer system	A device or collection of devices, including input and output support devices, containing programs, electronic instructions, data storage and retrieval and communication control
(na)	Cyber cafe	Any facility from where access to the internet is offered by any person in the ordinary course of business to members of the public
(nb)	Cyber security	Protecting information, equipment, devices, computer, computer resource, communication device and information stored therein from unauthorised access, use, disclosure, disruption, modification or destruction
(o)	Data	A representation of information, knowledge, facts, concepts or instructions prepared in a formalised manner and intended to be processed in a computer
(p)	Digital signature	Authentication of an electronic record by a subscriber by means of an electronic method or procedure in accordance with s.3
(r)	Electronic form	Information generated, sent, received or stored in media, magnetic, optical, computer memory, microfilm, computer-generated microfiche or similar device
(t)	Electronic record	Data, record or data generated, image or sound stored, received or sent in an electronic form or microfilm or computer-generated microfiche
(ta)	Electronic signature	Authentication of any electronic record by a subscriber by means of the electronic technique specified in the Second Schedule, and includes digital signature
(v)	Information	Includes data, message, text, images, sound, voice, codes, computer programmes, software and databases or microfilm or computer-generated microfiche

Clause	Term	Meaning (in brief)
(w)	Intermediary	With respect to any particular electronic record, any person who on behalf of another person receives, stores or transmits that record or provides any service with respect to that record, and includes telecom service providers, network service providers, internet service providers, web-hosting service providers, search engines, online payment sites, online auction sites, online market places and cyber cafes
(zg)	Subscriber	A person in whose name the electronic signature certificate is issued

6. The Information Technology (Amendment) Act, 2008

A. Background

1. The 2000 Act focused on e-commerce and had few cybercrime provisions.
2. New threats (identity theft, phishing, cyber terrorism, child sexual abuse material, data theft), the Bazee.com case (2004), and the 2008 Mumbai terror attacks led to the amendment.
3. An Expert Committee (2005) recommended changes; the Amendment Act was passed in December 2008 and came into force on 27 October 2009.

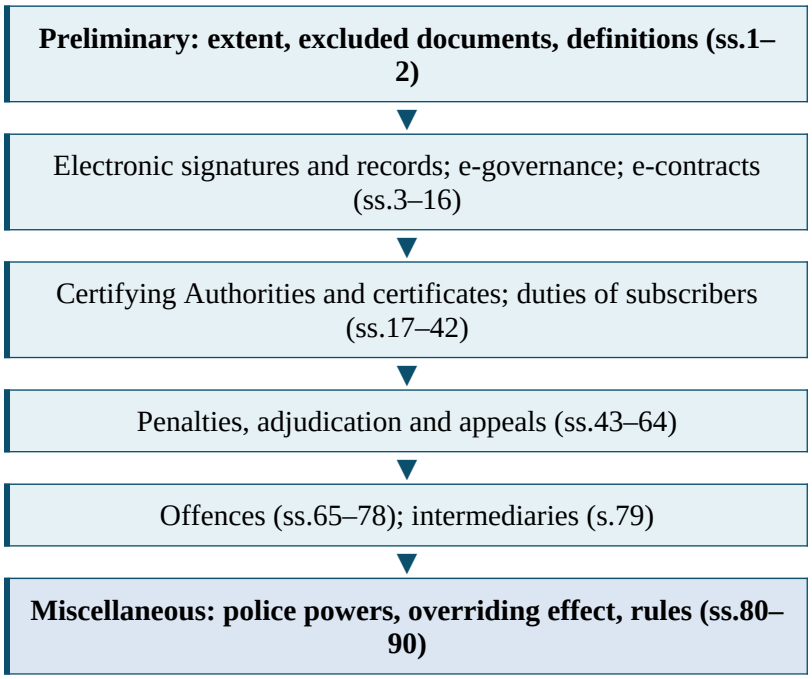
B. Main Changes

1. **Technology neutrality:** "digital signature" was replaced by the wider concept of **"electronic signature"** (s.3A), allowing other reliable techniques besides asymmetric cryptography (e.g. Aadhaar-based eSign).
2. **New definitions:** communication device, cyber cafe, cyber security, intermediary (expanded).
3. **Validity of e-contracts:** s.10A expressly recognised contracts formed through electronic means.
4. **Data protection:** s.43A (compensation for failure to protect sensitive personal data) and s.72A (punishment for disclosure of information in breach of lawful contract).
5. **New offences:** s.66 (computer-related offences, linked to s.43), ss.66A–66F (offensive messages, stolen computer resources, identity theft, cheating by personation, violation of privacy, cyber terrorism), ss.67A–67C (sexually explicit material, child sexual abuse material, preservation of information by intermediaries).
6. **Interception, monitoring and blocking powers:** s.69 (interception, monitoring, decryption), s.69A (blocking of public access), s.69B (monitoring traffic data).
7. **Cyber security institutions:** s.70A (National Critical Information Infrastructure Protection Centre, NCIIPC) and s.70B (Indian Computer Emergency Response Team, CERT-In).
8. **Intermediary liability:** s.79 was substituted to provide a "safe harbour" subject to due diligence.
9. **Examiner of electronic evidence:** s.79A.

10. **Procedural changes:** offences punishable with three years or more made cognizable; offences punishable with three years bailable (s.77B); certain offences made compoundable (s.77A); investigation by officers of the rank of **Inspector** instead of DSP (s.78).
11. **Removal of cap on compensation** under s.43 (earlier Rs. 1 crore); adjudicating officers' jurisdiction limited to claims up to Rs. 5 crore (s.46).
12. **Name change:** the Cyber Regulations Appellate Tribunal was renamed the Cyber Appellate Tribunal (later merged with TDSAT by the Finance Act, 2017).

Basis	IT Act, 2000 (original)	After the 2008 Amendment
Signature	Digital signature only (asymmetric crypto system)	Electronic signature (technology neutral), including digital signature
Data protection	No specific provision	ss.43A and 72A
Cyber offences	Few (ss.65–74)	Many new offences (ss.66A–66F, 67A–67C)
Investigation	DSP rank	Inspector rank
Intermediaries	Limited exemption for network service providers	Safe harbour subject to due diligence (s.79)
Cyber security	No institutions	CERT-In (s.70B), NCIIPC (s.70A)

Flowchart: Structure of the IT Act, 2000



Examples

- A sale deed for a flat cannot be executed electronically under the IT Act, because contracts for the sale of immovable property are excluded by the First Schedule.

- An electronic cheque is valid, since cheques were removed from the exclusion list.

Case Laws

There is no leading judgment on the scheme or the excluded transactions of the IT Act as such. The overriding effect of the IT Act as a special law (s.81) was explained in *Sharat Babu Digumarti v Government of NCT of Delhi* (2017), discussed in Topic 3.4. The validity of contracts formed by email was recognised in *Trimex International FZE v Vedanta Aluminium Ltd* (2010), discussed in Topic 2.6.

Important Points

- The IT Act is based on the UNCITRAL Model Law on Electronic Commerce, 1996 (UNGA Res. 51/162).
- In force: 17 October 2000; 2008 Amendment in force: 27 October 2009.
- 13 Chapters (plus Chapter XIIA) and two Schedules.
- Excluded: negotiable instruments other than cheques, powers of attorney, trusts, wills, contracts for sale or conveyance of immovable property.
- Key definitions: computer, computer resource, data, electronic record, electronic signature, intermediary.
- 2008 Amendment: electronic signatures, ss.43A and 72A, new offences, ss.69–69B, CERT-In, NCIIPC, new s.79, Inspector rank.

Remember: A will, a trust deed and a sale deed of immovable property cannot be made under the IT Act.

Topic 2.2: Legal recognition of electronic records and electronic signatures

Introduction

Traditional law requires many things to be "in writing", "signed" or "in original" form. If these requirements were applied strictly, electronic transactions would have no legal value. The IT Act solves this problem through the principle of **functional equivalence**: an electronic record or signature that performs the same function as a paper document or handwritten signature is given the same legal recognition. Chapter III (ss.4–10A) of the Act deals with this, together with Chapter IV (attribution, acknowledgment and despatch) and Chapter V (secure records and signatures).

Definition

> **Section 4 (Legal recognition of electronic records):** "Where any law provides that information or any other matter shall be in writing or in the typewritten or printed form, then, notwithstanding anything contained in such law, such requirement shall be deemed to have been satisfied if such information or matter is (a) rendered or made available in an electronic form; and (b) accessible so as to be usable for a subsequent reference."

> **Section 5 (Legal recognition of electronic signatures):** "Where any law provides that information or any other matter shall be authenticated by affixing the signature or any document shall be signed or bear the signature of any person, then, notwithstanding anything contained in such law, such requirement shall be deemed to have been satisfied, if such information or matter is authenticated by means of electronic signature affixed in such manner as may be prescribed by the Central Government."

Detailed Explanation

1. Legal Recognition of Electronic Records (Section 4)

1. **Conditions:**
   - The information is in **electronic form**.
   - It is **accessible** so as to be usable for subsequent reference.
2. **Effect:** any legal requirement of "writing" is satisfied by such an electronic record.
3. **Example:** an application, return or notice that must be "in writing" can be submitted by email or through an online portal.

2. Legal Recognition of Electronic Signatures (Section 5)

1. A requirement of signature is satisfied by an **electronic signature** affixed in the prescribed manner.
2. "Signed" includes affixing a handwritten signature or any mark, and in relation to electronic records, an electronic signature (Explanation to s.5).

3. Electronic Records and Signatures in Government (Sections 6–8)

1. **Section 6:** where any law requires the filing of forms, the issue of licences or permits, or the payment of money to or by the Government, it may be done electronically in the manner prescribed by the appropriate Government.
2. **Section 6A:** the Government may authorise **service providers** to deliver services electronically and collect service charges.
3. **Section 7:** a requirement to **retain** documents, records or information is satisfied by retaining them in electronic form, if they are accessible, retained in the format in which they were generated (or one which accurately represents the information) and include details of origin, destination, date and time.
4. **Section 7A:** a requirement of **audit** of documents applies also to electronic records.
5. **Section 8:** publication of rules, regulations and notifications in the **Electronic Gazette** is valid.
6. **Section 9:** sections 6, 7 and 8 do not confer a right to insist that a Government department accept documents in electronic form.
7. **Section 10:** the Central Government may make rules regarding electronic signatures.

4. Validity of Contracts Formed Electronically (Section 10A)

Where in a contract formation, the communication of proposals, acceptance, revocation of proposals and acceptances are expressed in electronic form or by means of an electronic record, such contract shall not be deemed unenforceable solely on the ground that such electronic form or means was used (see Topic 2.6).

5. Attribution, Acknowledgment and Despatch (Chapter IV, Sections 11–13)

1. **Section 11: Attribution:** an electronic record is attributed to the **originator** if it was sent by the originator himself, by a person authorised to act on his behalf, or by an information system programmed by or on behalf of the originator to operate automatically.
2. **Section 12: Acknowledgment of receipt:**
   - Where no particular method is agreed, acknowledgment may be given by any communication by the addressee (automated or otherwise) or any conduct sufficient to indicate receipt.
   - Where the originator has stipulated that the record is binding only on receipt of acknowledgment, it is deemed never to have been sent until acknowledgment is received.
3. **Section 13: Time and place of despatch and receipt:**
   - **Despatch:** when the record enters a computer resource outside the control of the originator.
   - **Receipt:** if the addressee has designated a computer resource, when the record enters it; if sent to another resource of the addressee, when he retrieves it; if no resource is designated, when it enters a computer resource of the addressee.
   - **Place:** an electronic record is deemed to be despatched at the place where the originator has his **place of business**, and received where the addressee has his place

of business (or usual place of residence if there is no place of business), irrespective of where the computer resource is located.

Event	Time	Place
Despatch	When the record enters a computer resource outside the originator's control	Originator's place of business
Receipt	When it enters the designated computer resource (or is retrieved, or enters the addressee's resource)	Addressee's place of business

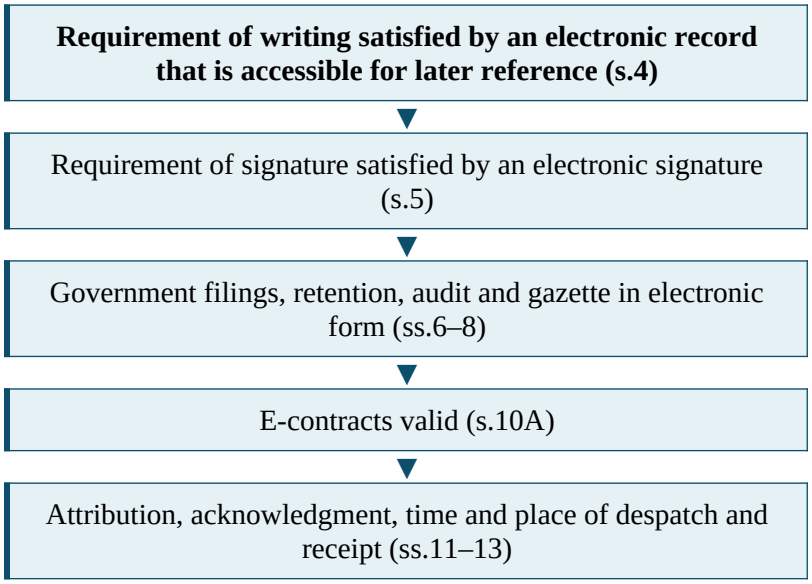
6. Secure Electronic Records and Signatures (Chapter V, Sections 14–16)

1. **Section 14:** where a security procedure has been applied to an electronic record at a specific point of time, it is deemed a **secure electronic record** from that time to the time of verification.
2. **Section 15:** an electronic signature is a **secure electronic signature** if (a) the signature creation data were, at the time of affixing, under the exclusive control of the signatory and no other person; and (b) the signature creation data were stored and affixed in the prescribed manner.
3. **Section 16:** the Central Government prescribes the security procedures and practices.

7. Electronic Records as Evidence

1. Under the **Bharatiya Sakshya Adhiniyam, 2023**, "document" includes electronic and digital records (s.2(1)(d)), and "evidence" includes electronic records.
2. **Sections 61–63, BSA** (formerly ss.65A and 65B of the Indian Evidence Act) lay down the conditions for admissibility of electronic records, including the certificate requirement (see Topic 3.5).
3. **Presumptions:** the BSA contains presumptions relating to electronic records, secure electronic records, secure electronic signatures, electronic signature certificates and electronic messages (corresponding to ss.81A, 85A, 85B, 85C, 88A and 90A of the old Evidence Act).

Flowchart: How the IT Act recognises electronic transactions



▼

Secure records and signatures (ss.14–16); admissibility under the BSA

Examples

- A company files its annual return with the Registrar of Companies online with the director's digital signature. This satisfies the requirements of writing and signature under ss.4 and 5.
- A supplier in Chennai emails an acceptance to a buyer in Mumbai, whose server is in Singapore. Under s.13, the acceptance is received in Mumbai (the buyer's place of business), not in Singapore.

Case Laws

Case Law: State of Maharashtra v Dr. Praful B. Desai

Year: 2003 **Court:** Supreme Court of India

Facts / Background: In a criminal trial for medical negligence, the prosecution wished to examine an important witness, a doctor in the USA, who was unwilling to come to India. The trial court allowed his evidence to be recorded by video-conferencing, but the High Court set aside the order, holding that the evidence must be recorded "in the presence of the accused" under s.273 CrPC.

Legal Issue: Can the evidence of a witness be recorded by video-conferencing, consistent with the requirement that evidence be taken in the presence of the accused?

Court's Decision / What the Court Held: The Supreme Court held that evidence can be recorded by video-conferencing, since "presence" does not mean only physical presence, and video-conferencing allows the witness and the accused to see and hear each other. The Court observed that "evidence" includes electronic records, and that the law must be interpreted to keep pace with technological developments.

Relevance to the Topic: The case reflects the principle of **functional equivalence** and the legal acceptance of electronic means in place of traditional forms (now reflected in the BNSS and BSA, which expressly allow electronic modes).

Important Points

- Section 4: electronic records satisfy the requirement of writing, if accessible for subsequent reference.
- Section 5: electronic signatures satisfy the requirement of signature.
- Sections 6–8: e-governance (filing, retention, audit, Electronic Gazette); s.9: no right to insist on electronic acceptance.
- Section 10A: validity of e-contracts.
- Sections 11–13: attribution, acknowledgment, time and place of despatch and receipt (place of business, not location of server).

- Sections 14–16: secure electronic records and signatures.
- *Praful B. Desai* (2003): technology-friendly interpretation of legal requirements.

Remember: The place of despatch and receipt is the place of business, irrespective of where the server is located (s.13).

Topic 2.3: Digital signature: cryptography, public key infrastructure, certificates

Introduction

In the electronic world, it is easy to create, copy and alter documents. To make electronic transactions trustworthy, there must be a way to verify **who** sent a document (authentication) and to ensure that it has **not been altered** (integrity). The **digital signature**, based on **asymmetric cryptography** and a **hash function**, provides this. Digital signatures work within a **public key infrastructure (PKI)**, in which trusted Certifying Authorities issue **digital signature certificates** linking public keys to persons.

Definition

> **Section 3(1), IT Act:** "any subscriber may authenticate an electronic record by affixing his digital signature."

> **Section 3(2):** "The authentication of the electronic record shall be effected by the use of asymmetric crypto system and hash function which envelop and transform the initial electronic record into another electronic record."

> **Cryptography:** the science of securing information by converting it into an unreadable form (encryption), which can be converted back only with a key (decryption).

Detailed Explanation

1. Cryptography: Basic Concepts

Term	Meaning
Plaintext	The original readable message
Ciphertext	The encrypted, unreadable message
Encryption	Converting plaintext into ciphertext using an algorithm and a key
Decryption	Converting ciphertext back into plaintext using a key
Key	A value (string of bits) used by the algorithm to encrypt or decrypt

2. Types of Cryptography

Basis	Symmetric (secret key) cryptography	Asymmetric (public key) cryptography
Keys	One key, shared by sender and receiver	A pair of keys: a private key and a public key
Key distribution	The secret key must be shared securely, which is difficult	The public key can be shared openly; the private key is kept secret

Basis	Symmetric (secret key) cryptography	Asymmetric (public key) cryptography
Speed	Fast	Slower
Use	Bulk encryption of data	Digital signatures, key exchange
Examples	AES, DES	RSA, ECC (elliptic curve)

Asymmetric crypto system (s.2(1)(f)): "a system of a secure key pair consisting of a private key for creating a digital signature and a public key to verify the digital signature".

3. Hash Function

1. **Meaning (Explanation to s.3):** a hash function is an algorithm that maps one sequence of bits into another, generally smaller, set (the "hash result"), such that:
   - the same input always gives the same hash result;
   - it is computationally infeasible to derive the original record from the hash result; and
   - it is computationally infeasible for two different records to produce the same hash result.
2. **Function:** the hash result is a "digital fingerprint" of the document; even a tiny change in the document changes the hash completely.
3. **Examples:** SHA-256, SHA-3.

4. How a Digital Signature Works

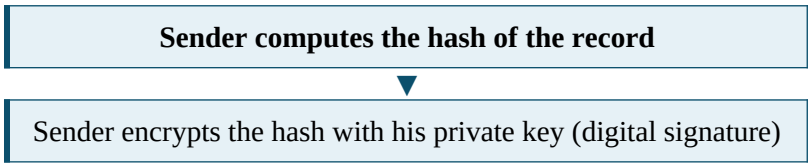
A. Signing (by the sender)

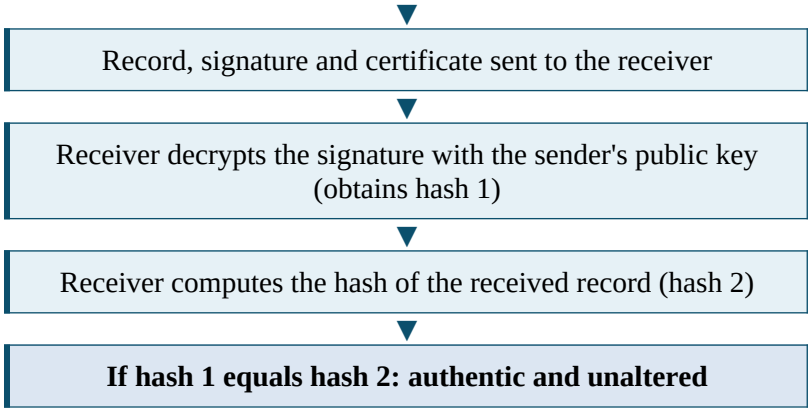
1. The sender's software computes the **hash** of the electronic record.
2. The hash is **encrypted with the sender's private key**. The encrypted hash is the digital signature.
3. The record and the digital signature are sent to the receiver (with the sender's digital signature certificate).

B. Verification (by the receiver)

1. The receiver **decrypts** the digital signature using the **sender's public key**, obtaining the original hash.
2. The receiver computes a **fresh hash** of the received record.
3. If the two hashes **match**, the signature is valid: the record was signed by the holder of the private key (authentication), and has not been altered (integrity).
4. **Section 3(4):** the private key and the public key are unique to the subscriber and constitute a functioning key pair.

Flowchart: Digital signature process





5. Functions of a Digital Signature

1. **Authentication:** identifies the signer.
2. **Integrity:** shows that the record has not been altered after signing.
3. **Non-repudiation:** the signer cannot later deny having signed.
4. **Legal validity:** satisfies the requirement of signature (s.5).

6. Public Key Infrastructure (PKI)

1. **Meaning:** PKI is the framework of policies, procedures, technology and institutions needed to create, manage, distribute, use, store and revoke digital certificates and public keys.
2. **Components:**

Component	Function
Controller of Certifying Authorities (CCA)	Licenses and regulates Certifying Authorities (s.17); acts as the Root Certifying Authority of India (RCAI)
Certifying Authority (CA)	Issues digital signature certificates (s.24), e.g. eMudhra, (n)Code Solutions, NIC, IDRBT
Registration Authority	Verifies the identity of applicants on behalf of the CA
Subscriber	The person to whom a certificate is issued
Repository	A database of certificates and certificate revocation lists, maintained by the Controller (s.20) and CAs
Certificate Revocation List (CRL)	A list of certificates that have been suspended or revoked

3. **Hierarchy of trust:** RCAI (Controller) → licensed CAs → subscribers. Users trust a certificate because it is issued by a CA licensed by the Controller.

7. Digital Signature Certificates (DSC)

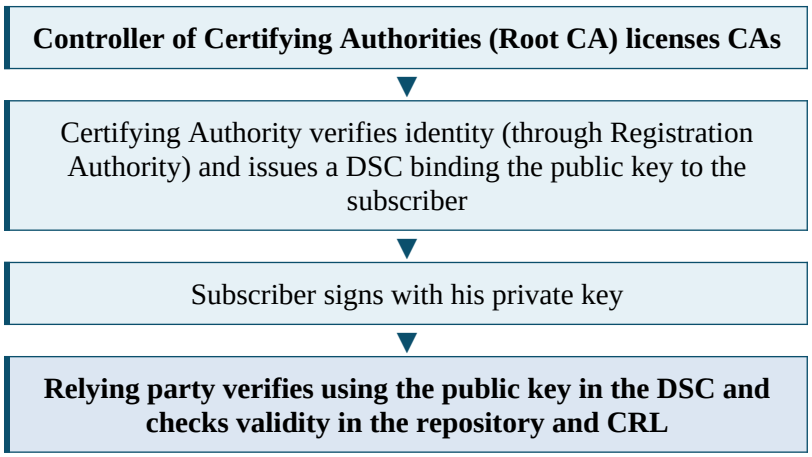
1. **Meaning:** an electronic signature certificate is an electronic document issued by a CA under s.35, which binds the subscriber's identity to his public key.
2. **Contents (generally, following the X.509 standard):** name of the subscriber, public key, name of the CA, serial number, validity period, digital signature of the CA.

3. **Classes:** certificates are issued at different assurance levels (e.g. Class 3 for higher-security uses such as e-tendering), as prescribed by the Controller's guidelines.
4. **Issue (s.35):** on application in the prescribed form and fee, after verifying the applicant's identity and that he holds the private key corresponding to the public key.
5. **Representations by the CA on issue (s.36):** that the subscriber holds the private key, that the public key and private key constitute a functioning key pair, that the information in the certificate is accurate, etc.
6. **Suspension (s.37):** on the request of the subscriber, or if the CA is of the opinion that it should be suspended in the public interest, for not more than 15 days unless the subscriber is heard.
7. **Revocation (s.38):** on the request of the subscriber, on his death, on dissolution or winding up of a company subscriber, or if a material fact in the certificate is false, the private key is compromised, or the subscriber is declared insolvent.
8. **Notice of suspension or revocation (s.39):** must be published in the repository.

8. Electronic Signatures (Technology Neutral)

1. **Section 3A:** a subscriber may authenticate an electronic record by an electronic signature or electronic authentication technique which is considered **reliable** and specified in the **Second Schedule**.
2. **Reliability conditions (s.3A(2)):** the signature creation data are linked to the signatory and no other person, are under his control, any alteration after affixing is detectable, and so on.
3. **Second Schedule:** includes "e-authentication technique using Aadhaar or other e-KYC services" (**eSign**), which allows a person to sign electronically using Aadhaar-based authentication through an eSign service provider.

Flowchart: Public key infrastructure



Examples

- A taxpayer signs his income tax return with a Class 3 DSC. The Income Tax Department verifies the signature using the public key in his certificate.
- A person signs a rental agreement using Aadhaar eSign, receiving an OTP on her registered mobile number.

Case Laws

There is no leading Indian judgment explaining the technical working of digital signatures or PKI. Courts generally accept digital signatures in accordance with the IT Act and the presumptions under the Bharatiya Sakshya Adhiniyam, 2023 (formerly ss.85A–85C of the Evidence Act), which presume that secure electronic records and secure electronic signatures are genuine, and that the information in an electronic signature certificate is correct unless the contrary is proved.

Important Points

- Digital signature: asymmetric crypto system and hash function (s.3).
- Private key signs; public key verifies; keys form a unique functioning pair (s.3(4)).
- Hash function: digital fingerprint; one-way; collision-resistant.
- Functions: authentication, integrity, non-repudiation.
- PKI: Controller (RCAI), Certifying Authorities, Registration Authorities, subscribers, repositories, CRLs.
- DSC: issue (s.35), representations (s.36), suspension (s.37), revocation (s.38), notice (s.39).
- Electronic signature (s.3A, Second Schedule) includes Aadhaar eSign.

Remember: The private key must be kept secret by the subscriber; the public key can be shared with everyone.

Topic 2.4: Controller and Certifying Authorities; duties of subscribers

Introduction

For digital signatures to be trusted, there must be trusted institutions that issue and manage digital signature certificates. The IT Act creates a regulatory structure headed by the **Controller of Certifying Authorities (CCA)**, who licenses and regulates **Certifying Authorities (CAs)**. CAs issue certificates to **subscribers**, who in turn have duties to keep their keys safe. Chapter VI (ss.17–34) deals with the regulation of CAs, Chapter VII (ss.35–39) with certificates, and Chapter VIII (ss.40–42) with the duties of subscribers.

Definition

> **Certifying Authority (s.2(1)(g)):** "a person who has been granted a licence to issue an electronic signature certificate under section 24."

> **Controller (s.2(1)(m)):** the Controller of Certifying Authorities appointed under s.17(1).

> **Subscriber (s.2(1)(zg)):** "a person in whose name the electronic signature certificate is issued."

Detailed Explanation

1. Controller of Certifying Authorities

A. Appointment (Section 17)

1. The Central Government appoints a Controller of Certifying Authorities, and may appoint Deputy and Assistant Controllers.
2. The Controller works under the general control and directions of the Central Government (in the Ministry of Electronics and Information Technology).

B. Functions (Section 18)

The Controller may perform all or any of the following functions:

1. exercise supervision over the activities of CAs;
2. certify the public keys of CAs;
3. lay down the standards to be maintained by CAs;
4. specify the qualifications and experience of employees of CAs;
5. specify the conditions subject to which CAs shall conduct their business;
6. specify the contents of written, printed or visual materials and advertisements distributed or used in respect of a certificate and the public key;
7. specify the form and content of an electronic signature certificate and the key;
8. specify the form and manner in which accounts shall be maintained by CAs;
9. specify the terms and conditions for the appointment of auditors;

10. facilitate the establishment of electronic systems by CAs;
11. specify the manner of conducting dealings with subscribers;
12. resolve conflicts of interest between CAs and subscribers;
13. lay down the duties of CAs;
14. maintain a database containing the disclosure record of every CA.

C. Other Powers

1. **Recognition of foreign CAs (s.19):** with the previous approval of the Central Government, the Controller may recognise any foreign CA as a CA for the purposes of the Act.
2. **Repository (s.20, now omitted; functions continue through s.18 and the rules):** the Controller maintains the Root CA and repository.
3. **Power to investigate contraventions (s.28)** and **access to computers and data (s.29)**.
4. **Power to give directions to CAs (s.68):** failure to comply is an offence (up to two years' imprisonment or fine up to Rs. 1 lakh, or both).

D. Root Certifying Authority of India (RCAI)

The CCA operates the RCAI, which issues certificates to licensed CAs, forming the top of the PKI hierarchy in India.

2. Certifying Authorities

A. Licence (Sections 21–26)

1. **Section 21:** any person may apply to the Controller for a licence to issue electronic signature certificates, if he fulfils the prescribed requirements of qualification, expertise, manpower, financial resources and infrastructure. A licence is valid for the prescribed period and is not transferable or heritable.
2. **Section 22:** application in the prescribed form, with a certification practice statement and identification documents, and the prescribed fee.
3. **Section 23:** renewal of licence.
4. **Section 24:** the Controller may grant or reject the application after considering the documents; no application shall be rejected without giving a reasonable opportunity of being heard.
5. **Section 25:** suspension of licence where the CA has made a false statement, failed to comply with the conditions, or contravened the Act. Suspension pending inquiry for up to 10 days without a hearing.
6. **Section 26:** notice of suspension or revocation to be published in the database.

B. Duties of Certifying Authorities (Sections 30–34)

1. **Section 30:** every CA shall:
   - make use of hardware, software and procedures that are secure from intrusion and misuse;
   - provide a reasonable level of reliability in its services;

- adhere to security procedures to ensure that the secrecy and privacy of electronic signatures are assured;
- be the repository of all electronic signature certificates issued under the Act;
- publish information regarding its practices, certificates and current status;
- observe the prescribed standards.

2. **Section 31:** ensure that every person employed by it complies with the Act, rules and regulations.
3. **Section 32:** display its licence at a conspicuous place.
4. **Section 33:** surrender its licence on suspension or revocation.
5. **Section 34:** disclose its electronic signature certificate, certification practice statement, notice of revocation or suspension, and other facts that materially affect the reliability of certificates.

C. Issue, Suspension and Revocation of Certificates (Sections 35–39)

See Topic 2.3.

3. Duties of Subscribers (Chapter VIII)

Section	Duty
40	**Generating key pair:** where a DSC whose public key corresponds to the private key of the subscriber is accepted by the subscriber, the subscriber shall generate the key pair by applying the prescribed security procedure
40A	**Duties relating to electronic signature certificates:** the subscriber shall perform such duties as may be prescribed
41	**Acceptance of certificate:** a subscriber is deemed to have accepted a certificate if he publishes or authorises its publication, or otherwise demonstrates approval. By accepting, he certifies to all who reasonably rely on it that he holds the private key, that all representations made to the CA are true, and that all information in the certificate within his knowledge is true
42	**Control of private key:** (1) every subscriber shall exercise reasonable care to retain control of the private key corresponding to the public key listed in his certificate and take all steps to prevent its disclosure; (2) if the private key has been compromised, the subscriber shall communicate the same without delay to the CA in the manner specified by the regulations

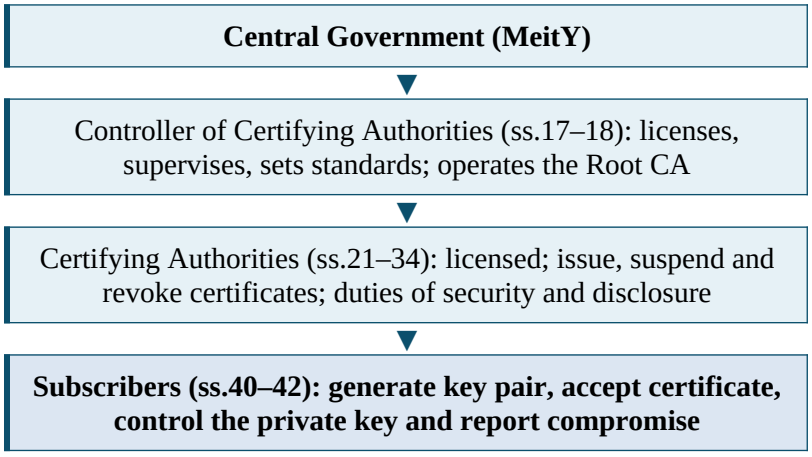
Explanation to s.42: for the removal of doubts, it is declared that the subscriber shall be **liable till he has informed the CA** that the private key has been compromised.

4. Liability for Misuse

1. **Section 71:** misrepresentation or suppression of material facts to the Controller or a CA to obtain a licence or certificate: imprisonment up to two years, or fine up to Rs. 1 lakh, or both.
2. **Section 73:** publishing a certificate known to be false in certain particulars (e.g. not issued by the CA, not accepted by the subscriber, or revoked or suspended): up to two years, or fine up to Rs. 1 lakh, or both.

3. **Section 74:** creating, publishing or making available a certificate for a fraudulent or unlawful purpose: up to two years, or fine up to Rs. 1 lakh, or both.

Flowchart: Regulatory structure for electronic signatures



Examples

- A company applies to the Controller for a licence to function as a CA. It must show adequate infrastructure, a certification practice statement and financial resources.
- A director's USB token with her private key is stolen. She must inform the CA without delay; until she does, she remains liable for its use (s.42).

Case Laws

There is no well-known reported judgment specifically on the powers of the Controller, the licensing of Certifying Authorities or the duties of subscribers under ss.40–42. The provisions are mainly administrative and are supplemented by the Information Technology (Certifying Authorities) Rules, 2000 and regulations and guidelines issued by the Controller.

Important Points

- The Controller of Certifying Authorities is appointed by the Central Government (s.17) and performs regulatory functions (s.18).
- The Controller may recognise foreign CAs (s.19) and gives directions (s.68).
- CAs are licensed (ss.21–24); licences can be suspended or revoked (s.25).
- Duties of CAs: security, reliability, repository, disclosure (ss.30–34).
- Duties of subscribers: generate key pair (s.40), accept certificate (s.41), control the private key and report compromise (s.42).
- Offences: ss.71, 73 and 74.

Remember: A subscriber remains liable until he informs the CA that his private key has been compromised (Explanation to s.42).

Topic 2.5: E-governance and e-commerce; UNCITRAL Model Laws

Introduction

The two main purposes of the IT Act are to promote **electronic governance (e-governance)**, i.e. the delivery of government services electronically, and **electronic commerce (e-commerce)**, i.e. the buying and selling of goods and services electronically. Both depend on legal recognition of electronic records and signatures. The IT Act draws its basic principles from the **UNCITRAL Model Laws**, which aim to harmonise e-commerce laws across the world.

Definition

E-governance: the use of information and communication technologies by the government to deliver services, exchange information, carry out transactions and integrate systems between government and citizens (G2C), government and businesses (G2B), and between government departments (G2G).

E-commerce: the buying and selling of goods and services, and the transfer of money or data, over electronic networks, primarily the internet.

Consumer Protection (E-Commerce) Rules, 2020, rule 3(1)(b): "e-commerce" means buying or selling of goods or services including digital products over digital or electronic network.

Detailed Explanation

1. E-Governance under the IT Act (Sections 4–10)

Section	Provision
4	Electronic records satisfy the requirement of writing
5	Electronic signatures satisfy the requirement of signature
6	Use of electronic records and signatures in Government and its agencies (filing of forms, issue of licences, receipt of payments)
6A	Delivery of services by authorised service providers (e.g. common service centres), with power to collect service charges
7	Retention of electronic records
7A	Audit of documents maintained in electronic form
8	Publication in the Electronic Gazette
9	No right to insist that the Government accept electronic documents
10	Power of the Central Government to make rules on electronic signatures

2. E-Governance Initiatives in India

1. **National e-Governance Plan (2006)** and **Digital India** programme (2015).
2. **Aadhaar** (digital identity), **DigiLocker** (digital documents; documents in DigiLocker are treated as equivalent to originals under the IT rules on DigiLocker), **UMANG** (unified services app).
3. **e-Courts**, e-filing, virtual hearings; **e-Procurement** (GeM portal); **GST Network**, income tax e-filing; **MCA21** (company filings).
4. **Common Services Centres** under s.6A.
5. **Unified Payments Interface (UPI)** and digital payments.

3. Advantages and Challenges of E-Governance

Advantages	Challenges
Speed and convenience	Digital divide (lack of access and literacy)
Transparency and reduced corruption	Privacy and data security risks
Lower cost	Cyber attacks on government systems
24-hour access to services	Exclusion due to authentication failures
Better record-keeping	Need for legal and institutional capacity

4. E-Commerce

A. Types of E-Commerce

Type	Meaning	Example
B2B (business to business)	Transactions between businesses	Wholesale online marketplaces
B2C (business to consumer)	Businesses selling to consumers	Online shopping websites
C2C (consumer to consumer)	Consumers selling to each other	Online classifieds and auctions
C2B (consumer to business)	Consumers offering services to businesses	Freelancing platforms
G2C / G2B	Government to citizens or businesses	Online tax payment, e-tendering

B. Legal Framework for E-Commerce in India

1. **IT Act, 2000:** legal recognition of records, signatures and contracts (ss.4, 5, 10A, 11–13); intermediary safe harbour (s.79).
2. **Indian Contract Act, 1872:** general principles of contract apply to online contracts.
3. **Consumer Protection Act, 2019:** s.2(16) defines "e-commerce"; s.2(17) "electronic service provider"; s.94 empowers rules for e-commerce.

4. **Consumer Protection (E-Commerce) Rules, 2020:** duties of e-commerce entities, marketplace and inventory models, grievance officers, disclosure of seller details, country of origin, no manipulation of prices, no unfair trade practices.
5. **Guidelines for Prevention and Regulation of Dark Patterns, 2023** (Central Consumer Protection Authority): prohibit manipulative design practices such as false urgency, basket sneaking, confirm shaming, forced action, subscription traps and drip pricing.
6. **FDI policy:** allows 100% FDI in the marketplace model, but not in the inventory model of B2C e-commerce.
7. **Payment and Settlement Systems Act, 2007** and RBI directions; **Digital Personal Data Protection Act, 2023**; **Goods and Services Tax** (tax collection at source by e-commerce operators).

5. UNCITRAL Model Laws

A. UNCITRAL Model Law on Electronic Commerce, 1996 (MLEC)

1. **Adopted** by UNCITRAL in 1996 and recommended by the UN General Assembly (Resolution 51/162, 30 January 1997).
2. **Key principles:**
   - **Non-discrimination:** information shall not be denied legal effect solely because it is in electronic form (Art 5).
   - **Functional equivalence:** electronic records satisfy requirements of writing (Art 6), signature (Art 7) and original (Art 8), if they fulfil the purposes of these requirements.
   - **Technology neutrality:** the law should not favour any particular technology.
   - **Party autonomy:** parties may vary many provisions by agreement.
3. **Other provisions:** admissibility of data messages as evidence (Art 9); retention (Art 10); formation and validity of contracts (Art 11); attribution (Art 13); acknowledgment of receipt (Art 14); time and place of despatch and receipt (Art 15).
4. **Influence on the IT Act:** ss.4, 5, 7, 10A and 11–13 of the IT Act are based on Arts 6, 7, 10, 11 and 13–15 of the MLEC.

B. UNCITRAL Model Law on Electronic Signatures, 2001 (MLES)

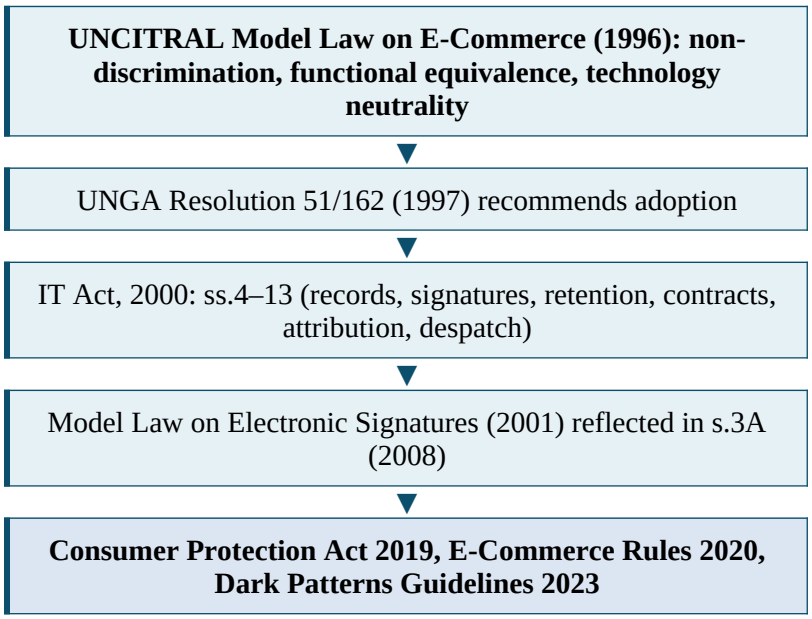
1. Provides criteria of **reliability** for electronic signatures (Art 6), conduct of the signatory (Art 8), the certification service provider (Art 9) and the relying party (Art 11), and recognition of foreign certificates (Art 12).
2. India's 2008 amendment, introducing technology-neutral electronic signatures (s.3A), reflects the approach of the MLES.

C. Other UNCITRAL Instruments

1. **UN Convention on the Use of Electronic Communications in International Contracts, 2005** (Electronic Communications Convention).
2. **Model Law on Electronic Transferable Records, 2017** (MLETR): electronic bills of lading, promissory notes, etc.
3. **Model Law on the Use and Cross-border Recognition of Identity Management and Trust Services, 2022.**

IT Act provision	Corresponding MLEC article
s.4 (writing)	Art 6
s.5 (signature)	Art 7
s.7 (retention)	Art 10
s.10A (contracts)	Art 11
s.11 (attribution)	Art 13
s.12 (acknowledgment)	Art 14
s.13 (time and place)	Art 15

Flowchart: From UNCITRAL to Indian e-commerce law



Examples

- A citizen applies for a passport online, pays the fee electronically and books an appointment. This is G2C e-governance.
- An online marketplace hides additional charges until the last step of checkout. This is "drip pricing", a dark pattern prohibited by the 2023 Guidelines.

Case Laws

Case Law: Amazon Seller Services Pvt Ltd v Amway India Enterprises Pvt Ltd

Year: 2020 **Court:** Delhi High Court (Division Bench)

Facts / Background: Direct selling companies (Amway, Oriflame, Modicare) sued e-commerce platforms to restrain the sale of their products on the platforms without their consent, relying on the Direct Selling Guidelines, 2016. A single judge granted an interim injunction, holding that the platforms were not mere intermediaries. The platforms

appealed.

Legal Issue: Are the Direct Selling Guidelines binding on e-commerce platforms, and do the platforms lose the safe harbour of s.79 by providing value-added services?

Court's Decision / What the Court Held: The Division Bench set aside the single judge's injunction, holding that the Direct Selling Guidelines were advisory and not law binding on e-commerce platforms, and that providing value-added services (such as warehousing, logistics and payment facilities) does not by itself deprive an e-commerce platform of its status as an intermediary, as long as it complies with the due diligence requirements. (Facts summarised briefly.)

Relevance to the Topic: An important decision on the **legal position of e-commerce marketplaces** in India.

Important Points

- E-governance provisions: ss.4–10 of the IT Act (records, signatures, Government use, service providers, retention, audit, Electronic Gazette).
- E-commerce: B2B, B2C, C2C, C2B, G2C, G2B.
- Legal framework: IT Act, Contract Act, Consumer Protection Act 2019, E-Commerce Rules 2020, Dark Patterns Guidelines 2023, FDI policy, DPDP Act.
- UNCITRAL MLEC 1996: non-discrimination, functional equivalence, technology neutrality; basis of the IT Act.
- UNCITRAL MLES 2001: reliability of electronic signatures; reflected in s.3A.
- Other instruments: Electronic Communications Convention 2005, MLETR 2017.

Remember: Functional equivalence means electronic records are treated as equal to paper records if they serve the same purpose.

Topic 2.6: E-contracts: click-wrap, shrink-wrap and browse-wrap; adjudication and appeal

Introduction

An **electronic contract (e-contract)** is a contract formed through electronic means, such as email, websites, mobile apps or electronic data interchange. Most online transactions today (buying goods, booking tickets, subscribing to services, installing software) involve e-contracts, usually on standard terms drafted by the business. The main forms are **click-wrap**, **shrink-wrap** and **browse-wrap** agreements. This topic also explains the **adjudication** of contraventions under the IT Act and the system of **appeals**.

Definition

> **E-contract:** a contract in which the offer, acceptance and other communications are made through electronic means.

> **Section 10A, IT Act:** "Where in a contract formation, the communication of proposals, the acceptance of proposals, the revocation of proposals and acceptances, as the case may be, are expressed in electronic form or by means of an electronic record, such contract shall not be deemed to be unenforceable solely on the ground that such electronic form or means was used for that purpose."

Detailed Explanation

1. Essentials of a Valid E-Contract

The general principles of the Indian Contract Act, 1872 apply:

1. **Offer and acceptance** (ss.3–9), communicated electronically.
2. **Intention to create legal relations.**
3. **Lawful consideration** and **lawful object** (ss.23–24).
4. **Capacity** of the parties (ss.11–12): a minor's agreement is void (this is a practical problem online, where age is difficult to verify).
5. **Free consent** (ss.13–22).
6. **Certainty** of terms.
7. **Not excluded** by the First Schedule of the IT Act (e.g. sale of immovable property).

2. Time and Place of Formation

1. **Communication of acceptance:** under s.4 of the Contract Act, communication of acceptance is complete against the proposer when it is put in a course of transmission to him so as to be out of the acceptor's power.
2. **IT Act s.13:** an electronic record is despatched when it enters a computer resource outside the control of the originator, and is received when it enters the designated computer resource of the addressee. The place is the place of business of the originator and addressee (see Topic 2.2).

3. Online contracts made through instantaneous communication (e.g. clicking "I agree" on a website) are generally treated like contracts made by telephone, where the contract is formed where the acceptance is received (*Bhagwandas Goverdhandas Kedia v Girdharilal Parshottamdas*, Supreme Court, 1966).

3. Types of E-Contracts

A. Click-Wrap Agreements

1. **Meaning:** the user must click a button such as "I agree" or "I accept", after being shown the terms (or a link to them), before he can proceed (e.g. creating an account, installing software, buying online).
2. **Enforceability:** generally **enforceable**, because the user has an opportunity to read the terms and gives a clear manifestation of assent.
3. **Example:** accepting the terms of service before signing up for an email account.

B. Shrink-Wrap Agreements

1. **Meaning:** the terms are enclosed inside the packaging of a product (typically software CDs). The terms state that by opening the shrink-wrap packaging, or by using the product, the buyer accepts the terms.
2. **Enforceability:** in the US, *ProCD Inc v Zeidenberg* (7th Circuit, 1996) held a shrink-wrap licence enforceable, since the buyer had notice of the terms and the opportunity to reject them by returning the product. Their validity depends on reasonable notice and an opportunity to reject.
3. Now less common, since software is mostly downloaded.

C. Browse-Wrap Agreements

1. **Meaning:** the terms are available through a hyperlink (usually at the bottom of a webpage), and the website states that by using it, the user accepts the terms. The user does not have to click "I agree".
2. **Enforceability:** often **not enforceable**, unless the user had actual or constructive notice of the terms. In *Specht v Netscape Communications Corp* (2nd Circuit, 2002), the court held that users who downloaded free software were not bound by licence terms that were visible only if they scrolled down below the download button, because there was no reasonable notice and no unambiguous assent.

Basis	Click-wrap	Shrink-wrap	Browse-wrap
How assent is given	Clicking "I agree"	Opening the package or using the product	Merely using the website
Notice of terms	Terms shown or linked before assent	Terms inside the package	Terms in a hyperlink, often unnoticed
Enforceability	Generally enforceable	Enforceable if notice and opportunity to reject	Doubtful; enforceable only with reasonable notice
Example	Software installation, app sign-up	Boxed software	Website footer "terms of use"

4. Standard Form Contracts and Fairness

1. E-contracts are usually **standard form contracts** with no negotiation.
2. Indian courts may refuse to enforce **unconscionable** terms in standard form contracts (*Central Inland Water Transport Corp v Brojo Nath Ganguly*, Supreme Court, 1986).
3. **Consumer Protection Act, 2019, s.2(46)** defines "unfair contract", and consumer commissions can declare unfair terms null and void.
4. Online arbitration clauses and choice of forum clauses are generally enforced if fair.

5. Electronic Contracts and Stamp Duty and Evidence

1. Several States have amended their stamp laws to provide for e-stamping and stamp duty on electronic agreements.
2. E-contracts are proved as electronic records under the BSA, 2023 (see Topic 3.5).

6. Adjudication under the IT Act

A. Adjudicating Officer (Section 46)

1. The Central Government appoints an officer not below the rank of a **Director** to the Government of India or an equivalent officer of a State Government (in practice, the **Secretary, Information Technology** of each State) as an **adjudicating officer**.
2. **Jurisdiction:** to adjudicate whether any person has committed a contravention of the Act, rules, regulations, directions or orders that renders him liable to pay **penalty or compensation**, where the claim for injury or damage **does not exceed Rs. 5 crore**.
3. Where the claim **exceeds Rs. 5 crore**, the jurisdiction vests in the **competent civil court**.
4. **Procedure:** the adjudicating officer must give the person a reasonable opportunity of making a representation, hold an inquiry, and may impose penalty or award compensation. He has the powers of a civil court under the CPC for certain purposes, and the proceedings are judicial proceedings (s.46(5)).

B. Factors to be Taken into Account (Section 47)

In adjudging the quantum of compensation, the adjudicating officer shall have regard to: (a) the amount of gain or unfair advantage made as a result of the default; (b) the amount of loss caused to any person; and (c) the repetitive nature of the default.

C. Civil Court Not to Have Jurisdiction (Section 61)

No civil court shall have jurisdiction to entertain any suit or proceeding in respect of any matter which an adjudicating officer or the Appellate Tribunal is empowered to determine (but a civil court has jurisdiction for claims above Rs. 5 crore).

7. Appeal

A. Appellate Tribunal

1. **Section 48:** originally the Cyber Regulations Appellate Tribunal, renamed the **Cyber Appellate Tribunal** in 2008.
2. By the **Finance Act, 2017**, the Cyber Appellate Tribunal was merged with the **Telecom Disputes Settlement and Appellate Tribunal (TDSAT)**, which now hears appeals under the IT Act.

B. Appeal to the Appellate Tribunal (Section 57)

1. Any person aggrieved by an order of the Controller or an adjudicating officer may appeal to the Appellate Tribunal.
2. No appeal lies from an order made by an adjudicating officer **with the consent of the parties**.
3. **Limitation:** within **45 days** from the date on which a copy of the order is received, extendable for sufficient cause.
4. The Tribunal shall endeavour to dispose of the appeal within six months.

C. Procedure of the Tribunal (Section 58)

The Tribunal is not bound by the CPC but is guided by the principles of natural justice, and has the powers of a civil court for certain purposes.

D. Appeal to the High Court (Section 62)

1. Any person aggrieved by any decision or order of the Appellate Tribunal may appeal to the **High Court** within **60 days** from the date of communication of the decision, on any question of fact or law arising out of such order.
2. The High Court may allow a further period of up to 60 days for sufficient cause.

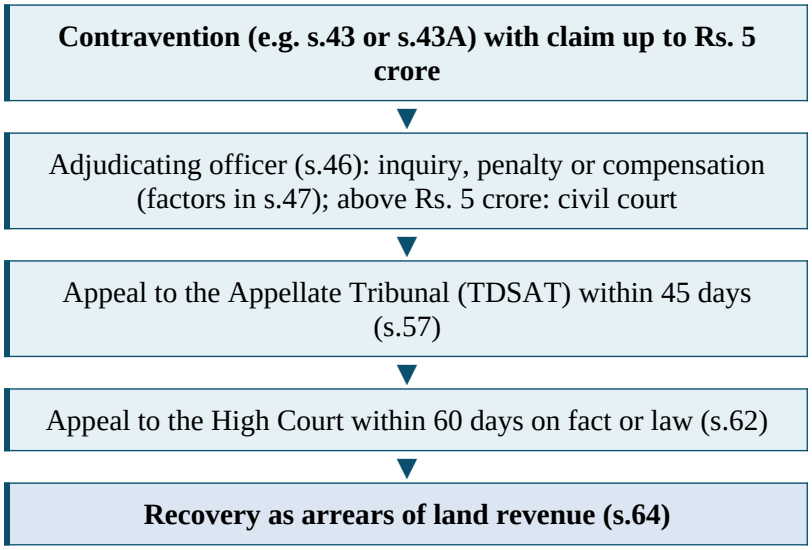
E. Compounding of Contraventions (Section 63)

Any contravention may be compounded by the Controller or an adjudicating officer, before or after the institution of adjudication proceedings, subject to conditions; but a person who commits the same or a similar contravention within three years of a previous compounding cannot have it compounded.

F. Recovery of Penalty or Compensation (Section 64)

A penalty imposed or compensation awarded, if not paid, shall be recovered as an **arrear of land revenue**, and the licence or the electronic signature certificate may be suspended till payment.

Flowchart: Adjudication and appeal under the IT Act



Examples

- A user signs up for an online course by clicking "I agree" to the terms, which include a clause on refunds. This click-wrap agreement is binding.
- A bank customer loses Rs. 10 lakh due to unauthorised transactions caused by the bank's poor security practices. He can file a complaint before the State's adjudicating officer under ss.43 and 43A; an appeal lies to the TDSAT and then to the High Court.

Case Laws

Case Law: Trimex International FZE Ltd v Vedanta Aluminium Ltd

Year: 2010 **Court:** Supreme Court of India

Facts / Background: Trimex, a Dubai company, offered by email to supply bauxite to Vedanta. The parties exchanged emails on the price, quantity, shipment and terms of payment, and Vedanta conveyed its acceptance by email. A formal written contract was never signed. When disputes arose, Trimex invoked the arbitration clause in the terms exchanged by email, and Vedanta argued that there was no concluded contract.

Legal Issue: Can a binding contract (including an arbitration agreement) be concluded through emails, without a formal signed contract?

Court's Decision / What the Court Held: The Court held that a valid and binding contract had been concluded through the exchange of emails, since the essential terms had been agreed upon. The mere fact that a formal contract was to be prepared and signed later did not affect the validity of the concluded contract. The Court appointed an arbitrator.

Relevance to the Topic: The leading Indian case confirming that **contracts formed by email are valid**, consistent with s.10A of the IT Act.

Case Law: Specht v Netscape Communications Corp

Year: 2002 **Court:** US Court of Appeals for the Second Circuit

Facts / Background: Netscape offered free software ("SmartDownload") on its website. Users could download it by clicking a "Download" button. The licence terms, including an arbitration clause, could be seen only if users scrolled down below the button to a link. Users sued Netscape for privacy violations, and Netscape sought to compel arbitration.

Legal Issue: Were the users bound by the licence terms, which they had not seen and to which they had not expressly agreed?

Court's Decision / What the Court Held: The court held that the users were not bound by the licence terms, because they had no reasonable notice of the terms and did not unambiguously manifest assent to them. A reference to terms on a submerged screen was not sufficient.

Relevance to the Topic: The leading case on the **unenforceability of browse-wrap agreements** without reasonable notice.

Important Points

- Section 10A: e-contracts are not unenforceable merely because they are electronic.
- Essentials of the Contract Act apply; time and place under s.13 IT Act.
- Click-wrap: generally enforceable; shrink-wrap: enforceable with notice and opportunity to reject (*ProCD*); browse-wrap: doubtful (*Specht v Netscape*).
- *Trimex v Vedanta* (2010): contracts by email are valid.
- Unfair terms: *Brojo Nath Ganguly* (1986); CPA 2019 s.2(46).
- Adjudicating officer (s.46): claims up to Rs. 5 crore; civil court beyond.
- Appeals: TDSAT within 45 days (s.57); High Court within 60 days (s.62).

Remember: The Cyber Appellate Tribunal was merged with the TDSAT by the Finance Act, 2017.

UNIT 3
Cyber Crimes

Topics in this Unit

3.1 Cyber contravention and cybercrime distinguished
3.2 Compensation and penalties under the Information Technology Act, 2000
3.3 Cyber offences under the Information Technology Act, 2000
3.4 Cyber offences under the Bharatiya Nyaya Sanhita, 2023 and other laws
3.5 Reporting and investigation of cybercrimes

Topic 3.1: Cyber contravention and cybercrime distinguished

Introduction

The IT Act deals with wrongful acts in cyberspace in two different ways. Some acts are treated as **cyber contraventions**: they are civil wrongs for which the wrongdoer must pay **penalty or compensation**, decided by an adjudicating officer (Chapter IX, ss.43–47). Other acts are treated as **cybercrimes (cyber offences)**: they are criminal offences punishable with **imprisonment and/or fine**, tried by criminal courts (Chapter XI, ss.65–78). The same act may be both a contravention and an offence; for example, unauthorised access is a contravention under s.43, and becomes an offence under s.66 if done "dishonestly or fraudulently".

Definition

Cyber contravention: a violation of the provisions of the IT Act, rules or directions that attracts civil liability in the form of penalty or compensation, adjudicated by an adjudicating officer.

Cybercrime: "any unlawful act where a computer, communication device or computer network is used to commit or facilitate the commission of crime" (definition used by the National Cyber Crime Reporting Portal). In the IT Act, cybercrimes are the offences in Chapter XI, and they also include offences under the BNS and other laws committed using computers.

Detailed Explanation

1. Cyber Contraventions (Chapter IX)

1. **Section 43:** damage to a computer, computer system or network (unauthorised access, downloading, introducing viruses, damage, disruption, denial of access, assisting unauthorised access, charging services to another's account, destroying or altering information, stealing source code): compensation to the person affected.
2. **Section 43A:** failure by a body corporate to protect sensitive personal data through reasonable security practices, causing wrongful loss or gain: compensation.
3. **Section 44:** failure to furnish information, returns or documents, or to maintain books of account: penalty.
4. **Section 45:** residuary penalty for contravention of rules or regulations for which no separate penalty is provided.

2. Cybercrimes (Chapter XI and Other Laws)

1. Offences under the IT Act: ss.65–67C, 68–74 (e.g. tampering with source documents, computer-related offences, identity theft, cheating by personation, violation of privacy, cyber terrorism, obscenity, child sexual abuse material).
2. Offences under the BNS committed through computers (e.g. cheating, forgery, stalking, defamation, extortion, criminal intimidation).

3. Offences under special laws (e.g. POCSO Act, Copyright Act).

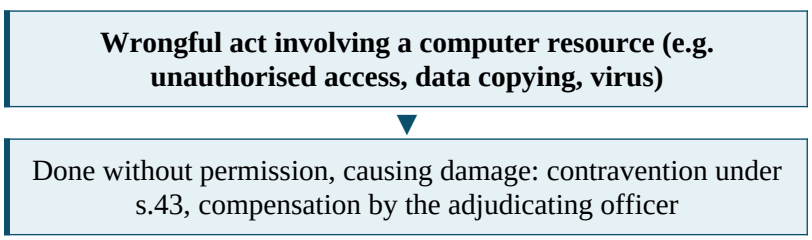
3. Distinction between Cyber Contravention and Cybercrime

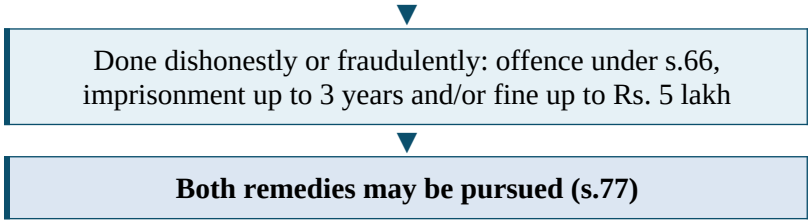
Basis	Cyber contravention	Cybercrime
Nature	Civil wrong	Criminal offence
Provisions of the IT Act	Chapter IX (ss.43–47)	Chapter XI (ss.65–78)
Mental element	Generally no mens rea required (e.g. s.43 applies to acts done "without permission")	Mens rea usually required (e.g. "dishonestly or fraudulently" in s.66)
Consequence	Penalty or compensation	Imprisonment and/or fine
Forum	Adjudicating officer (up to Rs. 5 crore) or civil court	Criminal courts (Magistrate or Sessions Court)
Procedure	Inquiry with opportunity of hearing (s.46); civil standard of proof	Investigation by police (Inspector rank, s.78) and trial under BNSS; proof beyond reasonable doubt
Initiation	Complaint by the affected person	FIR or complaint; State prosecutes
Appeal	To the Appellate Tribunal (TDSAT) and then the High Court	Under the BNSS to the Sessions Court or High Court
Compounding	By the Controller or adjudicating officer (s.63)	Compoundable only as allowed by s.77A
Example	Copying data from a computer without permission, without dishonest intent	Copying data dishonestly to sell it to a competitor (s.66)

4. Relationship: Section 43 and Section 66

1. Section 66 provides that if any person, **dishonestly or fraudulently**, does any act referred to in s.43, he shall be punishable with imprisonment up to three years, or fine up to Rs. 5 lakh, or both.
2. "Dishonestly" and "fraudulently" have the meanings given in the BNS (ss.2(7) and 2(9); formerly ss.24 and 25 IPC).
3. So the same act can lead to both civil liability (compensation under s.43) and criminal liability (s.66). Section 77 provides that compensation or penalty does not prevent punishment under any other law.

Flowchart: Contravention or offence?





5. Classification of Cybercrimes

Basis	Category	Examples
Target	Crimes against persons	Cyber stalking, harassment, defamation, morphing, sextortion, child sexual abuse material
	Crimes against property	Hacking, data theft, financial fraud, phishing, ransomware, IP theft
	Crimes against government/society	Cyber terrorism, attacks on critical infrastructure, spread of hate and misinformation
Role of the computer	Computer as target	Hacking, malware, denial of service
	Computer as tool	Online fraud, cyber stalking, distribution of obscene content
	Computer as incidental	Storing records of other crimes on a computer

Examples

- A student, out of curiosity, logs into his friend's email account without permission and reads emails, causing no loss. This is a contravention under s.43(a); he may be liable to pay compensation.
- An employee logs into his employer's server without authority and copies customer data to sell it to a competitor. This is an offence under s.66 (and possibly s.72A), as well as a contravention under s.43.

Case Laws

Case Law: Poona Auto Ancillaries Pvt Ltd v Punjab National Bank

Year: 2013 **Court:** Adjudicating Officer, Maharashtra (IT Secretary)

Facts / Background: The complainant company's bank account with Punjab National Bank was fraudulently accessed through internet banking after its official responded to a phishing email, and a large sum (about Rs. 80 lakh) was transferred out. The company filed a complaint under s.43 and s.43A of the IT Act before the adjudicating officer against the bank, alleging failure to follow security practices and know-your-customer norms. (Facts summarised briefly.)

Legal Issue: Is a bank liable to pay compensation under the IT Act for failure to maintain adequate security practices leading to a phishing fraud?

Court's Decision / What the Court Held: The adjudicating officer (Rajesh Aggarwal, Secretary IT, Maharashtra) held the bank liable for failure to follow due diligence and KYC norms and awarded substantial compensation to the company, reducing it partly for the company's own negligence in responding to the phishing email. (Orders of adjudicating officers are not binding precedents, but they illustrate how civil liability under the IT Act works.)

Relevance to the Topic: An example of **civil liability for a cyber contravention**, decided by an adjudicating officer, as distinct from criminal prosecution.

Important Points

- Cyber contraventions: civil wrongs; ss.43–45; penalty or compensation; adjudicating officer.
- Cybercrimes: criminal offences; ss.65–78 and the BNS; imprisonment and fine; criminal courts.
- Section 66 makes s.43 acts offences if done dishonestly or fraudulently.
- Section 77: compensation does not bar punishment under other laws.
- Classification: crimes against persons, property and government; computer as target, tool or incidental.

Remember: The key difference between s.43 and s.66 is the mental element: "dishonestly or fraudulently".

Topic 3.2: Compensation and penalties under the Information Technology Act, 2000

Introduction

Chapter IX of the IT Act (ss.43–47) provides **civil remedies** for wrongful acts involving computers and data. Its main purpose is to compensate persons who suffer loss because of unauthorised use of their computer systems or because a company fails to protect their personal data. These matters are decided by an **adjudicating officer** (for claims up to Rs. 5 crore) or a civil court (above Rs. 5 crore).

Definition

Compensation: an amount payable to the person affected by a contravention, to make good the loss suffered.

Penalty: an amount payable as a sanction for a contravention, not necessarily linked to the loss of a particular person.

Detailed Explanation

1. Section 43: Penalty and Compensation for Damage to Computer, Computer System, etc.

If any person, **without permission of the owner or any other person who is in charge** of a computer, computer system or computer network:

Clause	Act
(a)	Accesses or secures access to the computer, system, network or computer resource
(b)	Downloads, copies or extracts any data, database or information, including information held in removable storage
(c)	Introduces or causes to be introduced any computer contaminant or computer virus
(d)	Damages or causes to be damaged any computer, system, network, data, database or programme
(e)	Disrupts or causes disruption of any computer, system or network
(f)	Denies or causes the denial of access to any person authorised to access (e.g. denial of service attack)
(g)	Provides any assistance to any person to facilitate access in contravention of the Act
(h)	Charges the services availed of by a person to the account of another person by tampering with or manipulating the computer
(i)	Destroys, deletes or alters any information residing in a computer resource or diminishes its value or utility or affects it injuriously by any means
(j)	Steals, conceals, destroys or alters any computer source code used for a computer resource with an intention to cause damage

he shall be liable to pay **damages by way of compensation** to the person so affected. (The upper limit of Rs. 1 crore was removed by the 2008 Amendment.)

Explanations to s.43 define computer contaminant, computer database, computer virus, damage and computer source code.

2. Section 43A: Compensation for Failure to Protect Data

1. **Provision:** where a **body corporate**, possessing, dealing or handling any **sensitive personal data or information** in a computer resource which it owns, controls or operates, is **negligent** in implementing and maintaining **reasonable security practices and procedures**, and thereby causes **wrongful loss or wrongful gain** to any person, it shall be liable to pay damages by way of compensation to the person so affected.
2. **Body corporate:** includes a company, firm, sole proprietorship or other association of individuals engaged in commercial or professional activities.
3. **Reasonable security practices:** as agreed between the parties, or as specified by law, or as prescribed by the Central Government.
4. **Sensitive personal data or information (SPDI)** is defined in the **IT (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011** (Rule 3): passwords, financial information (bank account, credit/debit card details), physical, physiological and mental health condition, sexual orientation, medical records, and biometric information.
5. **Rule 8:** compliance with **ISO/IEC 27001** or an approved code of best practices is deemed to be reasonable security practices.
6. **Future position:** section 44(2) of the **Digital Personal Data Protection Act, 2023** omits s.43A from the IT Act. Under the phased commencement notified in November 2025, this omission is to take effect along with the main obligations of the DPDP Act (in the final phase, from May 2027); after that, protection of personal data will be governed by the DPDP Act and the Data Protection Board (see Unit V).

3. Section 44: Penalty for Failure to Furnish Information, Return, etc.

Penalties for failure to: (a) furnish documents, returns or reports to the Controller or a CA; (b) file returns or furnish information, books or documents within the time specified; (c) maintain books of account or records.

4. Section 45: Residuary Penalty

Whoever contravenes any rules or regulations made under the Act, for which no penalty has been separately provided, is liable to pay compensation to the person affected or a penalty.

5. Adjudication (Sections 46–47)

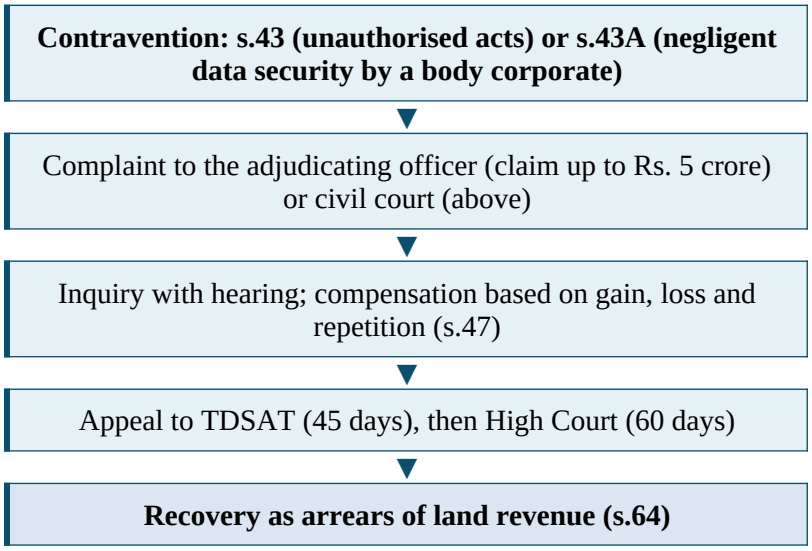
1. **Adjudicating officer (s.46):** appointed by the Central Government; for claims up to **Rs. 5 crore**. Beyond that, the civil court has jurisdiction.
2. **Factors for compensation (s.47):** (a) the gain or unfair advantage made as a result of the default; (b) the loss caused to any person; and (c) the repetitive nature of the default.
3. **Appeal:** to the Appellate Tribunal (TDSAT) within 45 days (s.57), and then to the High Court within 60 days (s.62).

6. Other Related Provisions

1. **Section 63:** compounding of contraventions.
2. **Section 64:** recovery of penalty or compensation as arrears of land revenue.
3. **Section 77:** compensation, penalties or confiscation under the Act do not prevent punishment or other penalties under any other law.
4. **Section 85:** where a contravention is committed by a company, every person in charge of and responsible for the conduct of its business is deemed guilty, unless he proves that the contravention took place without his knowledge or that he exercised all due diligence.

Section	Contravention	Liability
43	Unauthorised access, download, virus, damage, disruption, denial of access, etc.	Compensation (no upper limit)
43A	Body corporate's negligence in protecting sensitive personal data	Compensation (to be omitted under the DPDP Act)
44	Failure to furnish information or maintain records	Penalty
45	Residuary: contravention of rules or regulations	Compensation or penalty

Flowchart: Claiming compensation under the IT Act



Examples

- A person sends a virus by email to a company's computer system, which corrupts its database. The company can claim compensation under s.43(c) and (d).
- A hospital stores patients' medical records on an unsecured server, and they are leaked. Patients can claim compensation from the hospital under s.43A (until it is replaced by the DPDP Act regime).

Case Laws

There is no leading Supreme Court decision on the quantum of compensation under ss.43 and 43A. Most decisions come from adjudicating officers of the States and the Appellate Tribunal.

An example is *Poona Auto Ancillaries Pvt Ltd v Punjab National Bank* (2013), decided by the adjudicating officer in Maharashtra, in which a bank was held liable for a phishing fraud due to inadequate security and KYC practices (see Topic 3.1).

Important Points

- Section 43: ten kinds of unauthorised acts; compensation without upper limit.
- Section 43A: body corporate's negligent security of SPDI; SPDI Rules, 2011; ISO 27001.
- Section 43A will be omitted by s.44(2) of the DPDP Act, in the final phase of its commencement.
- Sections 44 and 45: penalties for failure to furnish information and residuary penalty.
- Adjudication: s.46 (up to Rs. 5 crore); s.47 factors.
- Section 77: civil liability does not bar criminal punishment.

Remember: Section 43 does not require any dishonest intention; acting "without permission" is enough.

Topic 3.3: Cyber offences under the Information Technology Act, 2000

Introduction

Chapter XI of the IT Act (ss.65–78) contains the **cyber offences**, which are punishable with imprisonment and/or fine. The original Act of 2000 contained only a few offences, but the IT (Amendment) Act, 2008 added many new ones, dealing with identity theft, cheating by personation, violation of privacy, cyber terrorism, sexually explicit material and child sexual abuse material. The Chapter also contains procedural provisions on extra-territorial application, confiscation, compounding, cognizability and investigation.

Definition

> **Cyber offence:** an act or omission punishable under the IT Act (or another law) that involves the use of a computer, computer resource, communication device or network as a target, a tool or an incidental means.

Detailed Explanation

1. Table of Offences under the IT Act

Section	Offence	Punishment
65	**Tampering with computer source documents:** knowingly or intentionally concealing, destroying or altering any computer source code, when it is required to be kept or maintained by law	Up to 3 years, or fine up to Rs. 2 lakh, or both
66	**Computer-related offences:** dishonestly or fraudulently doing any act referred to in s.43	Up to 3 years, or fine up to Rs. 5 lakh, or both
66A	Sending offensive messages through a communication service	**Struck down** in *Shreya Singhal* (2015)
66B	Dishonestly receiving or retaining any **stolen computer resource** or communication device	Up to 3 years, or fine up to Rs. 1 lakh, or both
66C	**Identity theft:** fraudulently or dishonestly making use of the electronic signature, password or any other unique identification feature of any other person	Up to 3 years and fine up to Rs. 1 lakh
66D	**Cheating by personation** by means of any communication device or computer resource	Up to 3 years and fine up to Rs. 1 lakh
66E	**Violation of privacy:** intentionally or knowingly capturing, publishing or transmitting the image of a private area of any person without consent, in circumstances violating privacy	Up to 3 years, or fine up to Rs. 2 lakh, or both

Section	Offence	Punishment
66F	**Cyber terrorism** (see below)	Imprisonment which may extend to imprisonment for life
67	Publishing or transmitting **obscene material** in electronic form (lascivious, appeals to the prurient interest, or tends to deprave and corrupt)	First conviction: up to 3 years and fine up to Rs. 5 lakh; subsequent: up to 5 years and fine up to Rs. 10 lakh
67A	Publishing or transmitting material containing a **sexually explicit act** or conduct	First conviction: up to 5 years and fine up to Rs. 10 lakh; subsequent: up to 7 years and fine up to Rs. 10 lakh
67B	Publishing, transmitting, creating, collecting, browsing, downloading, advertising, promoting, exchanging or distributing material depicting **children in sexually explicit acts**; online enticement of children; facilitating abuse of children online; recording abuse	First conviction: up to 5 years and fine up to Rs. 10 lakh; subsequent: up to 7 years and fine up to Rs. 10 lakh
67C	Intermediary intentionally or knowingly contravening directions on **preservation and retention** of information	Up to 3 years and fine
68	Failure to comply with **directions of the Controller**	Up to 2 years, or fine up to Rs. 1 lakh, or both
69(4)	Failure of a subscriber, intermediary or person in charge to assist the agency in **interception, monitoring or decryption**	Up to 7 years and fine
69A(3)	Failure of an intermediary to comply with a **blocking** direction	Up to 7 years and fine
69B(4)	Intermediary failing to provide technical assistance for **monitoring traffic data**	Up to 3 years and fine
70(3)	Securing or attempting to secure access to a **protected system**	Up to 10 years and fine
70B(7)	Failure to provide information to or comply with directions of **CERT-In**	Up to 1 year, or fine up to Rs. 1 lakh, or both
71	**Misrepresentation** to the Controller or a CA	Up to 2 years, or fine up to Rs. 1 lakh, or both
72	**Breach of confidentiality and privacy** by a person who has secured access to records under powers conferred by the Act	Up to 2 years, or fine up to Rs. 1 lakh, or both
72A	**Disclosure of personal information in breach of lawful contract**, with intent to cause or knowing that it is likely to cause wrongful loss or gain	Up to 3 years, or fine up to Rs. 5 lakh, or both
73	Publishing a false electronic signature certificate	Up to 2 years, or fine up to Rs. 1 lakh, or both

Section	Offence	Punishment
74	Publication of a certificate for a fraudulent purpose	Up to 2 years, or fine up to Rs. 1 lakh, or both

2. Important Offences Explained

A. Section 66: Hacking and Computer-Related Offences

1. Covers all acts in s.43 (unauthorised access, data theft, viruses, denial of service, etc.) when done dishonestly or fraudulently.
2. The original s.66 (2000) defined "hacking" as destroying, deleting or altering information with intent to cause wrongful loss; the 2008 Amendment replaced the term "hacking" with "computer-related offences".

B. Sections 66C and 66D: Identity Theft and Cheating by Personation

1. **Section 66C** covers misuse of passwords, OTPs, electronic signatures, biometric data, etc.
2. **Section 66D** covers online frauds in which the offender pretends to be someone else, e.g. fake bank officials calling for OTPs, fake profiles on matrimonial sites, impersonation of police or customs officials ("digital arrest" scams).

C. Section 66E: Violation of Privacy

1. "Private area" means the naked or undergarment-clad genitals, pubic area, buttocks or female breast.
2. "Under circumstances violating privacy" means circumstances in which a person can have a reasonable expectation that he could disrobe in privacy, or that his private area would not be visible to the public.
3. Covers hidden cameras in changing rooms, upskirt photographs, and sharing such images.

D. Section 66F: Cyber Terrorism

1. **Section 66F(1)(A):** whoever, with intent to threaten the unity, integrity, security or sovereignty of India or to strike terror in the people or any section of the people:
   - denies or causes the denial of access to any person authorised to access a computer resource; or
   - attempts to penetrate or access a computer resource without authorisation or exceeding authorised access; or
   - introduces or causes to introduce any computer contaminant,

and by means of such conduct causes or is likely to cause death or injuries to persons, damage to or destruction of property, or disruption of supplies or services essential to the life of the community, or adversely affects critical information infrastructure.

2. **Section 66F(1)(B):** knowingly or intentionally penetrating or accessing a computer resource without authorisation and obtaining access to information that is restricted for reasons of the security of the State or foreign relations, with reason to believe that it may be used to cause injury to the interests of sovereignty, security, friendly relations with foreign States, public order, decency or morality, etc.
3. **Punishment:** imprisonment which may extend to imprisonment for life.

E. Sections 67, 67A and 67B: Obscenity and Sexual Content

1. **Section 67:** follows the language of the old test of obscenity ("lascivious", "appeals to the prurient interest", "deprave and corrupt"); courts now apply the **community standards** test (*Aveek Sarkar v State of West Bengal*, 2014).
2. **Section 67A:** sexually explicit acts (more serious than mere obscenity).
3. **Section 67B:** child sexual abuse material (CSAM), including browsing and downloading. In *Just Rights for Children Alliance v S. Harish* (2024), the Supreme Court held that even storing or possessing child sexual exploitative and abuse material without deleting or reporting it can be an offence under s.15 of the POCSO Act and s.67B of the IT Act, and suggested the term "child sexual exploitative and abuse material" (CSEAM) instead of "child pornography".
4. **Exception (s.67, 67A, 67B provisos):** publications justified as being for the public good, e.g. in the interest of science, literature, art or learning, or kept or used for bona fide heritage or religious purposes.

3. Procedural Provisions in Chapter XI

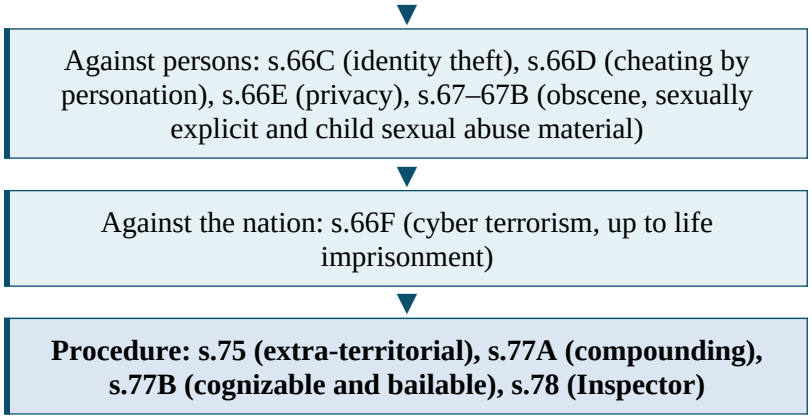
Section	Provision
75	The Act applies to offences or contraventions committed outside India by any person, if the act involves a computer system or network located in India
76	Confiscation of any computer, computer system, floppies, CDs, tape drives or accessories used in the contravention
77	Compensation, penalties or confiscation do not interfere with other punishment
77A	**Compounding of offences:** a court of competent jurisdiction may compound offences other than those for which punishment of life imprisonment or imprisonment exceeding three years is provided; but not where the accused is liable to enhanced punishment due to a previous conviction, or where the offence affects the socio-economic conditions of the country, or is committed against a child below 18 years or a woman
77B	Notwithstanding the CrPC (now BNSS), offences punishable with imprisonment of **three years and above** are **cognizable**, and offences punishable with imprisonment of **three years** are **bailable**
78	Offences may be investigated by a police officer not below the rank of **Inspector**

4. Section 85: Offences by Companies

Every person who, at the time of the contravention, was in charge of and responsible to the company for the conduct of its business is deemed guilty, unless he proves that it was committed without his knowledge or that he exercised all due diligence. Directors, managers and other officers are also liable if the offence was committed with their consent or connivance or due to their neglect.

Flowchart: Major IT Act offences by target

Against computer systems: s.65 (source code), s.66 (hacking and data theft), s.70 (protected systems)



Examples

- A person obtains another's net-banking password by phishing and transfers money from that account. This is identity theft (s.66C), cheating by personation (s.66D) and a computer-related offence (s.66), as well as cheating under the BNS.
- A person secretly records a woman changing clothes in a trial room and uploads the video. This is an offence under s.66E and s.67A of the IT Act, and voyeurism under s.77 BNS.

Case Laws

Case Law: Shreya Singhal v Union of India (on section 66A)

Year: 2015 **Court:** Supreme Court of India

Facts / Background: Section 66A punished sending offensive messages through communication services, using terms such as "grossly offensive", "menacing", "annoyance", "inconvenience" and "insult". It was widely misused to arrest people for social media posts (see Topic 1.2 for the full facts).

Legal Issue: Is s.66A constitutional?

Court's Decision / What the Court Held: The Court struck down s.66A as wholly unconstitutional, because its terms were vague and overbroad, it had no proximate relationship with public order, it did not fall within any ground in Article 19(2), and it had a chilling effect on free speech. In *People's Union for Civil Liberties v Union of India* (2021–2022), the Supreme Court noted with concern that cases were still being registered under s.66A, and directed that no person should be prosecuted under it and pending cases should be dropped.

Relevance to the Topic: The **most important judgment on the constitutionality of cyber offences**.

Case Law: State of Tamil Nadu v Suhas Katti

Year: 2004 **Court:** Court of the Additional Chief Metropolitan Magistrate, Egmore, Chennai

Facts / Background: The accused, a family friend of a divorced woman who had refused

his proposal of marriage, posted obscene, defamatory and annoying messages about her in Yahoo! message groups, along with her contact details, leading to harassing calls. The Chennai Cyber Crime Cell traced him using IP address evidence.

Legal Issue: Is the accused guilty of offences under the IT Act and the IPC for posting obscene and defamatory messages online?

Court's Decision / What the Court Held: The Court convicted the accused under s.67 of the IT Act and ss.469 and 509 of the IPC, and sentenced him to rigorous imprisonment and fine. The case was decided within about seven months of the FIR.

Relevance to the Topic: Widely regarded as the **first conviction under the IT Act** in India, showing the use of electronic evidence in cyber harassment cases.

Important Points

- Section 65: tampering with source code (3 years); s.66: computer-related offences (3 years, Rs. 5 lakh).
- Section 66A was struck down (*Shreya Singhal*, 2015).
- Sections 66B–66E: stolen computer resource, identity theft, cheating by personation, privacy violation.
- Section 66F: cyber terrorism, punishable up to life imprisonment.
- Sections 67, 67A, 67B: obscene, sexually explicit and child sexual abuse material; *Just Rights for Children Alliance* (2024).
- Sections 69–70B: offences of non-compliance with interception, blocking, monitoring, protected systems and CERT-In.
- Sections 72 and 72A: breach of confidentiality and disclosure in breach of contract.
- Procedure: ss.75, 76, 77A, 77B, 78; s.85 (companies).

Remember: Under s.77B, IT Act offences with three years or more are cognizable, and those with three years are bailable.

Topic 3.4: Cyber offences under the Bharatiya Nyaya Sanhita, 2023 and other laws

Introduction

The IT Act is not the only law that punishes cybercrimes. Many traditional offences, such as cheating, forgery, defamation, extortion, stalking and criminal intimidation, can be committed using computers and the internet. The **Bharatiya Nyaya Sanhita, 2023 (BNS)**, which replaced the Indian Penal Code from **1 July 2024**, expressly includes electronic means in many offences. Other laws such as the POCSO Act, the Copyright Act and the Telecommunications Act also cover cybercrimes. The relationship between these laws and the IT Act was explained by the Supreme Court in *Sharat Babu Digumarti* (2017).

Definition

Section 2(8), BNS: "document" means any matter expressed or described upon any substance by means of letters, figures or marks, or any other means, or by more than one of those means, intended to be used, or which may be used, as evidence of that matter, and includes electronic and digital records.

Section 2(39), BNS: words and expressions used but not defined in the BNS but defined in the IT Act, 2000 and the BNSS, 2023 shall have the meanings respectively assigned to them in those laws.

Detailed Explanation

1. Cyber-Related Offences under the BNS

BNS section	Offence	Old IPC section	Cyber relevance
1(5)	Extra-territorial application to any person committing an offence targeting a computer resource located in India	New	Foreign cyber attackers
77	Voyeurism	354C	Capturing or sharing private images
78	Stalking, including monitoring a woman's use of the internet, email or any other form of electronic communication	354D	Cyber stalking
111	Organised crime, which expressly includes cyber-crimes having severe consequences	New	Organised cyber fraud syndicates
152	Acts endangering sovereignty, unity and integrity of India, including by electronic communication	New (replaces sedition, 124A)	Online secessionist content

BNS section	Offence	Old IPC section	Cyber relevance
196	Promoting enmity between groups, including by electronic communication	153A	Online hate speech
197	Imputations prejudicial to national integration, including making or publishing false or misleading information	153B	Online misinformation
294	Sale, etc. of obscene books, including display of obscene content in electronic form	292	Online obscenity
303	Theft	378–379	Theft of devices
308	Extortion	383–384	Sextortion, ransomware
316	Criminal breach of trust	405–406	Misappropriation of data or funds
318	Cheating	415–420	Online fraud, phishing, investment scams
319	Cheating by personation	416, 419	Fake profiles, impersonation
335–336	Making a false document, including electronic record; forgery	463–464	Fake websites, forged emails
340	Forged document or electronic record used as genuine	471	Using forged e-documents
351	Criminal intimidation, including by anonymous communication	503, 506, 507	Online threats
353	Statements conducing to public mischief, including false information through electronic means	505	Fake news causing alarm
356	Defamation, including by electronic means	499–500	Online defamation

2. Other Laws Covering Cybercrimes

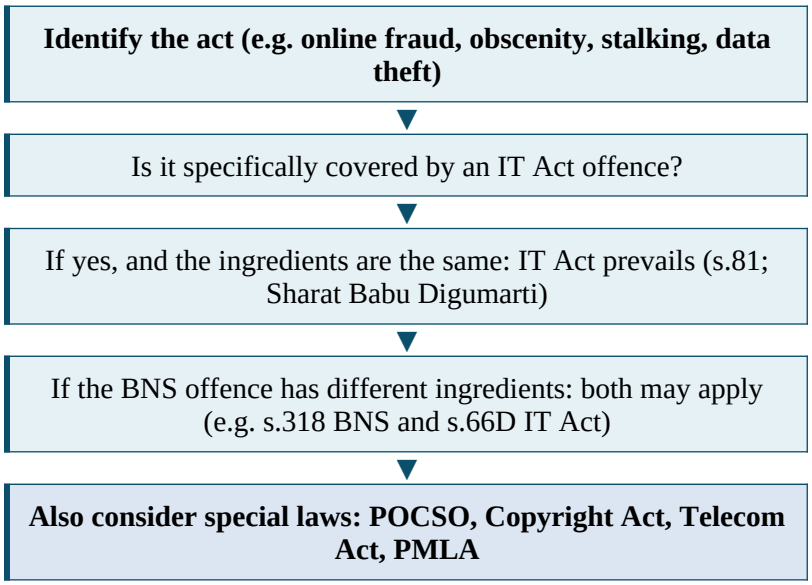
1. **Protection of Children from Sexual Offences Act, 2012:**
   - s.11: sexual harassment, including showing pornographic material to a child or stalking a child through electronic means.
   - s.13–15: use of a child for pornographic purposes; storage or possession of pornographic material involving a child (s.15, as amended in 2019).
2. **Copyright Act, 1957:** s.63 (infringement), s.65A (circumvention of technological protection measures), s.65B (removal of rights management information); software piracy.
3. **Trade Marks Act, 1999:** falsification of marks; counterfeit goods sold online.
4. **Indecent Representation of Women (Prohibition) Act, 1986.**

5. **Telecommunications Act, 2023:** s.42 makes it an offence to fraudulently obtain SIM cards or telecom identifiers, to use telecom services by impersonation, and to tamper with telecom identifiers.
6. **Payment and Settlement Systems Act, 2007** and **Prevention of Money Laundering Act, 2002** (laundering proceeds of cyber fraud).
7. **Unlawful Activities (Prevention) Act, 1967:** terrorist use of the internet.
8. **Promotion and Regulation of Online Gaming Act, 2025:** prohibits online money games and their advertisement and related banking transactions.

3. Relationship between the IT Act and the BNS (IPC)

1. **Section 81, IT Act:** the provisions of the IT Act shall have effect notwithstanding anything inconsistent therewith contained in any other law for the time being in force (with a proviso protecting rights under the Copyright Act and the Patents Act).
2. **Special law prevails:** where an act is specifically covered by an IT Act offence (e.g. electronic obscenity under s.67), the accused cannot be prosecuted for the same act under the general provision of the IPC/BNS (e.g. s.292 IPC/s.294 BNS) (*Sharat Babu Digumarti*, 2017).
3. **Where the offences are different:** where the BNS offence has different ingredients (e.g. cheating under s.318 and identity theft under s.66C), both may be invoked.
4. **Gagan Harsh Sharma v State of Maharashtra (Bombay High Court, 2018):** in a case of theft of data and software by employees, the High Court followed *Sharat Babu Digumarti* and held that where the act is covered by ss.43 and 66 of the IT Act, the accused could not also be charged under the IPC for theft and criminal breach of trust for the same act, since the IT Act has overriding effect.

Flowchart: Choosing the right law for a cybercrime



Examples

- A fraudster calls a victim pretending to be a bank official, obtains her OTP and empties her account. He is liable under ss.318 and 319 BNS (cheating and cheating by personation) and ss.66C and 66D of the IT Act.
- A man repeatedly sends threatening messages to a woman from fake accounts and tracks her online activity. He is liable for stalking (s.78 BNS) and criminal intimidation (s.351 BNS).

Case Laws

Case Law: Sharat Babu Digumarti v Government (NCT of Delhi)

Year: 2016 (reported in 2017) **Court:** Supreme Court of India

Facts / Background: In 2004, an obscene video clip involving two school students in Delhi was listed for sale on Bazee.com, an online auction website owned by an Indian subsidiary of eBay. The appellant was the senior manager of the website company. The Delhi Police registered a case, and he was charged under s.292 IPC (sale of obscene objects) and s.67 of the IT Act. The trial court dropped the charges under s.67 of the IT Act against him, but framed a charge under s.292 IPC.

Legal Issue: When the offence of electronic obscenity is specifically covered by s.67 of the IT Act, can the accused be prosecuted under s.292 IPC for the same act?

Court's Decision / What the Court Held: The Court held that once an offence is covered by s.67 of the IT Act, which is a special law dealing with obscenity in electronic form, the accused cannot be prosecuted under s.292 IPC for the same act. Section 81 of the IT Act gives it overriding effect, and the special law prevails over the general law. Since the charge under s.67 had been dropped against him, he could not be tried under s.292 IPC. The proceedings were quashed.

Relevance to the Topic: The leading case on the **relationship between the IT Act and the IPC (now BNS)**, and one of the landmark decisions prescribed for study.

Important Points

- The BNS (in force 1 July 2024) includes electronic means in many offences: ss.77, 78, 111, 152, 196, 197, 294, 318, 319, 335–336, 351, 353, 356.
- BNS s.1(5) applies to offences outside India targeting computer resources in India.
- Other laws: POCSO Act, Copyright Act, Trade Marks Act, Telecom Act 2023 (s.42), PMLA, UAPA, Online Gaming Act 2025.
- Section 81 IT Act: overriding effect.
- *Sharat Babu Digumarti* (2017): where s.67 IT Act applies, s.292 IPC cannot be invoked for the same act.

Remember: A special law (IT Act) prevails over a general law (BNS/IPC) for the same act with the same ingredients.

Topic 3.5: Reporting and investigation of cybercrimes

Introduction

Cybercrimes are difficult to investigate because offenders may be anonymous, located abroad, and able to destroy evidence quickly. Special reporting mechanisms, trained police units, digital forensic tools and legal rules on electronic evidence are needed. In India, cybercrimes can be reported to the police (including cyber police stations) and through the **National Cyber Crime Reporting Portal** and the helpline **1930**. Investigation is governed by the IT Act and the **Bharatiya Nagarik Suraksha Sanhita, 2023 (BNSS)**, and proof by the **Bharatiya Sakshya Adhiniyam, 2023 (BSA)**, as interpreted in *Anvar P.V.* (2014) and *Arjun Panditrao Khotkar* (2020).

Definition

Digital forensics: the scientific process of identifying, preserving, collecting, analysing and presenting digital evidence in a manner that is legally admissible.

Electronic evidence: information of probative value stored or transmitted in electronic form, such as emails, computer files, server logs, CCTV footage, call records and messages.

Detailed Explanation

1. Reporting Mechanisms

1. **National Cyber Crime Reporting Portal (cybercrime.gov.in):** run by the Ministry of Home Affairs; allows online complaints of all cybercrimes, with a special focus on crimes against women and children (anonymous reporting is allowed for child sexual abuse material and rape/gang rape content).
2. **National Cyber Crime Helpline 1930:** for immediate reporting of financial cyber fraud, so that the money can be frozen quickly through the **Citizen Financial Cyber Fraud Reporting and Management System**.
3. **Indian Cyber Crime Coordination Centre (I4C):** set up by the Ministry of Home Affairs (operational from 2020) to coordinate the efforts of law enforcement agencies; it operates the portal and the "Sahyog" portal for sending notices to intermediaries.
4. **Police stations and cyber police stations/cells:** an FIR can be registered at any police station; the BNSS provides for **Zero FIR** (s.173(1)) irrespective of jurisdiction, and **e-FIR** (information given by electronic communication, to be signed within three days).
5. **CERT-In:** cyber security incidents must be reported to CERT-In within six hours by service providers and organisations (see Topic 5.4).
6. **Banks:** under RBI directions, customers who report unauthorised electronic transactions promptly have limited or zero liability.

2. Investigation

A. Powers under the IT Act

1. **Section 78:** investigation by a police officer not below the rank of **Inspector**.
2. **Section 80:** any police officer not below the rank of Inspector, or any other officer authorised by the Central or State Government, may enter any **public place**, search and **arrest without warrant** any person found there who is reasonably suspected of having committed, or of committing, or of being about to commit any offence under the Act.
3. **Section 69:** interception, monitoring or decryption of information by authorised agencies, subject to the 2009 Interception Rules.
4. **Section 69B:** monitoring and collection of traffic data for cyber security.
5. **Section 67C:** intermediaries must preserve and retain information as prescribed.
6. **Section 79A:** the Central Government may notify **Examiners of Electronic Evidence** (e.g. certain forensic laboratories) to give expert opinion.

B. Powers under the BNSS, 2023

1. **Section 94 (s.91 CrPC):** summons to produce documents, including **electronic communication and devices** likely to contain digital evidence.
2. **Section 105: audio-video recording** of search and seizure, including preparation of the list of seized items, preferably on a mobile phone, to be forwarded to the Magistrate.
3. **Section 176(3):** for offences punishable with seven years or more, a **forensic expert** shall visit the crime scene to collect evidence, and the process shall be video-graphed.
4. **Section 185:** search by police officer; **Section 497:** disposal of property, including electronic devices.
5. **Electronic modes:** summons and warrants can be served electronically; trials, inquiries and proceedings may be held in electronic mode (s.530).

C. Steps in a Cybercrime Investigation

1. Receipt of complaint and registration of FIR.
2. Preservation of evidence: sending **preservation requests** to intermediaries and service providers.
3. Collection of data: IP addresses, logs, subscriber details, transaction records, CCTV, call detail records, through notices to intermediaries, banks and telecom companies.
4. Seizure of devices, following proper procedure (hash values, write blockers, chain of custody).
5. Forensic analysis by cyber forensic laboratories.
6. Identification and arrest of the accused.
7. International cooperation through **Mutual Legal Assistance Treaties (MLATs)**, letters rogatory (s.112 BNSS; s.166A CrPC), INTERPOL, and direct requests to foreign service providers.
8. Filing of the charge sheet with electronic evidence and the required certificates.

3. Proof of Electronic Evidence

A. The Law

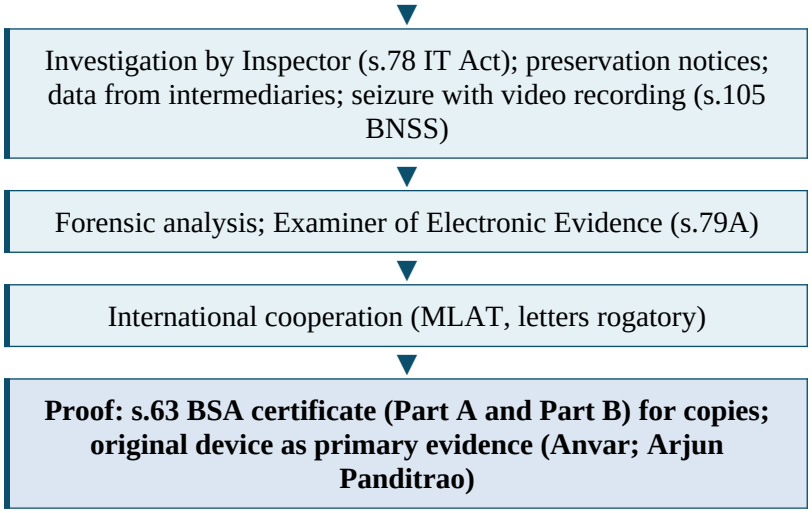
1. **Bharatiya Sakshya Adhiniyam, 2023 (in force 1 July 2024):**
   - s.2(1)(d): "document" includes electronic and digital records.
   - s.57: primary evidence; Explanations provide that an electronic or digital record is primary evidence in certain cases, e.g. where it is created or stored in multiple files, each is primary evidence; and where it is produced from proper custody.
   - **s.61:** an electronic or digital record shall not be denied admissibility on the ground that it is electronic, and it shall have the same legal effect as a paper document, subject to s.63.
   - **s.63 (formerly s.65B of the Indian Evidence Act):** any information contained in an electronic record which is printed, stored, recorded or copied in optical or magnetic media or semiconductor memory produced by a computer or communication device shall be deemed to be a document and admissible, if the conditions in s.63(2) are satisfied, and a **certificate** is produced in the form given in the Schedule. The BSA requires the certificate in two parts: **Part A** signed by the person in charge of the device and **Part B** signed by an **expert**.
2. **Conditions (s.63(2)), in brief:** the computer was used regularly to store or process information; information of the kind was regularly fed into it; the computer was operating properly; and the information reproduces or is derived from information fed into it in the ordinary course.

B. Judicial Interpretation

Case	Year	Holding
State (NCT of Delhi) v Navjot Sandhu (Parliament attack case)	2005	Electronic records could be proved by secondary evidence under ss.63 and 65 IEA even without a s.65B certificate
Anvar P.V. v P.K. Basheer	2014	Overruled *Navjot Sandhu* on this point: a s.65B certificate is mandatory for secondary electronic evidence; ss.65A and 65B are a complete code
Tomaso Bruno v State of UP	2015	Failure to produce CCTV footage can lead to an adverse inference against the prosecution
Shafhi Mohammad v State of HP	2018	Certificate not mandatory where the party is not in possession of the device (later overruled)
Arjun Panditrao Khotkar v Kailash Kushanrao Gorantyal	2020	Reaffirmed *Anvar*; overruled *Shafhi Mohammad*; certificate is a condition precedent; not needed if the original device is produced; can be produced at a later stage of trial

Flowchart: From complaint to proof

Report on cybercrime.gov.in, helpline 1930, or police (Zero FIR, e-FIR)



Examples

- A victim of a UPI fraud calls 1930 within an hour. The transaction is flagged, and the bank freezes the amount in the fraudster's account before it is withdrawn.
- The prosecution relies on a CD containing a copy of a WhatsApp chat. It must produce a certificate under s.63 BSA; otherwise the CD is not admissible.

Case Laws

Case Law: Anvar P.V. v P.K. Basheer

Year: 2014 **Court:** Supreme Court of India (three judges)

Facts / Background: In an election petition challenging the election of P.K. Basheer to the Kerala Legislative Assembly, the petitioner (Anvar P.V.) alleged that the returned candidate had indulged in corrupt practices by making defamatory speeches and songs about him. To prove this, he produced CDs containing recordings of speeches and songs. No certificate under s.65B of the Evidence Act was produced. The High Court dismissed the election petition.

Legal Issue: Can secondary evidence of an electronic record (such as a CD) be admitted without a certificate under s.65B(4) of the Evidence Act?

Court's Decision / What the Court Held: The Court held that ss.65A and 65B are a complete code for the admissibility of electronic records. Secondary evidence of an electronic record (such as a CD, VCD or pen drive) is admissible only if accompanied by a certificate under s.65B(4), identifying the electronic record, describing how it was produced, and certifying the conditions of s.65B(2), signed by a person occupying a responsible position. The certificate must accompany the electronic record when it is produced. Without the certificate, the CDs were inadmissible. If the primary evidence (the original device) is produced, the certificate is not needed, and it can be proved under s.62. The Court overruled *Navjot Sandhu* to the extent that it held otherwise.

Relevance to the Topic: The foundational case on the **mandatory certificate for electronic evidence**, and one of the landmark decisions prescribed for study. (The rule now applies under s.63 BSA.)

Case Law: Arjun Panditrao Khotkar v Kailash Kushanrao Gorantyal

Year: 2020 **Court:** Supreme Court of India (three judges)

Facts / Background: In an election dispute in Maharashtra, the question arose whether nomination papers had been filed within time. The election petitioner relied on video recordings from CCTV cameras at the office of the Returning Officer. The Election Commission officials did not issue a s.65B certificate despite requests. The High Court admitted the evidence relying on other evidence. Because of a conflict between *Anvar* (2014) and *Shafhi Mohammad* (2018), the matter was referred to a three-judge bench.

Legal Issue: Is a certificate under s.65B(4) mandatory, and what happens if the person who needs it cannot obtain it?

Court's Decision / What the Court Held: The Court held that the certificate under s.65B(4) is a **condition precedent** to the admissibility of secondary electronic evidence, reaffirming *Anvar*, and overruled *Shafhi Mohammad* and the earlier decision in *Tomaso Bruno* to the extent that it suggested otherwise. The certificate is not required if the original device itself is produced (e.g. the owner of a laptop or mobile phone steps into the witness box and proves it). Where a person has asked for the certificate but the authority refuses or fails to give it, the party may apply to the court to summon the person to produce it. The certificate may be produced at any stage of the trial, provided it does not cause prejudice to the accused. The Court also directed that service providers should preserve call records and other data and gave guidance for rules on retention of electronic records. On the facts, the High Court's judgment was upheld.

Relevance to the Topic: The **authoritative statement of the law on electronic evidence** in India, and one of the landmark decisions prescribed for study.

Important Points

- Reporting: cybercrime.gov.in; helpline 1930; I4C; Zero FIR (s.173 BNSS); CERT-In reporting.
- IT Act: s.78 (Inspector), s.80 (search and arrest in public places), s.69, 69B, 67C, s.79A (Examiner of Electronic Evidence).
- BNSS: s.94 (production of electronic documents), s.105 (video recording of search), s.176(3) (forensic team), s.530 (electronic proceedings).
- BSA: s.61 (electronic records admissible), s.63 (certificate in Part A and Part B).
- *Anvar* (2014): s.65B certificate mandatory for secondary electronic evidence.
- *Arjun Panditrao* (2020): certificate is a condition precedent; not needed for the original device; can be produced later.

Remember: Under s.63 of the BSA, the certificate must be signed by the person in charge of the device (Part A) and by an expert (Part B).

UNIT 4
Intermediaries, Content Regulation and Emerging Technologies

Topics in this Unit

4.1 Intermediary; safe harbour and due diligence
4.2 Intermediary Guidelines and Digital Media Ethics Code Rules, 2021
4.3 Freedom of speech in cyberspace; defamation, hate speech and obscenity
4.4 Deepfakes, misinformation and internet shutdowns
4.5 Artificial intelligence, algorithmic accountability and platform governance
4.6 Cloud computing, Internet of Things, blockchain and virtual digital assets

Topic 4.1: Intermediary; safe harbour and due diligence

Introduction

Most online content is created by users, but it is hosted, transmitted and made available by **intermediaries** such as internet service providers, social media platforms, search engines, messaging services and e-commerce marketplaces. If intermediaries were liable for every unlawful item posted by billions of users, the internet could not function. At the same time, they cannot be allowed to ignore illegal content. Section 79 of the IT Act resolves this by giving intermediaries a **"safe harbour"** (conditional immunity) from liability for third-party content, provided they act as neutral conduits and observe **due diligence**.

Definition

Section 2(1)(w), IT Act: "intermediary", with respect to any particular electronic record, means "any person who on behalf of another person receives, stores or transmits that record or provides any service with respect to that record and includes telecom service providers, network service providers, internet service providers, web-hosting service providers, search engines, online payment sites, online-auction sites, online-market places and cyber cafes."

Safe harbour: the legal protection given to an intermediary from liability for third-party information, data or communication links made available or hosted by it, subject to the conditions of s.79.

Detailed Explanation

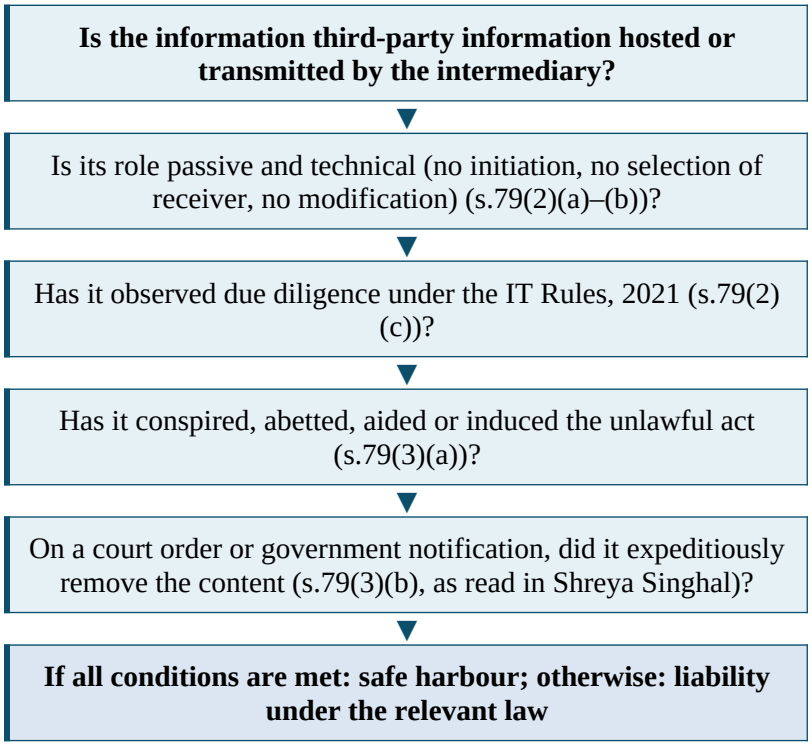
1. Section 79: Exemption from Liability of Intermediary in Certain Cases

1. **Section 79(1):** notwithstanding anything in any law for the time being in force, but subject to sub-sections (2) and (3), an intermediary shall not be liable for any **third party information, data or communication link** made available or hosted by it.
2. **Section 79(2): conditions for safe harbour.** The exemption applies if:
   - (a) the function of the intermediary is limited to providing access to a communication system over which third-party information is transmitted or temporarily stored or hosted; **or**
   - (b) the intermediary does not (i) initiate the transmission, (ii) select the receiver of the transmission, and (iii) select or modify the information contained in the transmission; **and**
   - (c) the intermediary observes **due diligence** while discharging its duties under the Act and observes the guidelines prescribed by the Central Government.
3. **Section 79(3): loss of safe harbour.** The exemption does not apply if:
   - (a) the intermediary has **conspired or abetted or aided or induced**, whether by threats or promise or otherwise, the commission of the unlawful act; or
   - (b) upon receiving **actual knowledge**, or on being notified by the appropriate Government or its agency, that any information, data or communication link residing

in or connected to a computer resource controlled by it is being used to commit an unlawful act, the intermediary fails to **expeditiously remove or disable access** to that material, without vitiating the evidence in any manner.

4. **Explanation:** "third party information" means any information dealt with by an intermediary in its capacity as an intermediary.

Flowchart: Does the intermediary get safe harbour under s.79?



2. Due Diligence

1. Due diligence obligations are prescribed by the **IT (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021** (replacing the Intermediaries Guidelines Rules, 2011): publishing rules and privacy policies, informing users not to host prohibited content, removing unlawful content on receiving orders, grievance redressal, assisting law enforcement, etc. (see Topic 4.2).
2. **Rule 7, IT Rules, 2021:** where an intermediary fails to observe the Rules, s.79(1) shall not apply, and it shall be liable for punishment under any law.

3. Shreya Singhal: Reading Down Section 79(3)(b)

1. The Supreme Court held that "actual knowledge" in s.79(3)(b) means knowledge through a **court order**, or a **notification by the appropriate Government** or its agency, and that the unlawful act must relate to the grounds in **Article 19(2)**.
2. **Reason:** intermediaries such as Google and Facebook receive millions of requests, and it would be very difficult for them to judge which requests are legitimate. Requiring them to act on private complaints would make them censors.
3. **Exception for intellectual property:** in *MySpace Inc v Super Cassettes Industries Ltd* (Delhi High Court, Division Bench, 2016), the Court held that for copyright infringement, a specific notice from the copyright owner identifying the infringing content is sufficient

to give actual knowledge, and a court order is not necessary; but intermediaries cannot be required to monitor all content in advance.

4. Active and Passive Intermediaries

1. **Passive intermediary:** merely provides a platform or transmission; does not control or modify content. Entitled to safe harbour.
2. **Active intermediary:** takes part in the unlawful activity, e.g. by selecting, modifying, guaranteeing, promoting or selling the goods or content. It may lose safe harbour (*Christian Louboutin v Nakul Bajaj*, 2018).
3. **E-commerce marketplaces:** in *Amazon Seller Services v Amway India* (Delhi High Court, 2020), value-added services alone were held not to take away intermediary status (see Topic 2.5).

5. Sahyog Portal and the 2025 Amendment

1. The **Sahyog portal**, run by the Indian Cyber Crime Coordination Centre (MHA), is used by authorised government agencies to send notices under s.79(3)(b) and Rule 3(1)(d) to intermediaries for removal of unlawful content.
2. In *X Corp v Union of India* (Karnataka High Court, single judge, 24 September 2025), the Court dismissed X Corp's challenge and held that s.79(3)(b) and Rule 3(1)(d) independently empower the Government to issue takedown notices, separate from blocking under s.69A, and that the Sahyog portal is a facilitation mechanism, not a tool of censorship. An appeal by X Corp is pending before a Division Bench.
3. **Amendment of Rule 3(1)(d) (notified 22 October 2025, in force 15 November 2025):** takedown intimations by the Government must be issued only by a senior officer (not below the rank of Joint Secretary, or Director where no Joint Secretary is appointed; in the police, not below the rank of Deputy Inspector General), must be reasoned and specify the legal basis and the content, and are subject to monthly review by a Secretary-level officer.

6. Comparative Position

Jurisdiction	Law	Approach
India	IT Act s.79; IT Rules, 2021	Conditional safe harbour subject to due diligence and takedown on order/notification
USA	Communications Decency Act, 1996, s.230; DMCA, 1998, s.512	Broad immunity for third-party content; notice-and-takedown for copyright
European Union	E-Commerce Directive, 2000 (Arts 12–15); Digital Services Act, 2022	Mere conduit, caching and hosting exemptions; no general monitoring obligation; added duties for very large online platforms

Examples

- A user uploads a defamatory video on a video-sharing platform. The platform is not liable if it removes the video expeditiously on receiving a court order.
- An online store that verifies, stores, packages and sells counterfeit goods, and guarantees their authenticity, is an active participant and may lose safe harbour.

Case Laws

Case Law: Shreya Singhal v Union of India (on section 79)

Year: 2015 **Court:** Supreme Court of India

Facts / Background: Along with s.66A, the petitioners challenged s.79(3)(b) of the IT Act and Rule 3(4) of the Intermediaries Guidelines Rules, 2011, which required intermediaries to remove content within 36 hours of "actual knowledge" (including by private complaints).

Legal Issue: Does s.79(3)(b), requiring intermediaries to remove content on "actual knowledge", violate Article 19(1)(a)?

Court's Decision / What the Court Held: The Court read down s.79(3)(b) and Rule 3(4) to mean that an intermediary must remove or disable access to content only upon receiving actual knowledge from a **court order**, or on being notified by the **appropriate Government** or its agency, and that such orders must be confined to the subjects in Article 19(2). Otherwise, intermediaries would have to judge the legitimacy of millions of requests.

Relevance to the Topic: The **foundation of intermediary liability law** in India, and one of the landmark decisions prescribed for study.

Case Law: Google India Pvt Ltd v Visaka Industries

Year: 2019 (reported in (2020) 4 SCC 162) **Court:** Supreme Court of India

Facts / Background: In 2008, articles were posted in a Google Groups discussion forum (by a coordinator of an anti-asbestos network) alleging that Visaka Industries, a manufacturer of asbestos cement sheets, was harming public health and was involved in wrongdoing. Visaka issued a notice to Google India asking it to remove the content, but it was not removed. Visaka filed a criminal complaint for defamation (ss.500, 501 and 120B IPC) against the author and Google India. Google India sought to quash the proceedings, claiming exemption as an intermediary. The Andhra Pradesh High Court refused.

Legal Issue: Could Google India claim exemption under s.79 of the IT Act (as it stood before the 2008 Amendment came into force in 2009) against criminal liability for defamation under the IPC?

Court's Decision / What the Court Held: The Court dismissed Google India's appeal. It held that the unamended s.79 (applicable since the acts took place in 2008) protected network service providers only from liability under the IT Act itself, and did not protect them from offences under other laws such as the IPC. It also observed that Google India had not acted on the notice to remove the content. The questions whether Google India was the correct party (as distinct from Google LLC) and whether it was in fact liable were left to be decided at trial. The Court noted that under the amended s.79 (from 2009), the protection extends to liability under any law, subject to the conditions of s.79(2) and (3).

Relevance to the Topic: An important decision on the **scope of safe harbour before and after the 2008 Amendment**, and one of the landmark decisions prescribed for study.

Case Law: Christian Louboutin SAS v Nakul Bajaj

Year: 2018 **Court:** Delhi High Court

Facts / Background: Christian Louboutin, a French luxury shoe brand famous for its red-soled shoes, sued the operator of Darveys.com, an e-commerce website that sold luxury products, including goods bearing the Louboutin name and marks. The website used the plaintiff's name as meta-tags, displayed the plaintiff's photographs, and claimed that the products were genuine and authentic. The defendant claimed to be an intermediary entitled to safe harbour under s.79.

Legal Issue: Is an e-commerce platform that takes an active part in the sale of goods an "intermediary" entitled to safe harbour under s.79 for trade mark infringement?

Court's Decision / What the Court Held: The Court (Justice Pratibha M. Singh) held that whether an e-commerce platform is an intermediary depends on its role. If it takes an active part in the sale, such as identifying sellers, guaranteeing authenticity, storing, packaging and delivering goods, or using the brand owner's marks to promote sales, it may be "conspiring, abetting, aiding or inducing" the unlawful act under s.79(3)(a), and loses safe harbour. The Court listed several factors (such as providing a guarantee of authenticity, invoicing in its own name, warehousing and packaging, advertising products and offering discounts) to decide whether a platform is a passive intermediary or an active participant. It directed the defendant to disclose complete seller details, obtain certificates of genuineness from sellers, and not to use the plaintiff's marks as meta-tags.

Relevance to the Topic: The leading case on **e-commerce platforms and loss of safe harbour**, and one of the landmark decisions prescribed for study.

Important Points

- Intermediary: s.2(1)(w); includes ISPs, search engines, marketplaces, cyber cafes.
- Section 79(1): no liability for third-party information, subject to s.79(2) and (3).
- Conditions: passive role (no initiation, selection or modification) and due diligence.
- Loss: conspiracy, abetment, aiding or inducing; failure to remove on actual knowledge.
- *Shreya Singhal* (2015): actual knowledge = court order or government notification, limited to Art 19(2).
- *MySpace* (2016): specific notice sufficient for copyright.
- *Google v Visaka* (2019/2020): pre-2009 s.79 did not protect against IPC liability.
- *Christian Louboutin* (2018): active e-commerce platforms lose safe harbour.
- *X Corp* (Karnataka HC, 2025): Sahyog portal upheld; Rule 3(1)(d) amended in 2025.

Remember: Safe harbour is conditional, not absolute: an intermediary must be neutral and diligent.

Topic 4.2: Intermediary Guidelines and Digital Media Ethics Code Rules, 2021

Introduction

The **Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021** (IT Rules, 2021) were notified on **25 February 2021** under ss.79(2)(c), 69A(2) and 87 of the IT Act, replacing the Intermediaries Guidelines Rules, 2011. They lay down the **due diligence** obligations of intermediaries (Part II) and a **Code of Ethics** and regulatory framework for **digital news publishers and online curated content (OTT) platforms** (Part III). The Rules have been amended several times: in 2022 (Grievance Appellate Committees), 2023 (online gaming and fact check unit), October 2025 (takedown procedure) and February 2026 (synthetically generated information).

Definition

Significant social media intermediary (SSMI): a social media intermediary having a number of registered users in India above the threshold notified by the Central Government, which is **50 lakh** (5 million) registered users.

Social media intermediary (rule 2(1)(w)): an intermediary which primarily or solely enables online interaction between two or more users and allows them to create, upload, share, disseminate, modify or access information using its services.

Detailed Explanation

1. Structure of the Rules

Part	Subject	Administered by
Part I	Preliminary and definitions	—
Part II	Due diligence by intermediaries and grievance redressal mechanism	Ministry of Electronics and Information Technology (MeitY)
Part III	Code of Ethics and procedure and safeguards in relation to digital media (news and current affairs publishers, and publishers of online curated content)	Ministry of Information and Broadcasting (MIB)

2. Part II: Due Diligence by All Intermediaries (Rule 3)

1. **Publish** rules and regulations, privacy policy and user agreement on the website or app, in English or any language in the Eighth Schedule.
2. **Inform users** not to host, display, upload, modify, publish, transmit, store, update or share information that (Rule 3(1)(b)), among other things:
   - belongs to another person without right;

- is obscene, pornographic, paedophilic, invasive of another's privacy (including bodily privacy), insulting or harassing on the basis of gender, racially or ethnically objectionable;
- is harmful to children;
- infringes patent, trade mark or copyright;
- deceives or misleads about the origin of a message or knowingly communicates misinformation which is patently false;
- impersonates another person;
- threatens the unity, integrity, defence, security or sovereignty of India, friendly relations with foreign States or public order;
- contains software viruses;
- violates any law.

3. **Make reasonable efforts** to cause users not to share such content (2022 amendment).
4. **Takedown on order (Rule 3(1)(d)):** on receiving actual knowledge by an order of a court or a notification from the appropriate Government or its authorised agency, remove or disable access to unlawful information. Under the **2026 amendment**, this must be done within **three hours** (earlier 36 hours). Under the **2025 amendment**, Government intimations must come from senior authorised officers and be reasoned.
5. **Retention:** preserve removed information and associated records for **180 days** for investigation (or longer if required by a court or agency).
6. **Registration information:** retain user registration information for 180 days after cancellation or withdrawal of registration.
7. **Assistance to law enforcement:** provide information or assistance to authorised government agencies for verification of identity, or for prevention, detection, investigation or prosecution of offences, within **72 hours** of a written order.
8. **Security:** take reasonable measures to secure the computer resource, and report cyber security incidents to CERT-In.
9. **No knowing hosting** of unlawful content.

3. Grievance Redressal (Rule 3(2))

1. Appoint a **Grievance Officer** and publish his name and contact details.
2. **Acknowledge** complaints within 24 hours.
3. **Resolve** complaints within **seven days** (reduced from 15 days by the 2026 amendment).
4. **Non-consensual intimate imagery:** where a complaint is made by an individual about content showing him or her in nudity, sexual acts or impersonation in electronic form, including artificially morphed images (deepfakes), the intermediary must remove or disable access within **two hours** (reduced from 24 hours by the 2026 amendment).
5. **Grievance Appellate Committees (GACs)** (rule 3A, inserted in 2022): a user aggrieved by the decision of the Grievance Officer may appeal to a GAC constituted by the Central Government within 30 days; the GAC must try to resolve the appeal within 30 days.

4. Additional Due Diligence by Significant Social Media Intermediaries (Rule 4)

1. **Appoint officers resident in India:**
   - a **Chief Compliance Officer**, responsible for compliance with the Act and Rules (with personal liability);
   - a **Nodal Contact Person** for 24x7 coordination with law enforcement agencies;
   - a **Resident Grievance Officer**.
2. **Monthly compliance reports** on complaints received and action taken, and content removed through proactive monitoring.
3. **Traceability (rule 4(2)):** an SSMI providing messaging services must enable identification of the **first originator** of information in India, on an order of a court or an order under s.69 of the IT Act, for serious offences (sovereignty, security, public order, rape, sexually explicit material, child sexual abuse material). It need not disclose the content of messages. WhatsApp and Facebook have challenged this rule in the Delhi High Court, arguing that it would break end-to-end encryption and violate privacy; the challenge is pending.
4. **Proactive monitoring (rule 4(4)):** an SSMI shall endeavour to deploy technology-based measures, including automated tools, to proactively identify information depicting rape, child sexual abuse or similar content.
5. **Physical address in India** and a voluntary user verification mechanism.
6. **Notice to users** before removing their content on its own, with an opportunity to dispute.
7. **Synthetically generated information (2026 amendment):** SSMIs must require users to declare whether content is synthetically generated, deploy technical measures to verify such declarations, and ensure labelling (see Topic 4.4).

5. Part III: Digital Media Ethics Code

1. **Applies to:** (a) publishers of **news and current affairs content** (digital news portals), and (b) publishers of **online curated content** (OTT platforms such as video streaming services).
2. **Code of Ethics (Appendix):**
   - News publishers must follow the Norms of Journalistic Conduct of the Press Council of India and the Programme Code under the Cable Television Networks (Regulation) Act, 1995.
   - OTT platforms must classify content by age: **U**, **U/A 7+**, **U/A 13+**, **U/A 16+** and **A** (adult), and provide parental locks and age verification for "A" content.
3. **Three-tier grievance redressal structure:**

Level	Mechanism
Level I	Self-regulation by the publisher: Grievance Officer to decide complaints within 15 days
Level II	Self-regulatory body of publishers, headed by a retired judge of the Supreme Court or High Court or an eminent person
Level III	Oversight mechanism by the Ministry of Information and Broadcasting, including an Inter-Departmental Committee

4. **Emergency blocking (rule 16):** in an emergency, the Secretary of the MIB may issue blocking directions, subject to review by the Inter-Departmental Committee.
5. **Judicial challenge:** in *AGIJ Promotion of Nineteenonea Media Pvt Ltd (The Leaflet) v Union of India* (Bombay High Court, August 2021), the Court stayed rule 9(1) and (3) (which required adherence to the Code of Ethics and the oversight mechanism), finding them prima facie beyond the IT Act and violative of Article 19(1)(a). The Madras High Court passed similar orders in 2021. The matters are pending.

6. Amendments to the Rules

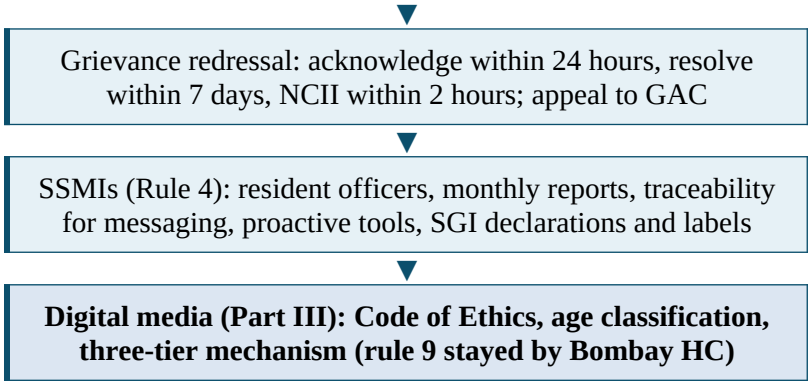
Year	Amendment
2022 (October)	Grievance Appellate Committees; intermediaries to "make reasonable efforts" to prevent prohibited content; respect for users' rights under Arts 14, 19 and 21
2023 (April)	Online gaming intermediaries and self-regulatory bodies (largely overtaken by the Promotion and Regulation of Online Gaming Act, 2025); **Fact Check Unit** (FCU) of the Central Government to identify "fake or false or misleading" information about the business of the Central Government
2025 (October, in force 15 November 2025)	Rule 3(1)(d) amended: takedown intimations only by senior authorised officers, reasoned, with monthly review
2026 (February, in force 20 February 2026)	Regulation of **synthetically generated information**: labelling, metadata, user declarations; takedown within 3 hours on orders; non-consensual intimate imagery within 2 hours; grievances within 7 days

7. The Fact Check Unit Case

1. **Kunal Kamra v Union of India:** the 2023 amendment to Rule 3(1)(b)(v), allowing a government Fact Check Unit to flag information about government business as "fake or false or misleading", was challenged by comedian Kunal Kamra, the Editors Guild of India and others.
2. A Division Bench of the Bombay High Court gave a split verdict (January 2024). The Supreme Court stayed the notification of the FCU in March 2024 until the High Court decided the matter.
3. In **September 2024**, the third (tie-breaker) judge (Justice A.S. Chandurkar) held the rule unconstitutional as violating Articles 14 and 19(1)(a) (vague, overbroad, making the Government a judge in its own cause), and the rule was struck down.
4. The Union of India has appealed to the Supreme Court, which has issued notice but has not stayed the High Court's judgment; the appeal is pending.

Flowchart: Obligations under the IT Rules, 2021

All intermediaries (Rule 3): publish rules, inform users of prohibited content, take down within 3 hours of order, retain data 180 days, assist agencies within 72 hours



Examples

- A woman finds a morphed nude image of herself on a social media platform. She complains to the platform, which must remove it within two hours.
- A messaging app with more than 50 lakh users in India receives a court order to identify the first originator of a message inciting violence. Under rule 4(2), it must provide the information, subject to the pending challenge.

Case Laws

Case Law: Kunal Kamra v Union of India

Year: 2024 **Court:** Bombay High Court (tie-breaker opinion of Justice A.S. Chandurkar, after a split verdict of a Division Bench)

Facts / Background: The IT (Amendment) Rules, 2023 amended Rule 3(1)(b)(v) to require intermediaries to make reasonable efforts not to host information "in respect of any business of the Central Government" that was identified as "fake or false or misleading" by a Fact Check Unit notified by the Central Government. Failure could lead to loss of safe harbour. Kunal Kamra, the Editors Guild of India and the Association of Indian Magazines challenged the rule.

Legal Issue: Is the Fact Check Unit rule constitutional, in light of Articles 14, 19(1)(a) and 19(1)(g)?

Court's Decision / What the Court Held: The Division Bench was divided (Justice G.S. Patel would strike down the rule; Justice Neela Gokhale would uphold it). The tie-breaker judge, Justice Chandurkar, held that the rule was unconstitutional: the expressions "fake, false or misleading" were vague and overbroad; the rule went beyond the restrictions permitted by Article 19(2) (which does not include "falsehood" as such); it had a chilling effect on speech and on intermediaries; and it made the Government the judge of truth about itself. Following his opinion, the rule was struck down in September 2024.

Relevance to the Topic: An important decision on the **limits of government regulation of online "fake news"** through the IT Rules. The Union's appeal is pending before the Supreme Court.

Important Points

- IT Rules, 2021: notified 25 February 2021; Part II (MeitY) and Part III (MIB).
- Due diligence: publish rules, prohibited content, takedown on order (3 hours after 2026), retention 180 days, assistance within 72 hours.
- Grievances: acknowledge 24 hours, resolve 7 days, NCII 2 hours; GAC appeal within 30 days.
- SSMIs (50 lakh users): Chief Compliance Officer, Nodal Contact Person, Resident Grievance Officer; monthly reports; traceability; proactive tools.
- Part III: Code of Ethics; OTT classification; three-tier structure; rule 9 stayed by Bombay HC (2021).
- Amendments: 2022, 2023, 2025, 2026.
- Fact Check Unit rule struck down (*Kunal Kamra*, Bombay HC, 2024); appeal pending.

Remember: Non-compliance with the Rules leads to loss of safe harbour under s.79 (Rule 7).

Topic 4.3: Freedom of speech in cyberspace; defamation, hate speech and obscenity

Introduction

The internet is the most powerful medium of free speech ever created. Anyone can publish to the whole world. **Article 19(1)(a)** of the Constitution guarantees the freedom of speech and expression, and this applies fully to speech on the internet. But the same freedom can be abused to defame, spread hatred and circulate obscene material. **Article 19(2)** allows the State to impose reasonable restrictions on certain grounds. This topic explains how Indian law balances free speech online with the protection of reputation, public order and decency.

Definition

> **Article 19(1)(a):** all citizens shall have the right to freedom of speech and expression.

> **Article 19(2):** nothing in Article 19(1)(a) shall prevent the State from making any law imposing reasonable restrictions on the exercise of the right in the interests of the **sovereignty and integrity of India, the security of the State, friendly relations with foreign States, public order, decency or morality, or in relation to contempt of court, defamation or incitement to an offence**.

Detailed Explanation

1. Freedom of Speech on the Internet

1. **Same protection online and offline:** speech on the internet enjoys the same constitutional protection as speech in print or broadcast (*Shreya Singhal*, 2015; *Anuradha Bhasin*, 2020).
2. **Discussion, advocacy and incitement (*Shreya Singhal*): mere discussion or even advocacy of a particular cause, however unpopular, is at the heart of Article 19(1)(a). Only when discussion or advocacy reaches the level of incitement** can it be restricted on the ground of public order.
3. **Vagueness and chilling effect:** laws restricting online speech must be clear and precise; vague laws that deter legitimate speech are unconstitutional (s.66A struck down).
4. **Prior restraint:** blocking or pre-censorship of content must have procedural safeguards.

2. Blocking of Online Content (Section 69A)

1. **Section 69A:** the Central Government (or an authorised officer) may, if satisfied that it is necessary or expedient in the interest of the sovereignty and integrity of India, defence of India, security of the State, friendly relations with foreign States or public order, or for preventing incitement to the commission of any cognizable offence relating to these, direct any agency or intermediary to block public access to any information, for reasons to be recorded in writing.
2. **Blocking Rules, 2009:** requests are examined by a **committee** (headed by a designated officer); the intermediary or originator is given a hearing where possible; **emergency**

blocking is allowed (rule 9), subject to confirmation; a **Review Committee** reviews orders; requests and actions are kept **confidential** (rule 16).

3. **Upheld in* Shreya Singhal*** because of the safeguards: reasons recorded in writing, hearing, review, and the possibility of challenge under Article 226.
4. **Twitter Inc v Union of India (Karnataka High Court, single judge, 2023):** the Court dismissed Twitter's challenge to several blocking orders, holding that the Government can block entire accounts, not merely specific tweets, and that notice to the user is not always mandatory; it imposed costs on Twitter. (An appeal was filed.)

3. Online Defamation

1. **Civil:** suit for damages and injunction (tort of defamation; see Law of Torts).
2. **Criminal: s.356, BNS** (ss.499–500 IPC): defamation by words spoken or intended to be read, by signs or by visible representations, including electronic means. Punishment: simple imprisonment up to two years, or fine, or both, or community service.
3. **Constitutional validity:** upheld in *Subramanian Swamy v Union of India* (2016), holding that reputation is part of Article 21.
4. **Online-specific issues:**
   - **Republication:** each share or retweet may be a fresh publication.
   - **Liability of intermediaries:** protected by s.79 if they act on court orders (*Shreya Singhal*), but not before 2009 (*Google v Visaka*).
   - **Global takedown:** in *Swami Ramdev v Facebook Inc* (Delhi High Court, 2019), the Court directed platforms to remove defamatory content globally (not merely geo-block it in India) where it was uploaded from India. (An appeal was filed.)
   - **Jurisdiction:** the place where the content is accessed and harm caused (*Dow Jones v Gutnick*, Topic 1.5; *Banyan Tree*).

4. Hate Speech

1. **No single definition** in Indian law. The Law Commission (267th Report, 2017) recommended new offences on incitement to hatred.
2. **Relevant offences in the BNS:**
   - **s.196** (s.153A IPC): promoting enmity between different groups on grounds of religion, race, place of birth, residence, language, etc., by words, signs, visible representations, **electronic communication** or otherwise.
   - **s.197** (s.153B IPC): imputations and assertions prejudicial to national integration, including making or publishing **false or misleading information** jeopardising sovereignty, unity and integrity.
   - **s.299** (s.295A IPC): deliberate and malicious acts intended to outrage religious feelings.
   - **s.302** (s.298 IPC): uttering words with deliberate intent to wound religious feelings.
   - **s.353** (s.505 IPC): statements conducing to public mischief, including through electronic means.
3. **Judicial guidance:**
   - *Pravasi Bhalai Sangathan v Union of India* (2014): existing laws are sufficient; the Court referred the question of defining hate speech to the Law Commission.

- *Tehseen S. Poonawalla v Union of India* (2018): directions to prevent mob lynching, including action against dissemination of irresponsible and explosive messages and videos on social media.
- *Amish Devgan v Union of India* (2020): explained the difference between free speech and hate speech, considering the content, the speaker's position, the context and the likely impact.
- *Shaheen Abdulla v Union of India* (2022–2023): the Supreme Court directed the police in all States to register cases of hate speech suo motu, without waiting for complaints.

4. **IT Rules:** intermediaries must inform users not to share content that is racially or ethnically objectionable or threatens public order (rule 3(1)(b)).

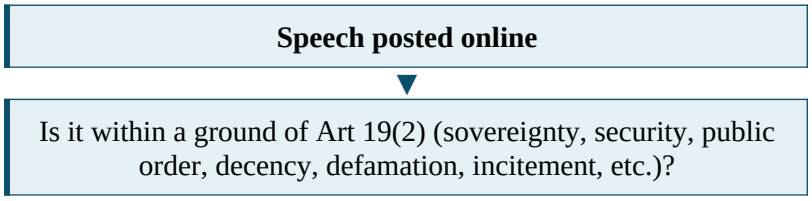
5. Obscenity

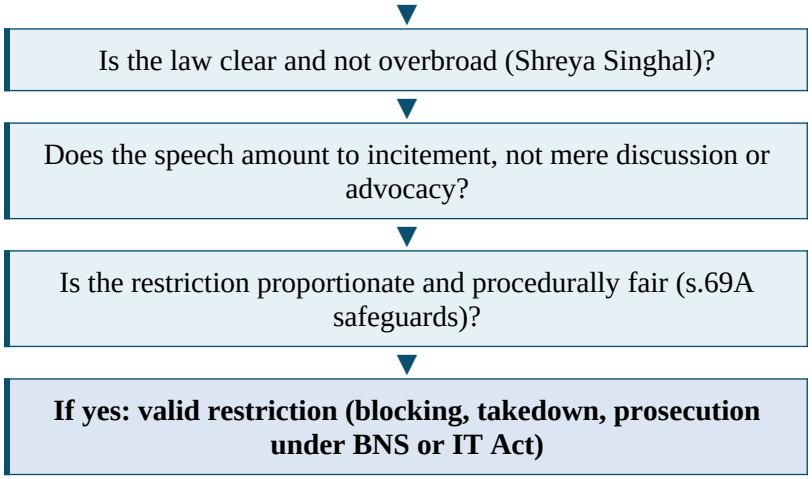
1. **Tests of obscenity:**

Test	Case	Meaning
Hicklin test	*R v Hicklin* (1868, England); adopted in *Ranjit D. Udeshi v State of Maharashtra* (1964)	Whether the matter tends to deprave and corrupt those whose minds are open to such immoral influences
Community standards test	*Aveek Sarkar v State of West Bengal* (2014)	Whether the average person, applying contemporary community standards, would find that the work, taken as a whole, appeals to the prurient interest

2. **Laws:**
   - **s.67, IT Act:** obscene material in electronic form.
   - **s.67A, IT Act:** sexually explicit material.
   - **s.67B, IT Act:** child sexual abuse material.
   - **s.294, BNS** (s.292 IPC): obscene books, pamphlets and representations, including display of obscene content in electronic form.
   - **Indecent Representation of Women (Prohibition) Act, 1986.**
3. **Profanity is not obscenity:** in *Apoorva Arora v State (Govt of NCT of Delhi)* (Supreme Court, 2024), concerning the web series *College Romance*, the Court held that the use of vulgar language and profanities does not by itself make content "obscene" under s.67 of the IT Act, since obscenity relates to material that arouses sexual and lustful thoughts; it quashed the criminal proceedings.
4. **Special law prevails:** where s.67 IT Act applies, s.292 IPC (s.294 BNS) cannot be invoked for the same act (*Sharat Babu Digumarti*, 2017).

Flowchart: Can online speech be restricted?





Examples

- A person posts a video calling for violence against members of a community. This is incitement and hate speech, punishable under s.196 BNS, and can be blocked under s.69A.
- A person writes a strongly worded blog criticising a government policy. This is protected speech, and cannot be restricted merely because it is critical.

Case Laws

Case Law: Shreya Singhal v Union of India (on free speech online)

Year: 2015 **Court:** Supreme Court of India

Facts / Background: See Topic 1.2. The petitions challenged s.66A (offensive messages), s.69A (blocking) and s.79 (intermediary liability) of the IT Act.

Legal Issue: How far can the State restrict online speech under Article 19(2)?

Court's Decision / What the Court Held: The Court held that: (i) s.66A was unconstitutional for vagueness and overbreadth and was not saved by Article 19(2); (ii) the right of the people to know and the market place of ideas are part of Article 19(1)(a); (iii) only speech that amounts to incitement, having a proximate connection with public disorder, may be restricted on the ground of public order; (iv) the internet may be treated as a separate class for creating separate offences, but restrictions must still fall within Article 19(2); (v) s.69A and the Blocking Rules are constitutional, because they are narrowly drawn, with safeguards; (vi) s.79(3)(b) must be read down.

Relevance to the Topic: The **leading case on free speech in cyberspace**, and one of the landmark decisions prescribed for study.

Case Law: Aveek Sarkar v State of West Bengal

Year: 2014 **Court:** Supreme Court of India

Facts / Background: A Bengali newspaper and a sports magazine reprinted a photograph,

originally published in a German magazine, of the famous tennis player Boris Becker posing nude with his fiancée (a dark-skinned actress), with his hand covering her breasts. The photograph was meant to protest against racism and apartheid. A complaint was filed under s.292 IPC.

Legal Issue: Was the photograph obscene, and what is the test of obscenity?

Court's Decision / What the Court Held: The Court held that the photograph was not obscene. It rejected the Hicklin test and adopted the **community standards test**, holding that a picture of a nude or semi-nude woman is not per se obscene unless it has a tendency to arouse feelings of sexual desire; the photograph must be judged as a whole, in its context, and considering the message it conveys (here, against racism). The proceedings were quashed.

Relevance to the Topic: The current **test of obscenity** in India, applicable to online content under s.67 IT Act and s.294 BNS.

Important Points

- Online speech has full protection of Art 19(1)(a), subject to Art 19(2).
- *Shreya Singhal*: discussion and advocacy protected; only incitement restricted; s.66A struck down; s.69A upheld.
- Section 69A and Blocking Rules, 2009: committee, hearing, emergency, review, confidentiality.
- Defamation: s.356 BNS; *Subramanian Swamy* (2016); global takedown (*Swami Ramdev*, 2019).
- Hate speech: ss.196, 197, 299, 302, 353 BNS; *Amish Devgan*; *Shaheen Abdulla*.
- Obscenity: Hicklin (*Ranjit Udeshi*, 1964) replaced by community standards (*Aveek Sarkar*, 2014); profanity is not obscenity (*Apoorva Arora*, 2024).

Remember: Article 19(2) does not include "falsehood" or "offensiveness" as such as a ground of restriction.

Topic 4.4: Deepfakes, misinformation and internet shutdowns

Introduction

Three modern problems threaten the information ecosystem. **Deepfakes**, created using artificial intelligence, can make anyone appear to say or do things they never did. **Misinformation** (false information) and **disinformation** (deliberately false information) spread faster online than corrections. In response to unrest or emergencies, governments sometimes order **internet shutdowns**, which cut off entire populations from communication, commerce and information. Each problem requires a careful balance between preventing harm and protecting rights.

Definition

> **Deepfake:** synthetic media (image, video or audio) created or altered using artificial intelligence, especially deep learning, to realistically depict a person saying or doing something that he or she did not say or do.

> **Synthetically generated information (IT Rules, 2021, as amended in 2026):** audio, visual or audio-visual information which is artificially or algorithmically created, generated, modified or altered using a computer resource, in a manner that such information reasonably appears to be authentic or true. (Text-only content and routine or good-faith editing are excluded.)

> **Internet shutdown:** an intentional disruption of internet or electronic communications, rendering them inaccessible or effectively unusable, for a specific population or within a location, often to exert control over the flow of information.

Detailed Explanation

A. Deepfakes

1. Harms Caused by Deepfakes

1. **Non-consensual intimate imagery:** morphed sexual images and videos, mostly targeting women.
2. **Fraud and impersonation:** voice cloning of family members or company executives to obtain money.
3. **Political manipulation:** fake videos of leaders before elections.
4. **Defamation and harassment.**
5. **Misuse of celebrities' identity** for advertising (personality rights).
6. **Erosion of trust:** people may disbelieve genuine evidence ("liar's dividend").

2. Legal Framework in India

Law	Provision	Application
IT Act	s.66C (identity theft), s.66D (cheating by personation), s.66E (privacy), s.67, 67A, 67B (obscene, sexually explicit, child abuse material)	Creating or sharing sexual deepfakes; impersonation frauds
BNS	s.77 (voyeurism), s.78 (stalking), s.318–319 (cheating, personation), s.336 (forgery of electronic records), s.351 (intimidation), s.353 (public mischief), s.356 (defamation)	Various deepfake harms
IT Rules, 2021 (as amended in 2026)	Labelling, metadata, user declarations, takedown timelines	Duties of intermediaries
DPDP Act, 2023	Processing of personal data (images, voice) requires a lawful basis	Use of a person's likeness as personal data
Copyright Act and Trade Marks Act; common law	Personality and publicity rights; passing off	Unauthorised commercial use of celebrities' likeness

3. The 2026 Amendment on Synthetically Generated Information

The IT (Intermediary Guidelines and Digital Media Ethics Code) Amendment Rules, 2026, notified on **10 February 2026** and in force from **20 February 2026**:

1. **Define** synthetically generated information (SGI), covering AI-created or altered audio, visual and audio-visual content that appears real; text is excluded, as are routine editing, accessibility improvements and ordinary document creation.
2. **Labelling:** SGI must carry a **prominent visible label** (for visual content) or an audio disclosure (for audio).
3. **Provenance:** permanent **metadata** or a unique identifier must be embedded, and must not be removable or modifiable.
4. **Duties of intermediaries offering generative tools:** must ensure labelling and provenance for content created with their tools, and use reasonable technical measures to prevent unlawful SGI.
5. **Duties of SSMIs:** require users to **declare** whether uploaded content is synthetically generated, deploy technical measures (including automated tools) to **verify** declarations, and label accordingly.
6. **Faster takedowns:** three hours for court or government orders; two hours for non-consensual intimate imagery including deepfake nudity.
7. **Consequence:** failure to comply risks loss of safe harbour under s.79.

4. Personality Rights Cases

1. **Amitabh Bachchan v Rajat Nagi (Delhi High Court, 2022):** injunction against unauthorised use of the actor's name, image and voice.
2. **Anil Kapoor v Simply Life India (Delhi High Court, 2023):** injunction against misuse of the actor's name, likeness, voice and persona, including through AI, deepfakes and face morphing, and his catchphrase.

3. Similar orders were passed for other celebrities, e.g. *Arijit Singh v Codible Ventures LLP* (Bombay High Court, 2024), which restrained AI voice cloning of the singer.

B. Misinformation

1. **Misinformation:** false information shared without intent to deceive. **Disinformation:** false information spread deliberately to deceive. **Malinformation:** true information shared to cause harm.
2. **Laws:**
   - **s.353, BNS** (s.505 IPC): statements conducing to public mischief, including false information through electronic means.
   - **s.197(1)(d), BNS:** making or publishing false or misleading information jeopardising the sovereignty, unity, integrity or security of India.
   - **Disaster Management Act, 2005, s.54:** punishment for false warnings about disasters.
   - **IT Rules, rule 3(1)(b):** prohibition on knowingly communicating misinformation that is patently false.
3. **Fact Check Unit:** the 2023 rule allowing the Government to identify "fake or false" information about its business was struck down by the Bombay High Court in *Kunal Kamra v Union of India* (2024) (see Topic 4.2).
4. **Constitutional concern:** "falsehood" as such is not a ground under Article 19(2); restrictions must relate to recognised grounds such as public order, defamation or incitement.

C. Internet Shutdowns

1. Legal Basis

Period	Law
Before 2017	s.144 CrPC (now s.163 BNSS) orders by District Magistrates; s.5(2) Indian Telegraph Act, 1885
2017–2024	Temporary Suspension of Telecom Services (Public Emergency or Public Safety) Rules, 2017, under s.7 of the Telegraph Act
From 2024	**Telecommunications Act, 2023, s.20(2)(b)**, and the **Telecommunications (Procedures and Safeguards for Temporary Suspension of Services) Rules, 2024**: suspension orders by the competent authority (Union or State Home Secretary), with reasons, a review committee, and a limit on duration (not exceeding fifteen days for an order)

2. Judicial Approach

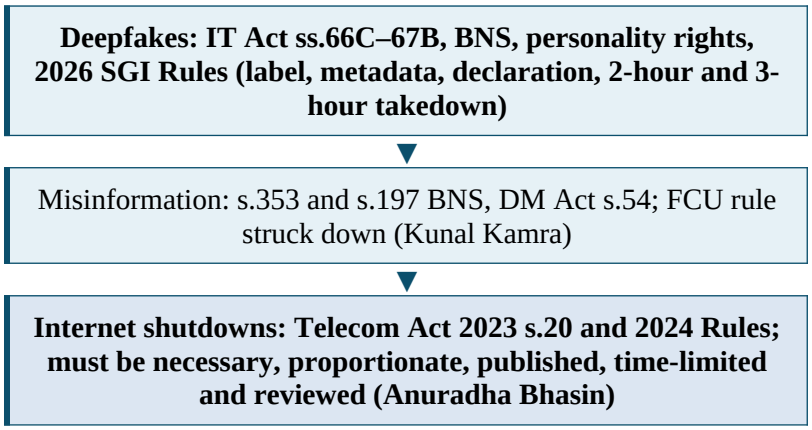
1. **Faheema Shirin R.K. v State of Kerala (Kerala High Court, 2019):** restriction on the use of mobile phones in a hostel was struck down; the Court held that the right to access the internet is part of the right to education and the right to privacy under Article 21.
2. **Anuradha Bhasin v Union of India (2020):** the leading case (see below).

3. **Foundation for Media Professionals v UT of Jammu and Kashmir (2020):** the Court directed a special committee to examine the need for continuing restrictions on 4G mobile internet in J&K.

3. Concerns about Shutdowns

1. They affect speech, trade, education, health services, banking and emergency communication.
2. They often affect entire regions, not only wrongdoers.
3. India has been reported by civil society trackers as having one of the highest numbers of internet shutdowns in the world.
4. The Parliamentary Standing Committee on Communications and IT (2021) recommended clear criteria and a review of their effectiveness.

Flowchart: Responding to online harms



Examples

- A deepfake video shows a film actress in an obscene scene. She complains to the platform, which must remove it within two hours, and the creator may be prosecuted under ss.66E and 67A of the IT Act.
- A District Magistrate suspends mobile internet in a district for three days after communal violence. Under *Anuradha Bhasin* and the 2024 Rules, the order must be reasoned, published, limited in time and reviewed.

Case Laws

Case Law: Anuradha Bhasin v Union of India

Year: 2020 **Court:** Supreme Court of India (three judges)

Facts / Background: On 4–5 August 2019, before and after the changes to the constitutional status of Jammu and Kashmir under Article 370, the Government imposed a complete communication shutdown (internet, mobile and landline services) and restrictions on movement under s.144 CrPC in many parts of J&K. Anuradha Bhasin, executive editor of the *Kashmir Times*, argued that journalists could not publish. Ghulam Nabi Azad also filed a petition.

Legal Issue: Are internet shutdowns and restrictions on movement constitutionally valid, and what safeguards apply?

Court's Decision / What the Court Held: The Court held that: (i) the freedom of speech and expression and the freedom to practise any profession or carry on any trade, business or occupation **over the medium of the internet** are constitutionally protected under Articles 19(1)(a) and 19(1)(g); (ii) restrictions must satisfy the test of **proportionality** (legitimate aim, suitability, necessity, least restrictive measure, balance); (iii) the Government must **publish** suspension orders, so that they can be challenged; (iv) an **indefinite suspension** of internet services is impermissible; suspension can be only for a temporary duration, and the Review Committee must review orders every **seven days**; (v) s.144 CrPC cannot be used to suppress legitimate expression of opinion or grievance. The Court did not declare access to the internet a fundamental right in itself, and directed the authorities to review all orders.

Relevance to the Topic: The **leading case on internet shutdowns**, and one of the landmark decisions prescribed for study.

Case Law: Anil Kapoor v Simply Life India

Year: 2023 **Court:** Delhi High Court

Facts / Background: The actor Anil Kapoor sued several defendants who were using his name, image, voice, likeness and the catchphrase "jhakaas" without his consent, including through AI-generated content, deepfakes, morphed videos, GIFs, emojis and ringtones, for commercial purposes.

Legal Issue: Can a celebrity restrain the unauthorised use of his persona, including through AI and deepfakes?

Court's Decision / What the Court Held: The Court granted an ex parte interim injunction restraining the defendants from using the actor's name, likeness, image, voice, persona or any other attributes for commercial purposes, including through AI tools, deepfakes and face morphing, holding that such misuse violates his personality and publicity rights and that technology cannot be allowed to be used to exploit a person's persona without consent. It also directed the blocking of infringing websites and links.

Relevance to the Topic: An important decision on **personality rights against AI-generated deepfakes**.

Important Points

- Deepfakes: AI-generated synthetic media; harms include NCII, fraud, political manipulation, defamation.
- Laws: IT Act ss.66C, 66D, 66E, 67–67B; BNS; DPDP Act; personality rights (*Anil Kapoor*, 2023).
- 2026 SGI Rules: label, metadata, declaration, verification; takedown in 3 hours (orders) and 2 hours (NCII).

- Misinformation: s.353 and s.197 BNS; DM Act s.54; FCU rule struck down (*Kunal Kamra*, 2024).
- Internet shutdowns: Telecom Act 2023 s.20 and 2024 Rules; *Anuradha Bhasin* (2020): proportionality, publication, no indefinite shutdown, review every 7 days.

Remember: *Anuradha Bhasin* protects speech and trade **through** the internet, but did not declare internet access itself a fundamental right.

Topic 4.5: Artificial intelligence, algorithmic accountability and platform governance

Introduction

Artificial intelligence (AI) systems now recommend content, approve loans, screen job applicants, detect fraud, assist doctors, generate text and images, and drive vehicles. They offer great benefits, but also raise legal concerns: bias and discrimination, lack of transparency ("black box" decisions), privacy, safety, liability for harm, copyright, and misuse such as deepfakes. Large digital **platforms** use algorithms to decide what billions of people see. **Algorithmic accountability** and **platform governance** seek to make these systems transparent, fair and answerable.

Definition

> **Artificial intelligence:** a machine-based system that, for explicit or implicit objectives, infers from the input it receives how to generate outputs such as predictions, content, recommendations or decisions that can influence physical or virtual environments (based on the OECD definition, 2023).

> **Algorithmic accountability:** the principle that those who design, deploy or use algorithms are responsible for their outcomes, and that algorithmic decisions affecting people should be transparent, explainable, fair and subject to oversight and remedy.

> **Platform governance:** the rules and institutions (laws, regulators, platform policies, and user and civil society participation) that govern how digital platforms moderate content, use data, design services and compete.

Detailed Explanation

1. Legal Issues Raised by AI

Issue	Explanation
Bias and discrimination	AI trained on biased data may discriminate (e.g. in hiring, credit, policing), raising issues under Arts 14 and 15
Transparency and explainability	Complex models make decisions that are difficult to explain
Privacy and data protection	AI relies on large amounts of personal data
Liability	Who is liable when AI causes harm: developer, deployer, user?
Intellectual property	Can AI be an author or inventor? Is training on copyrighted data infringement?
Safety and security	Errors in autonomous vehicles, medical AI; misuse for cyber attacks

Issue	Explanation
Misinformation and deepfakes	Generative AI produces realistic false content
Employment and competition	Displacement of jobs; concentration of power in a few firms

2. India's Approach to AI Regulation

1. **No specific AI law** in India as of these notes. AI is governed by existing laws: IT Act and Rules, DPDP Act, Consumer Protection Act, Copyright Act, Competition Act, sector-specific regulations (RBI, SEBI, medical device rules).
2. **NITI Aayog:** *National Strategy for Artificial Intelligence* (#AIForAll, 2018) and *Principles for Responsible AI* (2021): safety and reliability, equality, inclusivity and non-discrimination, privacy and security, transparency, accountability, and protection and reinforcement of positive human values.
3. **IndiaAI Mission (approved March 2024):** to build AI computing infrastructure, datasets, foundation models and skills.
4. **MeitY advisories (2023–2024):** on deepfakes and on AI models, asking intermediaries to label AI-generated content and to ensure that AI models do not permit bias, discrimination or threats to the integrity of the electoral process.
5. **India AI Governance Guidelines (released by MeitY in November 2025):** a principles-based, "light-touch" framework that relies on existing laws, voluntary measures and institutional coordination rather than a new AI statute, with principles such as trust, people-centricity, fairness, accountability and safety.
6. **IT Rules amendment on synthetically generated information (2026):** labelling and provenance duties for AI-generated media (see Topic 4.4).

3. International Approaches

Jurisdiction	Approach
European Union	**AI Act** (Regulation (EU) 2024/1689), in force from 1 August 2024 with phased application: a **risk-based** approach: unacceptable-risk practices banned (e.g. social scoring, certain biometric surveillance); high-risk systems subject to strict obligations (risk management, data quality, documentation, human oversight); transparency duties for limited-risk systems (e.g. chatbots, deepfakes); minimal-risk systems largely free; obligations for general-purpose AI models
United States	No comprehensive federal AI law; sectoral regulation, executive policy and State laws
China	Specific rules on recommendation algorithms (2022), deep synthesis (2023) and generative AI (2023)
International	OECD AI Principles (2019, updated 2024); UNESCO Recommendation on the Ethics of AI (2021); Council of Europe Framework Convention on AI (2024)

4. Algorithmic Accountability

1. **Transparency:** disclosure that AI is being used and how decisions are made.
2. **Explainability:** giving reasons for decisions affecting individuals.
3. **Fairness audits:** testing for bias.
4. **Human oversight:** a human can review or override automated decisions.
5. **Right to challenge:** a remedy against wrongful automated decisions. (The GDPR, Art 22, gives a right not to be subject to solely automated decisions with legal or similarly significant effects; India's DPDP Act does not contain such a specific right.)
6. **Impact assessments:** e.g. Data Protection Impact Assessments for Significant Data Fiduciaries under the DPDP Act (s.10).
7. **Constitutional standards:** State use of algorithms (e.g. facial recognition, welfare eligibility) must satisfy legality, necessity and proportionality (*Puttaswamy*, 2017).

5. Platform Governance

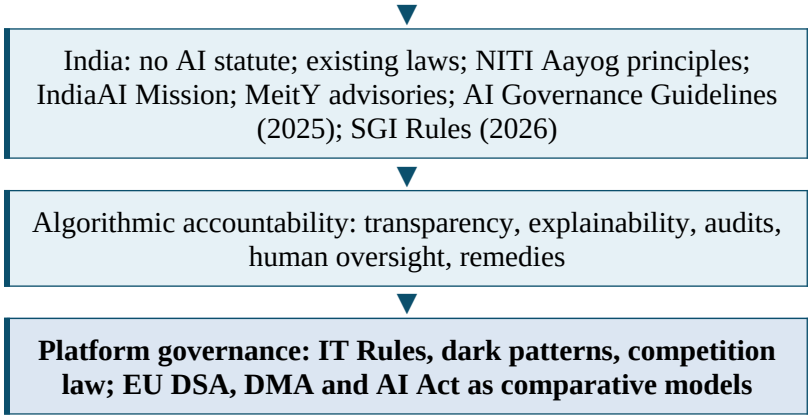
1. **Content moderation:** platforms decide what content to remove, rank or recommend, using community standards and algorithms, subject to the IT Rules, 2021 (grievance officers, GACs, transparency reports).
2. **Consumer protection:** the **Guidelines for Prevention and Regulation of Dark Patterns, 2023** (CCPA) prohibit manipulative design.
3. **Competition law:** the Competition Commission of India (CCI) has acted against dominant platforms, e.g. it imposed a penalty of about Rs. 1,337.76 crore on Google in 2022 for abuse of dominance in the Android mobile ecosystem (largely upheld by the NCLAT in 2023). A draft **Digital Competition Bill (2024)** proposed ex ante regulation of "systemically significant digital enterprises"; it has not been enacted.
4. **EU model:** the Digital Services Act (2022) imposes risk assessments, audits, transparency about recommender systems and data access for researchers on very large online platforms; the Digital Markets Act (2022) regulates "gatekeepers".
5. **Self-regulation:** platform oversight boards and transparency reports.

6. AI and Intellectual Property

1. **Inventorship:** in *Thaler v Comptroller-General of Patents, Designs and Trade Marks* (UK Supreme Court, 2023), the Court held that an AI system (DABUS) cannot be named as an inventor under the UK Patents Act, 1977, since an inventor must be a natural person. Courts and patent offices in the USA, EU and other countries have taken the same view.
2. **Authorship:** copyright law generally requires a human author; the Indian Copyright Act, s.2(d)(vi), treats the person who causes a computer-generated work to be created as its author.
3. **Training on copyrighted works:** in *ANI Media Pvt Ltd v OpenAI* (Delhi High Court), the question whether training AI models on copyrighted news content is infringement is pending.

Flowchart: Governing AI and platforms

AI systems: benefits and risks (bias, opacity, privacy, liability, IP, deepfakes)



Examples

- A bank uses an AI system that rejects loan applications from women more often than men with similar profiles. This raises issues of discrimination and the need for fairness audits.
- A social media platform's recommendation algorithm repeatedly promotes harmful content to teenagers. Platform governance rules may require risk assessment and transparency.

Case Laws

Case Law: Thaler v Comptroller-General of Patents, Designs and Trade Marks

Year: 2023 **Court:** Supreme Court of the United Kingdom

Facts / Background: Dr. Stephen Thaler filed patent applications for two inventions (a food container and a flashing light) that he said had been created autonomously by his AI system "DABUS". He named DABUS as the inventor and claimed the right to the patents as its owner.

Legal Issue: Can an AI system be named as an inventor, and can its owner obtain a patent for its inventions?

Court's Decision / What the Court Held: The UK Supreme Court unanimously held that under the Patents Act, 1977, an "inventor" must be a natural person; DABUS was not a person and could not be an inventor. Dr. Thaler was not entitled to the patents merely because he owned DABUS. The applications were treated as withdrawn. The Court did not decide whether AI-generated inventions should be patentable as a matter of policy.

Relevance to the Topic: An important decision on the **legal status of AI** in intellectual property law.

Case Law: Internet and Mobile Association of India v Reserve Bank of India (on regulatory proportionality in new technologies)

Year: 2020 **Court:** Supreme Court of India

Facts / Background: See Topic 4.6. The RBI prohibited regulated entities from dealing in

virtual currencies or providing services to persons dealing in them.

Legal Issue: Was the RBI's measure against a new technology proportionate?

Court's Decision / What the Court Held: The Court held that the RBI had the power to regulate, but the measure was disproportionate, since there was no evidence of harm to regulated entities and less intrusive measures had not been considered.

Relevance to the Topic: Shows that regulation of **emerging technologies** must be evidence-based and proportionate. (Full case in Topic 4.6.)

Important Points

- AI issues: bias, transparency, privacy, liability, IP, safety, deepfakes, competition.
- India: no AI statute; NITI Aayog (2018, 2021); IndiaAI Mission (2024); MeitY advisories; AI Governance Guidelines (2025); SGI Rules (2026).
- EU AI Act (2024): risk-based approach.
- Algorithmic accountability: transparency, explainability, audits, human oversight, remedies; DPIA under DPDP s.10.
- Platform governance: IT Rules, Dark Patterns Guidelines 2023, CCI v Google (2022), draft Digital Competition Bill (2024); EU DSA and DMA.
- *Thaler* (UK SC, 2023): AI cannot be an inventor.

Remember: The DPDP Act, unlike the GDPR, does not give a specific right against solely automated decision-making.

Topic 4.6: Cloud computing, Internet of Things, blockchain and virtual digital assets

Introduction

New technologies continue to reshape cyberspace. **Cloud computing** moves data and software from personal devices to remote servers. The **Internet of Things (IoT)** connects everyday objects to the internet. **Blockchain** creates tamper-resistant shared ledgers, and supports **virtual digital assets (VDAs)** such as cryptocurrencies and non-fungible tokens (NFTs). Each raises legal questions about data protection, security, jurisdiction, liability, contracts and regulation.

Definition

Cloud computing: a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort (based on the US NIST definition).

Internet of Things (IoT): a network of physical objects ("things") embedded with sensors, software and connectivity that collect and exchange data over the internet.

Blockchain: a distributed, decentralised digital ledger in which transactions are recorded in blocks linked cryptographically in a chain, making records difficult to alter.

Virtual digital asset (s.2(47A), Income-tax Act, 1961, inserted by the Finance Act, 2022): any information, code, number or token (not being Indian or foreign currency), generated through cryptographic means or otherwise, providing a digital representation of value exchanged with or without consideration, with the promise or representation of having inherent value, or functioning as a store of value or a unit of account, including its use in any financial transaction or investment; it includes non-fungible tokens and other notified tokens.

Detailed Explanation

A. Cloud Computing

1. Service and Deployment Models

Service model	Meaning	Example
Infrastructure as a Service (IaaS)	Virtual servers, storage and networks	Renting computing capacity
Platform as a Service (PaaS)	Platforms to develop and run applications	Application development platforms

Service model	Meaning	Example
Software as a Service (SaaS)	Software delivered over the internet	Online email, office software

Deployment models: public cloud, private cloud, hybrid cloud, community cloud.

2. Legal Issues

1. **Data protection and privacy:** cloud providers process personal data on behalf of customers, and act as **Data Processors** under the DPDP Act, with the Data Fiduciary responsible for compliance (s.8).
2. **Data localisation and cross-border transfer:** sectoral rules require some data to be stored in India (e.g. RBI's 2018 direction on storage of payment system data in India); the DPDP Act follows a negative-list approach (s.16).
3. **Jurisdiction and law enforcement access:** data may be stored in several countries.
4. **Security:** the CERT-In Directions, 2022 require cloud service providers to maintain logs and report incidents; VPN, cloud and data centre providers must retain certain customer information for five years.
5. **Contracts:** service level agreements, limitation of liability, data ownership, exit and portability.
6. **Government use:** MeitY empanels cloud service providers for government departments (MeghRaj initiative).

B. Internet of Things

1. Examples

Smart homes (smart speakers, cameras, locks), wearables (fitness trackers), connected cars, smart meters, medical devices, industrial sensors, smart cities.

2. Legal Issues

1. **Privacy:** IoT devices continuously collect personal data, including audio, video, location and health data; DPDP Act duties apply.
2. **Security:** many devices have weak security and can be hijacked into **botnets** (e.g. the Mirai botnet attack, 2016) for DDoS attacks; s.43 and s.66 of the IT Act apply to such attacks.
3. **Liability:** for defects causing harm, under product liability (Consumer Protection Act, 2019, ss.82–87) and negligence.
4. **Standards and certification:** the Telecommunication Engineering Centre has issued security guidelines for IoT devices, and the Telecommunications Act, 2023 and the Telecom Cyber Security Rules, 2024 apply to telecom equipment.

C. Blockchain

1. Features

1. **Decentralisation:** no single central authority.
2. **Immutability:** records cannot easily be altered.

3. **Transparency:** in public blockchains, transactions are visible to all.
4. **Consensus mechanisms:** e.g. proof of work, proof of stake.
5. **Smart contracts:** self-executing code on the blockchain that performs actions automatically when conditions are met.

2. Legal Issues

1. **Validity of smart contracts:** a smart contract can be a valid contract if it satisfies the Contract Act (offer, acceptance, consideration, free consent, lawful object), and s.10A IT Act recognises contracts formed electronically; but code may not capture all legal terms, and disputes about errors and remedies may arise.
2. **Evidence:** blockchain records may be proved as electronic records under the BSA, 2023.
3. **Privacy:** immutability conflicts with the right to erasure (DPDP s.12; GDPR Art 17).
4. **Jurisdiction and liability:** decentralised networks have no single operator.
5. **Uses in governance:** land records, supply chains, certificates.

D. Virtual Digital Assets

1. Regulatory Position in India

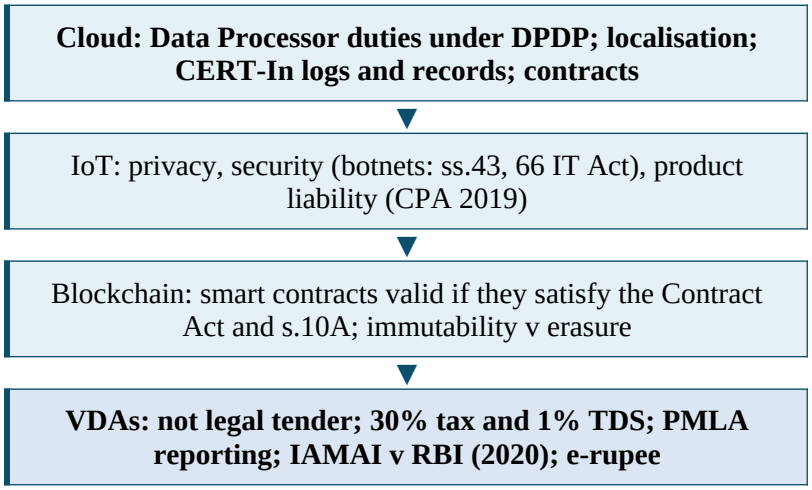
Aspect	Position
Legal tender	VDAs are not legal tender in India
Banking	The RBI's 2018 circular prohibiting banks from dealing with VDA businesses was set aside in *Internet and Mobile Association of India v RBI* (2020)
Taxation	Finance Act, 2022: tax at **30%** on income from transfer of VDAs (s.115BBH), no set-off of losses, and **1% TDS** on transfers above a threshold (s.194S), under the Income-tax Act, 1961; corresponding provisions continue under the Income-tax Act, 2025 (in force from 1 April 2026)
Anti-money laundering	Notification of 7 March 2023: VDA service providers (exchanges, custodians, wallet providers) are "reporting entities" under the **Prevention of Money Laundering Act, 2002**, and must register with the Financial Intelligence Unit-India
Cyber security	CERT-In Directions, 2022: VDA exchanges and wallet providers must maintain KYC and transaction records for five years
Legislation	A draft Cryptocurrency and Regulation of Official Digital Currency Bill was listed in 2021 but not introduced; there is no comprehensive VDA law
Central Bank Digital Currency	The **e-rupee** (digital rupee) issued by the RBI (pilots since 2022), after the RBI Act was amended by the Finance Act, 2022 to include digital currency in the definition of "bank note"

2. Legal Issues

1. **Investor protection:** volatility, fraud, Ponzi schemes and exchange failures.
2. **Money laundering and terror financing.**
3. **Cybercrime:** hacking of exchanges, ransomware demanding cryptocurrency.

4. **Nature of VDAs:** whether they are property, goods, securities or currency (relevant for theft, insolvency and taxation). Courts in some countries have held that cryptocurrency is property capable of being held in trust.
5. **NFTs:** intellectual property and consumer protection issues.

Flowchart: Emerging technologies and the law



Examples

- A hospital stores patient records with a cloud provider. The hospital is the Data Fiduciary and remains responsible under the DPDP Act for ensuring that the provider protects the data.
- A person sells Bitcoin at a profit of Rs. 2 lakh. He must pay tax at 30% on the gain, and 1% TDS is deducted on the transfer.

Case Laws

Case Law: Internet and Mobile Association of India v Reserve Bank of India

Year: 2020 **Court:** Supreme Court of India (three judges)

Facts / Background: On 6 April 2018, the RBI issued a circular directing all entities regulated by it (banks, NBFCs, payment system providers) not to deal in virtual currencies or provide services to any person or business dealing in or settling virtual currencies. This effectively cut off cryptocurrency exchanges from the banking system. The Internet and Mobile Association of India (representing exchanges) challenged the circular.

Legal Issue: Did the RBI have the power to issue the circular, and was it a proportionate restriction on the freedom to carry on trade or business under Article 19(1)(g)?

Court's Decision / What the Court Held: The Court held that the RBI had the power to regulate activities that might affect the payment and financial system, including virtual currencies. However, the circular was **disproportionate**: the RBI had not shown that any entity regulated by it had suffered loss or damage due to virtual currency exchanges, and it had not considered less intrusive measures. Virtual currencies were not banned by the Government. The circular was set aside on the ground of proportionality.

Relevance to the Topic: The leading Indian case on the **regulation of virtual digital assets** and the application of proportionality to restrictions on new technologies.

Important Points

- Cloud: IaaS, PaaS, SaaS; providers as Data Processors; localisation; CERT-In Directions; MeghRaj.
- IoT: privacy, security (botnets), product liability, standards.
- Blockchain: decentralised, immutable ledger; smart contracts; conflict with erasure.
- VDA defined in s.2(47A) Income-tax Act, 1961 (Finance Act, 2022); 30% tax (s.115BBH); 1% TDS (s.194S); continued under the Income-tax Act, 2025.
- VDA service providers are reporting entities under PMLA (2023).
- *IAMAI v RBI* (2020): RBI's banking ban on VDAs set aside as disproportionate.
- VDAs are not legal tender; e-rupee is the RBI's digital currency.

Remember: In India, cryptocurrencies are taxed and regulated for anti-money laundering, but are neither banned nor legal tender.

UNIT 5
Privacy, Data Protection and Cyber Security

Topics in this Unit

5.1 Concept of privacy; constitutional framework; judicial interpretation in India

5.2 Digital Personal Data Protection Act, 2023: scope, definitions and consent

5.3 Obligations of Data Fiduciaries; rights and duties of Data Principals

5.4 Cyber security: confidentiality, integrity and availability; threat landscape and risk management; Critical Information Infrastructure; CERT-In and the Directions of 2022

5.5 Data Protection Board of India; penalties, exemptions and cross-border transfer

5.6 Comparative regimes: GDPR and CCPA; international cyber security cooperation

Topic 5.1: Concept of privacy; constitutional framework; judicial interpretation in India

Introduction

Privacy is the right of an individual to be left alone, to control information about himself, and to make personal choices without unjustified interference. In the digital age, privacy is threatened by mass collection of personal data by companies and governments, surveillance, profiling and data breaches. The Constitution of India does not expressly mention privacy, but in **Justice K.S. Puttaswamy (Retd.) v Union of India** (2017), a nine-judge bench of the Supreme Court unanimously held that **privacy is a fundamental right** protected by Article 21 and Part III. This judgment led to the Digital Personal Data Protection Act, 2023.

Definition

> **Warren and Brandeis, "The Right to Privacy" (Harvard Law Review, 1890):** the "right to be let alone".

> **Alan Westin (1967):** privacy is "the claim of individuals, groups, or institutions to determine for themselves when, how, and to what extent information about them is communicated to others".

> **Puttaswamy (2017):** privacy is the constitutional core of human dignity; it includes the preservation of personal intimacies, the sanctity of family life, marriage, procreation, the home and sexual orientation, and the right to be left alone, and it extends to informational privacy and the autonomy of individual choices.

Detailed Explanation

1. Dimensions of Privacy

Dimension	Meaning	Example
Spatial (territorial) privacy	Privacy of the home and personal spaces	Protection against unlawful searches
Bodily privacy	Integrity of the body	Protection against forced medical tests
Informational privacy	Control over personal information	Protection against misuse of personal data
Decisional privacy (autonomy)	Freedom to make intimate personal choices	Choice of partner, sexual orientation, reproductive choices
Communicational privacy	Privacy of communications	Protection against unlawful phone tapping and interception

2. Constitutional Framework

1. **Article 21:** no person shall be deprived of his life or personal liberty except according to procedure established by law. Privacy is part of "life" and "personal liberty".
2. **Article 19:** freedoms of speech, movement, association and profession require privacy for their meaningful exercise (e.g. anonymity in speech).
3. **Article 14:** arbitrary invasions of privacy violate equality.
4. **Article 20(3):** protection against self-incrimination is linked to privacy of the mind.
5. **International law:** Article 12 of the Universal Declaration of Human Rights and Article 17 of the International Covenant on Civil and Political Rights (to which India is a party) protect privacy.

3. Judicial Development before Puttaswamy

Case	Year	Holding
M.P. Sharma v Satish Chandra	1954	8 judges: search and seizure did not violate Art 20(3); the Constitution does not recognise a right to privacy analogous to the US Fourth Amendment
Kharak Singh v State of UP	1962	6 judges: domiciliary visits at night by police violated Art 21 (personal liberty); but the majority held that privacy is not a guaranteed right; Subba Rao J. (dissenting) recognised privacy as part of personal liberty
Gobind v State of MP	1975	Assumed a right to privacy as an emanation of Arts 19 and 21, subject to compelling State interest
R. Rajagopal v State of Tamil Nadu	1994	Right to privacy (right to be let alone) is implicit in Art 21; the press cannot publish private matters without consent, except matters of public record
People's Union for Civil Liberties v Union of India (telephone tapping case)	1997	Telephone conversations are protected by privacy; tapping only under s.5(2) Telegraph Act with procedural safeguards (later codified in Rule 419A of the Telegraph Rules)
District Registrar and Collector v Canara Bank	2005	Privacy relates to persons, not places; unrestricted inspection of bank records violated privacy
Selvi v State of Karnataka	2010	Involuntary narco-analysis, polygraph and brain mapping tests violate Arts 20(3) and 21

4. K.S. Puttaswamy v Union of India (2017): The Privacy Judgment

1. A nine-judge bench was constituted because of the conflicting decisions in *M.P. Sharma* and *Kharak Singh* (larger benches) and later smaller benches recognising privacy.
2. **Unanimous holding:** the right to privacy is protected as an intrinsic part of the right to life and personal liberty under Article 21 and as a part of the freedoms guaranteed by Part III.

3. *M.P. Sharma* and *Kharak Singh* were **overruled** to the extent that they held that privacy is not protected by the Constitution.
4. **Test for restrictions** (plurality opinion of Chandrachud J.): any invasion of privacy must satisfy:
   - **legality:** existence of a law;
   - **legitimate aim:** a legitimate State purpose;
   - **proportionality:** a rational nexus between the means and the aim, and the least intrusive measure.

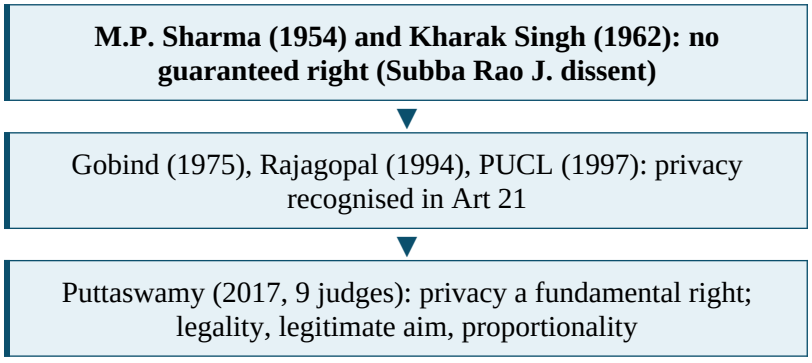
(Kaul J. added procedural guarantees against abuse.)

5. **Informational privacy:** recognised the dangers of data collection by State and non-State actors, and recommended that the Government put in place a robust data protection regime.
6. **Other observations:** privacy includes the right to sexual orientation (criticising *Suresh Kumar Koushal*, 2013); the judgment in *ADM Jabalpur v Shivkant Shukla* (1976) was described as seriously flawed (Chandrachud J.).

5. Developments after Puttaswamy

1. **K.S. Puttaswamy v Union of India (Aadhaar case, 2018):** the Aadhaar Act was upheld as a reasonable restriction for targeted delivery of welfare benefits, but s.57 (allowing private entities to use Aadhaar for authentication) was struck down, and Aadhaar was held not mandatory for bank accounts, mobile connections and school admissions.
2. **Navtej Singh Johar v Union of India (2018):** decriminalised consensual homosexual acts, relying on privacy and dignity.
3. **Joseph Shine v Union of India (2018):** struck down adultery (s.497 IPC), relying on privacy and autonomy.
4. **Justice B.N. Srikrishna Committee (2018)** drafted the Personal Data Protection Bill, 2018; the PDP Bill, 2019 was referred to a Joint Parliamentary Committee and withdrawn in August 2022; the **Digital Personal Data Protection Act, 2023** was enacted on 11 August 2023.
5. **Manohar Lal Sharma v Union of India (Pegasus case, 2021):** on allegations of surveillance using the Pegasus spyware, the Court held that the State does not get a free pass every time national security is raised, and appointed an independent technical committee to inquire.

Flowchart: Development of the right to privacy in India





Examples

- A State government installs CCTV cameras with facial recognition in all public places without any law. This may fail the legality and proportionality tests of *Puttaswamy*.
- A mobile app collects users' contact lists and location data without telling them. This violates informational privacy and the DPDP Act.

Case Laws

Case Law: Justice K.S. Puttaswamy (Retd.) v Union of India

Year: 2017 **Court:** Supreme Court of India (nine judges)

Facts / Background: Justice K.S. Puttaswamy, a retired judge of the Karnataka High Court, and others challenged the Aadhaar scheme, which involved the collection of biometric data (fingerprints and iris scans) and demographic data of residents, as violating the right to privacy. The Union argued that the Constitution did not recognise a fundamental right to privacy, relying on *M.P. Sharma* (8 judges, 1954) and *Kharak Singh* (6 judges, 1962). A bench of five judges, and then a bench of nine judges, was constituted to decide whether privacy is a fundamental right.

Legal Issue: Is the right to privacy a fundamental right under the Constitution of India?

Court's Decision / What the Court Held: The nine-judge bench unanimously held that the right to privacy is a fundamental right, protected as an intrinsic part of the right to life and personal liberty under Article 21 and as part of the freedoms guaranteed by Part III. It overruled *M.P. Sharma* and *Kharak Singh* to the extent that they held otherwise. Privacy is not absolute; restrictions must satisfy the tests of legality, legitimate aim and proportionality. Informational privacy is a facet of the right to privacy, and the State must create a robust data protection regime balancing individual interests and legitimate concerns of the State. The validity of Aadhaar itself was left to a smaller bench (decided in 2018).

Relevance to the Topic: The **foundational privacy judgment**, which forms the constitutional basis of the DPDP Act, 2023, and one of the landmark decisions prescribed for study.

Case Law: People's Union for Civil Liberties v Union of India (Telephone Tapping Case)

Year: 1997 **Court:** Supreme Court of India

Facts / Background: Following a CBI report on the tapping of politicians' telephones, PUCL filed a petition challenging the constitutional validity of s.5(2) of the Indian Telegraph Act, 1885, which allowed the Government to intercept messages on the

occurrence of a public emergency or in the interest of public safety, without procedural safeguards.

Legal Issue: Does telephone tapping violate the right to privacy, and what safeguards are required?

Court's Decision / What the Court Held: The Court held that the right to hold a telephone conversation in the privacy of one's home or office without interference is part of the right to privacy under Article 21, and telephone tapping is a serious invasion of privacy. Since no rules had been framed, the Court laid down procedural safeguards: tapping orders only by the Home Secretary of the Union or a State; recording of reasons; consideration of whether the information could be obtained by other means; time limits (orders valid for two months, extendable up to six months); and review by a committee. These were later incorporated in Rule 419A of the Telegraph Rules.

Relevance to the Topic: The leading case on **communicational privacy** and surveillance, relevant to interception under s.69 of the IT Act and the Telecommunications Act, 2023.

Important Points

- Privacy: right to be let alone; control over personal information; autonomy.
- Dimensions: spatial, bodily, informational, decisional, communicational.
- Constitutional basis: Arts 21, 19, 14; UDHR Art 12; ICCPR Art 17.
- *M.P. Sharma* (1954) and *Kharak Singh* (1962): overruled on privacy.
- *Gobind* (1975), *Rajagopal* (1994), *PUCL* (1997): privacy recognised.
- *Puttaswamy* (2017): privacy a fundamental right; legality, legitimate aim, proportionality.
- *Aadhaar* (2018): s.57 struck down; *Pegasus* (2021): technical committee.

Remember: *Puttaswamy* (2017) was a unanimous decision of nine judges.

Topic 5.2: Digital Personal Data Protection Act, 2023: scope, definitions and consent

Introduction

The **Digital Personal Data Protection Act, 2023** (DPDP Act) is India's first comprehensive law on the protection of personal data. It received the President's assent on **11 August 2023**. It aims to balance the right of individuals to protect their personal data with the need to process personal data for lawful purposes. The **Digital Personal Data Protection Rules, 2025** were notified in **November 2025**, and the Act and Rules are being brought into force in three phases: the Data Protection Board provisions immediately (November 2025), the Consent Manager provisions after one year (November 2026), and the main obligations after eighteen months (May 2027).

Definition

> **Preamble:** "An Act to provide for the processing of digital personal data in a manner that recognises both the right of individuals to protect their personal data and the need to process such personal data for lawful purposes and for matters connected therewith or incidental thereto."

> **Personal data (s.2(t)):** "any data about an individual who is identifiable by or in relation to such data."

Detailed Explanation

1. Structure of the Act

Chapter	Sections	Subject
I	1–3	Preliminary: short title, definitions, application
II	4–10	Obligations of Data Fiduciaries (grounds, notice, consent, legitimate uses, general obligations, children, Significant Data Fiduciaries)
III	11–15	Rights and duties of Data Principals
IV	16–17	Special provisions: cross-border transfer and exemptions
V	18–26	Data Protection Board of India
VI	27–28	Powers, functions and procedure of the Board
VII	29–32	Appeal and alternate dispute resolution
VIII	33–34	Penalties and adjudication
IX	35–44	Miscellaneous (including blocking, consistency with other laws, amendments to the IT Act and RTI Act)
Schedule	—	Penalties

2. Important Definitions (Section 2)

Term	Meaning
Child (s.2(f))	An individual who has not completed eighteen years of age
Consent Manager (s.2(g))	A person registered with the Board, who acts as a single point of contact to enable a Data Principal to give, manage, review and withdraw consent through an accessible, transparent and interoperable platform
Data (s.2(h))	A representation of information, facts, concepts, opinions or instructions suitable for communication, interpretation or processing by human beings or by automated means
Data Fiduciary (s.2(i))	Any person who alone or in conjunction with other persons determines the purpose and means of processing of personal data
Data Principal (s.2(j))	The individual to whom the personal data relates; for a child, includes the parents or lawful guardian; for a person with disability, includes the lawful guardian acting on his behalf
Data Processor (s.2(k))	Any person who processes personal data on behalf of a Data Fiduciary
Digital personal data (s.2(n))	Personal data in digital form
Personal data (s.2(t))	Any data about an individual who is identifiable by or in relation to such data
Personal data breach (s.2(u))	Any unauthorised processing of personal data or accidental disclosure, acquisition, sharing, use, alteration, destruction or loss of access to personal data, that compromises its confidentiality, integrity or availability
Processing (s.2(x))	A wholly or partly automated operation or set of operations performed on digital personal data, including collection, recording, organisation, structuring, storage, adaptation, retrieval, use, alignment or combination, indexing, sharing, disclosure by transmission, dissemination or otherwise making available, restriction, erasure or destruction
Significant Data Fiduciary (s.2(z))	Any Data Fiduciary or class of Data Fiduciaries notified by the Central Government under s.10

3. Scope and Application (Section 3)

1. **Territorial application:** the Act applies to the processing of digital personal data **within India**, where the personal data is collected in **digital form**, or in **non-digital form and digitised** subsequently.
2. **Extra-territorial application:** it also applies to processing of digital personal data **outside India**, if such processing is in connection with any activity related to **offering goods or services to Data Principals within India**.
3. **Exclusions:** the Act does not apply to:
   - personal data processed by an individual for any **personal or domestic purpose**; and
   - personal data that is made or caused to be made **publicly available** by the Data Principal herself, or by any other person under a legal obligation to make it publicly available.

4. **Non-digital data** that is never digitised (e.g. paper records) is outside the Act.

4. Grounds for Processing (Section 4)

A person may process personal data only in accordance with the Act and for a **lawful purpose**:

1. for which the Data Principal has given her **consent**; or
2. for certain **legitimate uses** (s.7).

5. Notice (Section 5)

1. Every request for consent must be accompanied or preceded by a **notice** informing the Data Principal of:
   - the personal data and the purpose of processing;
   - the manner in which she may exercise her rights (withdrawal of consent, grievance redressal); and
   - the manner in which she may complain to the Board.
2. For consent given before the commencement of the Act, notice must be given as soon as reasonably practicable.
3. The notice must be available in English or any language in the Eighth Schedule.
4. **Rule 3, DPDP Rules, 2025:** the notice must be presented independently of other information, be clear and plain, and give an itemised description of the personal data and specified purposes, and a link or means to withdraw consent and exercise rights.

6. Consent (Section 6)

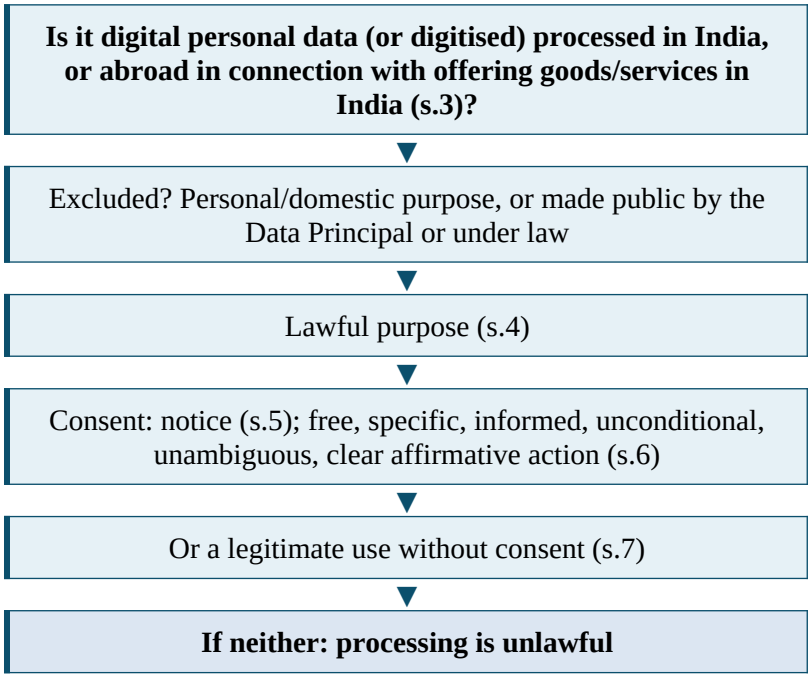
1. **Requirements:** consent must be **free, specific, informed, unconditional and unambiguous with a clear affirmative action**, and signify an agreement to the processing of personal data for the **specified purpose**, limited to such personal data as is **necessary** for that purpose.
2. **Invalid parts:** any part of the consent that constitutes an infringement of the Act or any other law is invalid to that extent (e.g. a clause waiving the right to complain to the Board).
3. **Withdrawal (s.6(4)):** the Data Principal may withdraw consent at any time, with ease comparable to the ease with which it was given. The consequences of withdrawal are borne by her, but withdrawal does not affect the legality of processing before it.
4. **After withdrawal (s.6(6)):** the Data Fiduciary must cease processing (and cause its Data Processors to cease) within a reasonable time, unless processing without consent is required or authorised by law.
5. **Consent Managers (s.6(7)–(9)):** a Data Principal may give, manage, review or withdraw consent through a Consent Manager registered with the Board. Under the Rules, Consent Manager provisions come into force one year after notification (November 2026).
6. **Burden of proof (s.6(10)):** where consent is the basis of processing and a question arises, the **Data Fiduciary** must prove that notice was given and consent was obtained.

7. Certain Legitimate Uses (Section 7)

A Data Fiduciary may process personal data **without consent** for the following uses:

1. for the specified purpose for which the Data Principal has **voluntarily provided** her personal data and has not indicated that she does not consent (e.g. sharing a phone number with a shop to receive a bill);
2. for the State to provide **subsidies, benefits, services, certificates, licences or permits**;
3. for the performance of any **function of the State** under law, or in the interest of sovereignty, integrity or security of India;
4. for fulfilling any **legal obligation** to disclose information to the State;
5. for compliance with a **judgment, decree or order**;
6. for responding to a **medical emergency** involving a threat to life;
7. for providing medical treatment or health services during an **epidemic** or threat to public health;
8. for ensuring safety or providing services during any **disaster** or breakdown of public order;
9. for the purposes of **employment** or safeguarding the employer from loss or liability (e.g. preventing corporate espionage, maintaining confidentiality).

Flowchart: Can personal data be processed under the DPDP Act?



Examples

- An online store asks a customer to agree to a single checkbox allowing it to use her data "for any purpose". This consent is not specific or limited to necessary data, and is invalid.
- A hospital processes the data of an unconscious accident victim to treat him. This is a legitimate use (medical emergency) under s.7, and consent is not required.

Case Laws

The DPDP Act is new, and its main provisions come into force in phases up to May 2027, so there are no leading judgments interpreting it yet. Its constitutional foundation is *Justice K.S.*

Puttaswamy v Union of India (2017) (Topic 5.1). Challenges to some provisions of the Act (especially s.44(3), which amends s.8(1)(j) of the Right to Information Act to exempt all personal information from disclosure, and the wide exemptions for the State in s.17) have been raised by civil society, and petitions on these issues have been filed before the Supreme Court; their outcome is awaited.

Important Points

- DPDP Act: assent 11 August 2023; DPDP Rules notified November 2025; phased commencement (November 2025, November 2026, May 2027).
- Key definitions: personal data, Data Fiduciary, Data Principal, Data Processor, Consent Manager, personal data breach, processing, child (under 18).
- Scope (s.3): digital personal data in India; abroad if offering goods/services to Indians; excludes personal/domestic and publicly available data.
- Grounds (s.4): consent or legitimate uses.
- Notice (s.5) and consent (s.6): free, specific, informed, unconditional, unambiguous, clear affirmative action; withdrawal as easy as giving; burden on the Data Fiduciary.
- Legitimate uses (s.7): voluntary provision, State benefits and functions, legal obligation, court orders, medical emergency, epidemic, disaster, employment.

Remember: Under s.6(10), the burden of proving notice and consent is on the Data Fiduciary.

Topic 5.3: Obligations of Data Fiduciaries; rights and duties of Data Principals

Introduction

The DPDP Act places the main responsibility for protecting personal data on the **Data Fiduciary**, the entity that decides why and how personal data is processed. It must follow certain general obligations, take special care of children's data, and, if notified as a **Significant Data Fiduciary**, comply with additional obligations. On the other hand, the **Data Principal** (the individual) has rights to information, correction, erasure, grievance redressal and nomination, and also certain duties.

Definition

> **Data Fiduciary (s.2(i)):** any person who alone or in conjunction with other persons determines the purpose and means of processing of personal data.

> **Data Principal (s.2(j)):** the individual to whom the personal data relates.

Detailed Explanation

1. General Obligations of Data Fiduciaries (Section 8)

Sub-section	Obligation
8(1)	Responsible for complying with the Act in respect of any processing undertaken by it or on its behalf by a Data Processor, irrespective of any agreement to the contrary or failure of the Data Principal to perform her duties
8(2)	May engage a Data Processor only under a valid contract
8(3)	Ensure the **completeness, accuracy and consistency** of personal data if it is likely to be used to make a decision affecting the Data Principal or disclosed to another Data Fiduciary
8(4)	Implement appropriate **technical and organisational measures** to ensure effective observance of the Act
8(5)	Protect personal data by taking **reasonable security safeguards** to prevent personal data breach
8(6)	In the event of a personal data breach, give **intimation** to the Board and to each affected Data Principal
8(7)	**Erase** personal data (and cause its Data Processors to erase it) when the Data Principal withdraws consent or when it is reasonable to assume that the specified purpose is no longer being served, whichever is earlier, unless retention is necessary for compliance with law
8(8)	The purpose is deemed no longer served if the Data Principal does not approach the Data Fiduciary for the specified purpose or exercise her rights for the prescribed period

Sub-section	Obligation
8(9)	Publish the business contact information of a **Data Protection Officer** (if applicable) or a person able to answer questions about processing
8(10)	Establish an effective mechanism to **redress grievances** of Data Principals

2. Requirements under the DPDP Rules, 2025 (Final Phase, May 2027)

1. **Security safeguards (Rule 6):** encryption, obfuscation, masking or virtual tokens; access controls; visibility over access through logs and monitoring; measures for continued processing (backups); retention of logs and personal data related to processing for at least **one year** for detecting breaches; contractual obligations on Data Processors.
2. **Breach intimation (Rule 7):** on becoming aware of a breach, the Data Fiduciary must:
   - inform each affected Data Principal **without delay**, in plain language, describing the breach, its likely consequences, mitigation measures, safety steps she can take and a contact person; and
   - inform the **Board without delay**, and within **72 hours** (or such longer period as the Board allows) submit a detailed report on the facts, circumstances, reasons, mitigation measures, findings about the person responsible and remedial measures.
3. **Erasure of inactive data (Rule 8 and Third Schedule):** large e-commerce entities, online gaming intermediaries and social media intermediaries (above user thresholds specified in the Rules) must erase personal data if the Data Principal has not approached them or exercised her rights for **three years**, after giving her **48 hours'** notice.
4. **Grievance redressal (Rule 14):** grievances must be responded to within a period not exceeding **90 days**.

3. Processing of Children's Personal Data (Section 9)

1. Before processing the personal data of a **child** (under 18) or a person with disability who has a lawful guardian, the Data Fiduciary must obtain the **verifiable consent of the parent or lawful guardian**.
2. The Data Fiduciary shall not undertake processing that is likely to cause any **detrimental effect on the well-being** of a child.
3. The Data Fiduciary shall not undertake **tracking or behavioural monitoring** of children or **targeted advertising** directed at children.
4. The Central Government may exempt certain classes of Data Fiduciaries or purposes (e.g. healthcare, educational institutions, child safety) from some of these obligations, and may notify an age above which some exemptions apply for Data Fiduciaries that process data in a verifiably safe manner.
5. **Rule 10:** verifiable parental consent may be confirmed through reliable identity and age details already available with the Data Fiduciary, or voluntarily provided, or through a virtual token issued by an authorised entity (e.g. through DigiLocker).

4. Significant Data Fiduciaries (Section 10)

1. **Notification:** the Central Government may notify any Data Fiduciary or class as a Significant Data Fiduciary (SDF), considering: the volume and sensitivity of personal

data processed; risk to the rights of Data Principals; potential impact on the sovereignty and integrity of India; risk to electoral democracy; security of the State; and public order.

2. **Additional obligations:**
   - appoint a **Data Protection Officer** based in India, responsible to the Board of Directors or similar governing body, who shall be the point of contact for grievances;
   - appoint an **independent data auditor** to evaluate compliance;
   - undertake periodic **Data Protection Impact Assessments** and periodic audits;
   - other measures as prescribed.
3. **Rule 13:** annual DPIA and audit with reports to the Board; due diligence to verify that **algorithmic software** used does not pose a risk to the rights of Data Principals; and measures to ensure that personal data specified by the Government on the recommendation of a committee is processed subject to the restriction that it is not transferred outside India.

5. Rights of Data Principals (Chapter III)

Section	Right	Content
11	Right to access information about personal data	To obtain a summary of the personal data being processed and the processing activities; the identities of all other Data Fiduciaries and Data Processors with whom it has been shared; and other information as prescribed (not applicable where sharing is with other Data Fiduciaries authorised by law for prevention, detection or investigation of offences)
12	Right to correction and erasure	To correct inaccurate or misleading data, complete incomplete data, update data, and erase data no longer necessary for the purpose (unless retention is required by law)
13	Right of grievance redressal	To have readily available means of grievance redressal provided by the Data Fiduciary or Consent Manager; she must **exhaust** this remedy before approaching the Board
14	Right to nominate	To nominate any other individual to exercise her rights in the event of her death or incapacity

6. Duties of Data Principals (Section 15)

A Data Principal shall:

1. comply with the provisions of all applicable laws while exercising her rights;
2. ensure not to **impersonate** another person while providing her personal data for a specified purpose;
3. ensure not to **suppress any material information** while providing her personal data for any document, unique identifier, proof of identity or address issued by the State;
4. ensure not to register a **false or frivolous grievance or complaint** with a Data Fiduciary or the Board;

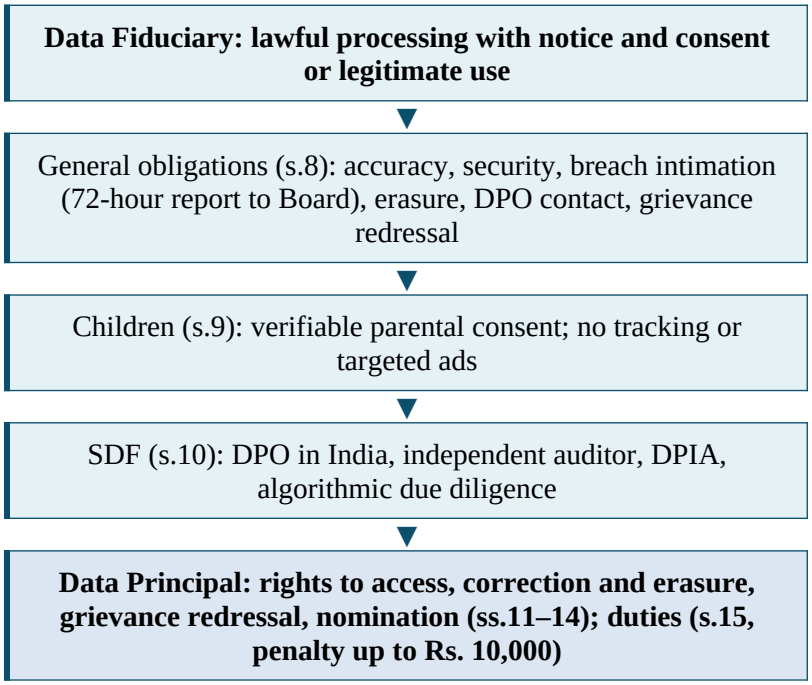
5. furnish only information that is **verifiably authentic** while exercising the right to correction or erasure.

Penalty: breach of these duties may attract a penalty of up to **Rs. 10,000** (Schedule, item 5).

7. Comparison: Obligations and Rights

Data Fiduciary obligations	Corresponding Data Principal rights
Notice and consent (ss.5–6)	Right to withdraw consent (s.6(4))
Accuracy (s.8(3))	Right to correction (s.12)
Erasure when purpose is served (s.8(7))	Right to erasure (s.12)
Grievance redressal mechanism (s.8(10))	Right of grievance redressal (s.13)
Breach intimation (s.8(6))	Right to be informed of a breach (Rule 7)
Disclosure of sharing	Right to access information (s.11)

Flowchart: Duties and rights under the DPDP Act



Examples

- A social media company suffers a breach exposing users' phone numbers. It must inform affected users without delay and the Board immediately, followed by a detailed report within 72 hours.
- A 15-year-old student wants to open an account on a gaming app. The app must obtain verifiable consent from her parent and cannot show her targeted advertisements.

Case Laws

There are no judicial decisions yet interpreting ss.8–15 of the DPDP Act, since the main obligations come into force in the final phase (May 2027). Before the Act, a similar duty to

protect sensitive personal data was enforced under s.43A of the IT Act and the SPDI Rules, 2011 (Topic 3.2). The constitutional principles from *Justice K.S. Puttaswamy v Union of India* (2017), including informational privacy and proportionality, will guide the interpretation of these obligations (Topic 5.1).

Important Points

- Section 8: Data Fiduciary responsible even for processing by Data Processors; accuracy, security, breach intimation, erasure, DPO contact, grievance redressal.
- Rules 2025: security safeguards (logs for one year); breach report to the Board within 72 hours; erasure after three years of inactivity for large platforms (48-hour notice); grievances within 90 days.
- Section 9: children (under 18): verifiable parental consent; no harmful processing, tracking or targeted ads.
- Section 10: SDFs: DPO based in India, independent auditor, DPIA; Rule 13: algorithmic due diligence and possible localisation.
- Rights: access (s.11), correction and erasure (s.12), grievance redressal (s.13), nomination (s.14).
- Duties (s.15): no impersonation, no suppression, no frivolous complaints; penalty up to Rs. 10,000.

Remember: A Data Principal must first use the Data Fiduciary's grievance mechanism before complaining to the Board (s.13(3)).

Topic 5.4: Cyber security: confidentiality, integrity and availability; threat landscape and risk management; Critical Information Infrastructure; CERT-In and the Directions of 2022

Introduction

Cyber security is the protection of computers, networks, data and digital services from attacks, damage and unauthorised access. As banking, health care, power, telecom, transport and government services depend on digital systems, cyber attacks can cause enormous harm. Indian law addresses cyber security through the IT Act (ss.43, 66, 66F, 70, 70A, 70B), the **CERT-In Directions of 2022**, the protection of **Critical Information Infrastructure** by the NCIIPC, sectoral regulations and the DPDP Act's security safeguards.

Definition

> **Cyber security (s.2(1)(nb), IT Act):** "protecting information, equipment, devices, computer, computer resource, communication device and information stored therein from unauthorised access, use, disclosure, disruption, modification or destruction."

> **Cyber security incident (s.2(1)(nb) read with CERT-In Rules):** any real or suspected adverse event in relation to cyber security that violates an explicitly or implicitly applicable security policy, resulting in unauthorised access, denial of service or disruption, unauthorised use of a computer resource, or changes to data or information without authorisation.

> **Critical Information Infrastructure (Explanation to s.70(1)):** "the computer resource, the incapacitation or destruction of which, shall have debilitating impact on national security, economy, public health or safety."

Detailed Explanation

1. The CIA Triad

Principle	Meaning	Threat	Safeguard
Confidentiality	Information is accessible only to authorised persons	Data theft, eavesdropping, breaches	Encryption, access controls, authentication
Integrity	Information is accurate and has not been altered without authorisation	Tampering, malware, forged records	Hashing, digital signatures, audit logs
Availability	Systems and data are available to authorised users when needed	Denial of service attacks, ransomware, outages	Backups, redundancy, DDoS protection

Other principles often added: **authentication**, **non-repudiation** and **accountability**.

2. Threat Landscape

Threat	Description
Malware	Malicious software: viruses, worms, trojans, spyware
Ransomware	Encrypts the victim's data and demands payment (often in cryptocurrency) for decryption
Phishing and social engineering	Fraudulent emails, messages or calls to obtain passwords or OTPs; vishing (voice), smishing (SMS)
Denial of service (DoS/DDoS)	Overwhelming a system with traffic to make it unavailable
Advanced persistent threats (APTs)	Long-term targeted attacks, often by State-sponsored groups
Insider threats	Misuse by employees or contractors
Supply chain attacks	Compromising software or hardware suppliers to reach their customers
Zero-day exploits	Attacks using unknown vulnerabilities
Data breaches	Unauthorised access to large amounts of personal data
Attacks on IoT and industrial control systems	Targeting connected devices and critical infrastructure

Examples in India: the **Cosmos Co-operative Bank** malware attack (2018), in which about Rs. 94 crore was withdrawn through cloned cards and SWIFT transfers; the **AIIMS, New Delhi ransomware attack** (November 2022), which disrupted hospital services for days; and numerous large data leaks.

3. Risk Management

1. **Steps:**
   - **Identify** assets, threats and vulnerabilities.
   - **Assess** risks (likelihood and impact).
   - **Treat** risks: avoid, reduce (controls), transfer (insurance) or accept.
   - **Monitor and review**, including incident response and recovery.
2. **Standards and frameworks:** ISO/IEC 27001 (information security management systems); NIST Cybersecurity Framework (identify, protect, detect, respond, recover; "govern" added in version 2.0, 2024).
3. **Legal recognition:** under the SPDI Rules, 2011 (Rule 8), ISO 27001 compliance is deemed to be reasonable security practices; the DPDP Rules, 2025 (Rule 6) prescribe minimum security safeguards.
4. **Sectoral requirements:** RBI cyber security frameworks for banks; SEBI's Cybersecurity and Cyber Resilience Framework (2024) for market entities; IRDAI guidelines for insurers; Telecom Cyber Security Rules, 2024.

4. Critical Information Infrastructure (CII)

1. **Protected systems (s.70):** the appropriate Government may, by notification, declare any computer resource that directly or indirectly affects the facility of CII to be a **protected system**, and authorise persons to access it.
   - **Section 70(3):** any person who secures access or attempts to secure access to a protected system in contravention of s.70 is punishable with imprisonment up to **10 years** and fine.
2. **National Critical Information Infrastructure Protection Centre (NCIIPC) (s.70A):** the Central Government designates a national nodal agency for the protection of CII. The NCIIPC (notified in 2014, under the National Technical Research Organisation) is responsible for research, planning, guidance and protection of CII.
3. **CII sectors identified by NCIIPC:** power and energy; banking, financial services and insurance; telecom; transport; government; strategic and public enterprises.
4. **Cyber terrorism (s.66F)** includes conduct that adversely affects CII.

5. CERT-In (Section 70B)

1. **Indian Computer Emergency Response Team (CERT-In):** the national agency for incident response, under MeitY.
2. **Functions (s.70B(4)):**
   - collection, analysis and dissemination of information on cyber incidents;
   - forecasts and alerts of cyber security incidents;
   - emergency measures for handling cyber security incidents;
   - coordination of cyber incident response activities;
   - issuing guidelines, advisories, vulnerability notes and white papers on information security practices, procedures, prevention, response and reporting;
   - other functions as prescribed.
3. **Powers (s.70B(6)):** CERT-In may call for information and give directions to service providers, intermediaries, data centres, body corporates and any other person.
4. **Penalty (s.70B(7)):** failure to provide information or comply with directions is punishable with imprisonment up to **one year**, or fine up to Rs. 1 lakh, or both.
5. **Rules:** Information Technology (The Indian Computer Emergency Response Team and Manner of Performing Functions and Duties) Rules, 2013.

6. CERT-In Directions of 28 April 2022

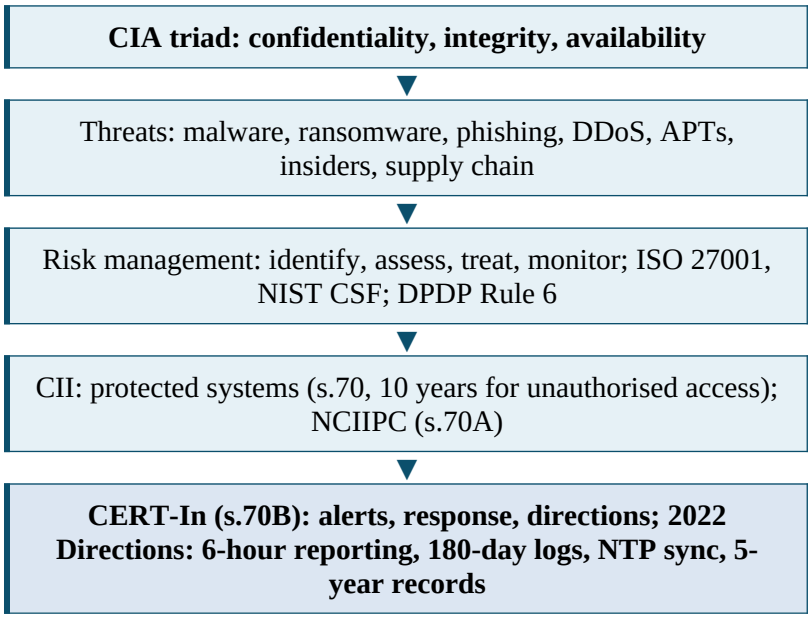
Issued under s.70B(6), effective (after an extension for some entities) in 2022:

No.	Direction
1	**Clock synchronisation:** all service providers, intermediaries, data centres, body corporates and government organisations must connect to the Network Time Protocol (NTP) server of the National Informatics Centre (NIC) or the National Physical Laboratory (NPL), or to NTP servers traceable to them

No.	Direction
2	**Six-hour reporting:** cyber incidents of the types listed in Annexure I (e.g. targeted scanning, compromise of critical systems, unauthorised access to IT systems or data, defacement of websites, malicious code attacks, attacks on servers, identity theft, data breaches and leaks, attacks on IoT devices, attacks on applications such as e-governance and e-commerce, attacks through drones or other new technologies) must be reported to CERT-In within **six hours** of noticing them
3	**Point of contact:** entities must designate a point of contact to interface with CERT-In
4	**Logs:** all ICT system logs must be maintained securely for a rolling period of **180 days** within Indian jurisdiction, and provided to CERT-In on request
5	**Data centres, VPS, cloud and VPN providers:** must register and maintain accurate information (validated names of subscribers, period of hire, IP addresses allotted, email addresses and contact numbers, purpose of hiring, validated addresses, ownership pattern) for **five years** or longer as required by law, even after cancellation
6	**Virtual asset service providers, exchanges and custodian wallet providers:** must maintain KYC information and records of financial transactions for **five years**
7	**Compliance:** failure to comply is punishable under s.70B(7)

Criticism: industry and civil society criticised the short six-hour window, the breadth of reportable incidents, the burden of log retention and the KYC requirements for VPN providers (some VPN providers removed their physical servers from India). CERT-In issued FAQs clarifying some points.

Flowchart: India's cyber security framework



Examples

- A hospital's systems are encrypted by ransomware. It must report the incident to CERT-In within six hours, and, if personal data is affected, intimate the Board and patients under the DPDP Act and Rules.
- A person tries to break into a computer system of a power grid that has been notified as a protected system. He is punishable under s.70(3) with up to ten years' imprisonment.

Case Laws

There is no leading judicial decision interpreting ss.70, 70A and 70B or the CERT-In Directions of 2022. The Directions were challenged in some forums by industry bodies and VPN providers, but as of these notes no court has struck them down. Cyber security incidents generally come before courts in the form of prosecutions under ss.43, 66 and 66F of the IT Act and the BNS, or in claims for compensation under s.43A.

Important Points

- Cyber security: s.2(1)(nb) IT Act.
- CIA triad: confidentiality, integrity, availability.
- Threats: malware, ransomware, phishing, DDoS, APTs, insider, supply chain, zero-day; examples: Cosmos Bank (2018), AIIMS (2022).
- Risk management: identify, assess, treat, monitor; ISO 27001; NIST CSF 2.0.
- CII: protected systems (s.70; 10 years); NCIIPC (s.70A, 2014).
- CERT-In (s.70B): functions, directions (s.70B(6)), penalty (s.70B(7): 1 year or Rs. 1 lakh).
- CERT-In Directions (28 April 2022): 6-hour reporting; NTP sync; 180-day logs in India; 5-year records for VPN, cloud, data centres and VDA providers.

> **Remember:** Under the CERT-In Directions, 2022, cyber incidents must be reported within **six hours**; under the DPDP Rules, 2025, a detailed breach report goes to the Board within **72 hours**.

Topic 5.5: Data Protection Board of India; penalties, exemptions and cross-border transfer

Introduction

The DPDP Act is enforced by the **Data Protection Board of India**, an independent body that inquires into personal data breaches and complaints, and imposes **monetary penalties** of up to Rs. 250 crore. The Act also contains wide **exemptions**, especially for the State and for certain purposes, and a **negative-list approach** to **cross-border transfers** of personal data. The Board was established in November 2025, when the first phase of the Act came into force.

Definition

> **Data Protection Board of India (s.18):** the body corporate established by the Central Government under the DPDP Act to perform functions of inquiry and adjudication and to impose penalties.

Detailed Explanation

1. The Data Protection Board

A. Establishment and Composition (Sections 18–26)

1. **Establishment (s.18):** by the Central Government, by notification; it is a body corporate with perpetual succession and a common seal; headquarters as notified (the National Capital Region).
2. **Composition (s.19):** a Chairperson and such number of other Members as the Central Government may notify (the Board has been constituted with four members under the 2025 notification).
3. **Qualifications:** persons of ability, integrity and standing with special knowledge or practical experience in data governance, administration or implementation of laws relating to social or consumer protection, dispute resolution, information and communication technology, digital economy, law, regulation or techno-regulation; at least one Member must be an expert in law.
4. **Term (s.20):** two years, eligible for re-appointment.
5. **Officers and employees (s.24)**, **members to act in good faith**, and removal (s.22).

B. Powers and Functions (Section 27)

1. On intimation of a personal data breach, direct urgent remedial or mitigation measures, and inquire into the breach and impose penalty.
2. On a complaint by a Data Principal about a breach of obligations by a Data Fiduciary, or in relation to her rights, inquire and impose penalty.
3. On a complaint about a Consent Manager, inquire and impose penalty; and suspend or cancel the registration of a Consent Manager.
4. On a reference by the Central Government or a State Government or in compliance with a court's direction, inquire into a breach by a Data Fiduciary.

5. Inquire into a breach of a voluntary undertaking.
6. Issue directions after giving the person a hearing.

C. Procedure (Section 28)

1. The Board functions as an independent body and, as far as practicable, as a **digital office**, with receipt of complaints, hearings and pronouncement of decisions by digital means (techno-legal measures).
2. On receiving a complaint, it determines whether there are sufficient grounds to proceed; if not, it may close the proceedings (and may warn or impose costs for a false or frivolous complaint).
3. It has the powers of a civil court for summoning, examination on oath, receiving evidence on affidavit, and requiring discovery and inspection of documents, but it cannot prevent access to premises or take custody of equipment where this would adversely affect day-to-day functioning.
4. It may issue interim orders after giving a hearing.

D. Appeal and ADR (Sections 29–32)

1. **Appeal (s.29):** a person aggrieved by an order of the Board may appeal to the **Appellate Tribunal (TDSAT)** within **60 days**; the Tribunal shall endeavour to dispose of the appeal within six months.
2. **Orders as decrees (s.30):** orders of the Appellate Tribunal are executable as decrees of a civil court.
3. **Mediation (s.31):** if the Board considers that a complaint can be resolved by mediation, it may direct the parties to attempt mediation.
4. **Voluntary undertaking (s.32):** the Board may accept a voluntary undertaking from a person in respect of compliance, which bars proceedings on the matter, unless the person fails to adhere to it.

E. Bar of Civil Court and Blocking

1. **Section 39:** no civil court shall have jurisdiction in respect of any matter for which the Board is empowered, and no injunction shall be granted by any court in respect of action taken by the Board.
2. **Section 37:** on a reference by the Board that it has imposed penalties on a Data Fiduciary in **two or more instances** and advised blocking, the Central Government may, after giving a hearing, direct the blocking of public access to the Data Fiduciary's information or platform, in the interests of the general public.

2. Penalties (Section 33 and the Schedule)

Item	Breach	Maximum penalty
1	Failure to take reasonable security safeguards to prevent personal data breach (s.8(5))	Rs. 250 crore
2	Failure to give intimation of a personal data breach to the Board or affected Data Principals (s.8(6))	Rs. 200 crore

Item	Breach	Maximum penalty
3	Non-fulfilment of additional obligations in relation to children (s.9)	Rs. 200 crore
4	Non-fulfilment of additional obligations of a Significant Data Fiduciary (s.10)	Rs. 150 crore
5	Breach of duties of a Data Principal (s.15)	Rs. 10,000
6	Breach of any term of a voluntary undertaking accepted by the Board (s.32)	Up to the extent applicable for the breach in respect of which proceedings were instituted
7	Breach of any other provision of the Act or the Rules	Rs. 50 crore

Factors for determining the penalty (s.33(2)): the nature, gravity and duration of the breach; the type and nature of personal data affected; the repetitive nature of the breach; whether the person realised a gain or avoided a loss; whether the person took any action to mitigate the effects and its timeliness and effectiveness; whether the penalty is proportionate and effective; and the likely impact on the person.

Section 34: all sums realised by way of penalties are credited to the **Consolidated Fund of India**.

Note: the DPDP Act provides **only monetary penalties**, not compensation to affected individuals and not criminal punishment. (Compensation under s.43A of the IT Act will no longer be available once s.43A is omitted.)

3. Exemptions (Section 17)

A. Exemptions for Certain Purposes (s.17(1))

The provisions of Chapter II (except s.8(1) and (5)), Chapter III and s.16 do not apply where personal data is processed:

1. for enforcing any **legal right or claim**;
2. by any **court, tribunal** or other body entrusted with judicial or quasi-judicial functions;
3. in the interest of **prevention, detection, investigation or prosecution** of any offence;
4. of Data Principals **not within India**, under a contract with a person outside India, by a person in India;
5. for a scheme of **compromise, arrangement, merger or amalgamation** approved by a court or tribunal;
6. for ascertaining the financial information and assets of a person who has **defaulted** on a loan.

B. Exemptions for the State and Research (s.17(2))

1. The Central Government may exempt any **instrumentality of the State** from the Act, by notification, in the interests of sovereignty and integrity of India, security of the State, friendly relations with foreign States, maintenance of public order, or preventing incitement to any cognizable offence.

2. The Act does not apply to processing necessary for **research, archiving or statistical purposes**, if the personal data is not used to take any decision specific to a Data Principal and is carried out in accordance with the prescribed standards (Rule 16).

C. Other Exemptions

1. **Section 17(3):** the Government may notify certain Data Fiduciaries or classes, including **startups**, to which ss.5, 8(3), 8(7), 10 and 11 shall not apply, considering the volume and nature of data processed.
2. **Section 17(4):** for the State and its instrumentalities, the obligation to erase data (s.8(7)) and the right to erasure (s.12(3)) do not apply where processing is for a purpose that does not include making a decision about the Data Principal.
3. **Section 17(5):** within five years of the commencement of the Act, the Government may declare that any provision shall not apply to a Data Fiduciary or class of Data Fiduciaries for a specified period.

Criticism: the wide exemptions for the State, especially s.17(2)(a), have been criticised as inconsistent with the proportionality requirement of *Puttaswamy*, since they do not require independent oversight.

4. Cross-Border Transfer (Section 16)

1. **Negative list approach (s.16(1)):** the Central Government may, by notification, **restrict** the transfer of personal data by a Data Fiduciary for processing to such country or territory outside India as may be notified. In other words, transfers are generally permitted, except to countries notified as restricted.
2. **Sectoral laws prevail (s.16(2)):** nothing in s.16 restricts the applicability of any law that provides for a higher degree of protection or restriction on transfer (e.g. the RBI's payment data localisation direction of 2018).
3. **Rule 15, DPDP Rules, 2025:** transfers of personal data outside India are subject to such requirements as the Central Government may specify by general or special order, in respect of making the data available to a foreign State or its agencies.
4. **Significant Data Fiduciaries (Rule 13):** personal data specified by the Government on the recommendation of a committee may be required not to be transferred outside India.

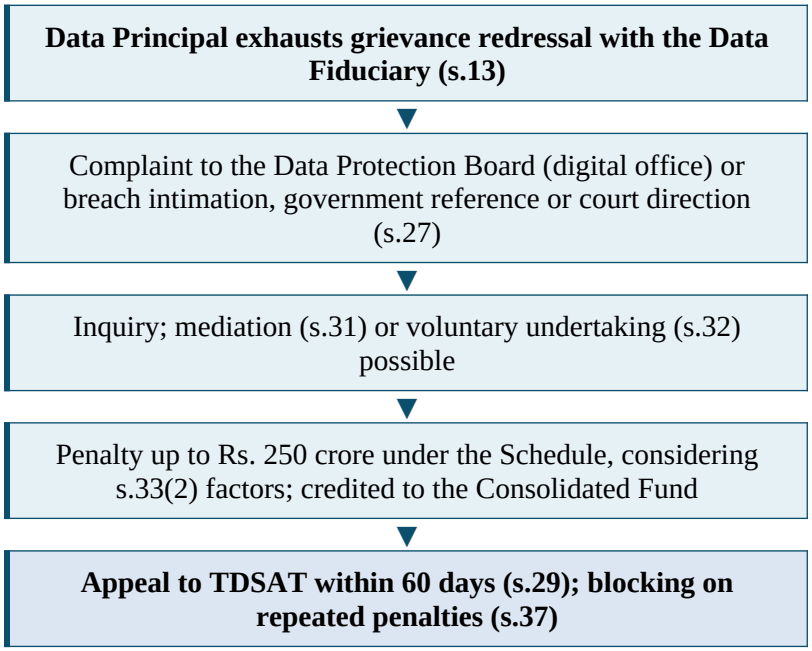
Approach	Meaning	Example
Positive list (adequacy)	Transfers allowed only to approved countries	GDPR adequacy decisions (Art 45)
Negative list	Transfers allowed except to notified (blacklisted) countries	DPDP Act, s.16
Localisation	Data must be stored in the country	RBI payment data direction (2018)

5. Consistency with Other Laws and Amendments

1. **Section 38:** the Act is in addition to, and not in derogation of, any other law; but in case of conflict, the DPDP Act prevails to the extent of the conflict.

2. **Section 44(2):** amends the IT Act (omits s.43A and the rule-making power for it; adds a proviso to s.81).
3. **Section 44(3):** amends s.8(1)(j) of the **Right to Information Act, 2005**, so that all information relating to personal information is exempt from disclosure, removing the earlier public interest override in that clause. This has been criticised as weakening transparency.

Flowchart: Enforcement under the DPDP Act



Examples

- A fintech company fails to encrypt customer data, and hackers steal it. The Board may impose a penalty of up to Rs. 250 crore for failure to take reasonable security safeguards.
- The police process personal data while investigating a murder. Under s.17(1)(c), most obligations of the Act (such as consent) do not apply.

Case Laws

There are no decisions of the Data Protection Board or the courts yet on penalties, exemptions or cross-border transfers under the DPDP Act, since the Board was constituted only in November 2025 and the substantive obligations come into force in May 2027. Students should follow future decisions of the Board and the TDSAT, and the pending challenges to s.17 and s.44(3).

Important Points

- Board: established under s.18; Chairperson and Members; two-year term; digital office (s.28); powers in s.27.
- Appeal: TDSAT within 60 days (s.29), to be disposed of within six months as far as possible.
- Mediation (s.31) and voluntary undertakings (s.32).

- Penalties: up to Rs. 250 crore (security), Rs. 200 crore (breach intimation, children), Rs. 150 crore (SDF), Rs. 50 crore (others), Rs. 10,000 (Data Principal); to the Consolidated Fund.
- No compensation to individuals and no criminal punishment under the DPDP Act.
- Exemptions: s.17(1) purposes; s.17(2) State instrumentalities and research; s.17(3) startups.
- Cross-border: negative list (s.16); sectoral laws prevail; Rule 15; SDF localisation (Rule 13).
- Section 44: omits s.43A IT Act; amends RTI s.8(1)(j).

Remember: The maximum penalty under the DPDP Act is Rs. 250 crore, for failure to take reasonable security safeguards.

Topic 5.6: Comparative regimes: GDPR and CCPA; international cyber security cooperation

Introduction

Data protection and cyber security are global issues. The **European Union's General Data Protection Regulation (GDPR)** is the most influential data protection law in the world, and many countries, including India, have drawn on it. In the United States, there is no single federal data protection law, but the **California Consumer Privacy Act (CCPA)** is the leading State law. Comparing these regimes with India's DPDP Act helps to understand its strengths and gaps. Cyber security also requires **international cooperation**, because attacks cross borders and evidence lies in other countries.

Definition

> **GDPR (Regulation (EU) 2016/679):** the EU regulation on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, applicable from **25 May 2018**.

> **CCPA:** the California Consumer Privacy Act, 2018, in force from 1 January 2020, as amended and expanded by the **California Privacy Rights Act (CPRA), 2020** (operative from 1 January 2023).

Detailed Explanation

A. The GDPR

1. Key Features

1. **Scope (Art 3):** applies to processing by controllers and processors established in the EU, and to those outside the EU who offer goods or services to, or monitor the behaviour of, persons in the EU (extra-territorial reach).
2. **Principles (Art 5):** lawfulness, fairness and transparency; purpose limitation; data minimisation; accuracy; storage limitation; integrity and confidentiality; and accountability.
3. **Lawful bases (Art 6):** consent; contract; legal obligation; vital interests; public task; and **legitimate interests**.
4. **Special categories of data (Art 9):** racial or ethnic origin, political opinions, religious beliefs, trade union membership, genetic and biometric data, health, sex life and sexual orientation, with stricter conditions.
5. **Rights of data subjects (Arts 12–22):** information; access (Art 15); rectification (Art 16); **erasure ("right to be forgotten", Art 17)**; restriction (Art 18); **data portability (Art 20)**; objection (Art 21); and rights related to **automated decision-making and profiling (Art 22)**.
6. **Obligations:** privacy by design and by default (Art 25); records of processing; Data Protection Impact Assessments (Art 35); Data Protection Officers (Art 37); **breach**

notification to the supervisory authority within 72 hours (Art 33) and to data subjects where there is high risk (Art 34).

7. **Cross-border transfers (Chapter V):** allowed to countries with an **adequacy decision** (Art 45), or with appropriate safeguards such as **Standard Contractual Clauses** or **Binding Corporate Rules** (Art 46).
8. **Enforcement:** independent supervisory authorities in each Member State; the European Data Protection Board; the one-stop-shop mechanism; administrative fines of up to **€20 million or 4% of total worldwide annual turnover**, whichever is higher (Art 83); and a right to compensation for material or non-material damage (Art 82).

2. Leading Cases

1. **Google Spain SL v Agencia Española de Protección de Datos (CJEU, 2014):** recognised the "right to be forgotten" against search engines.
2. **Data Protection Commissioner v Facebook Ireland and Maximillian Schrems (Schrems II) (CJEU, 2020):** invalidated the EU–US Privacy Shield because of US surveillance laws; Standard Contractual Clauses remain valid but require case-by-case assessment.
3. **Large fines:** for example, the Irish Data Protection Commission fined Meta €1.2 billion in 2023 for transfers of personal data to the US.

B. The CCPA (as amended by the CPRA)

1. Key Features

1. **Applies to** for-profit businesses that do business in California and meet thresholds such as annual gross revenue above a specified amount (originally USD 25 million, adjusted for inflation), buying, selling or sharing the personal information of 100,000 or more consumers or households, or deriving 50% or more of annual revenue from selling or sharing personal information.
2. **Consumer rights:**
   - right to **know** what personal information is collected, used, shared or sold;
   - right to **delete** personal information;
   - right to **correct** inaccurate information (CPRA);
   - right to **opt out of the sale or sharing** of personal information ("Do Not Sell or Share My Personal Information" link);
   - right to **limit the use and disclosure of sensitive personal information** (CPRA);
   - right to **non-discrimination** for exercising rights.
3. **Approach:** mainly an **opt-out** model (unlike the opt-in consent model of the GDPR and DPDP Act), focused on the sale and sharing of data.
4. **Enforcement:** the **California Privacy Protection Agency** (created by the CPRA) and the Attorney General; civil penalties per violation; a limited **private right of action** for certain data breaches resulting from failure to maintain reasonable security.

C. Comparison: GDPR, CCPA and DPDP Act

Basis	GDPR (EU)	CCPA/CPRA (California)	DPDP Act (India)
Data covered	All personal data (digital and manual filing systems)	Personal information of California residents, including household data	Digital personal data (and digitised data)
Lawful bases	Six bases, including legitimate interests	No lawful basis requirement; notice and opt-out	Consent or specified legitimate uses; no general "legitimate interests"
Sensitive data	Special categories with stricter rules	Sensitive personal information; right to limit	No separate category (children's data has special protection)
Consent model	Opt-in for consent-based processing	Opt-out of sale and sharing	Opt-in consent with notice
Rights	Access, rectification, erasure, restriction, portability, objection, automated decisions	Know, delete, correct, opt out, limit, non-discrimination	Access, correction, erasure, grievance redressal, nomination
Data portability	Yes (Art 20)	Yes (as part of the right to know, in portable format)	No
Automated decisions	Art 22 right	Regulations on automated decision-making technology	No specific right
Children	Parental consent below 16 (Member States may lower to 13)	Opt-in consent required for selling data of consumers under 16	Verifiable parental consent below 18; no tracking or targeted ads
Breach notification	Supervisory authority within 72 hours	Notification under California breach law	Board and Data Principals without delay; detailed report within 72 hours
Cross-border transfers	Adequacy, SCCs, BCRs	No general restriction	Negative list (s.16)
Regulator	Independent supervisory authorities	California Privacy Protection Agency; Attorney General	Data Protection Board of India
Maximum sanction	€20 million or 4% of worldwide turnover	Civil penalties per violation	Rs. 250 crore per instance
Damages for individuals	Yes (Art 82)	Limited private right of action for breaches	No

Basis	GDPR (EU)	CCPA/CPRA (California)	DPDP Act (India)
State exemptions	Limited, subject to necessity and proportionality	Not applicable to government agencies	Wide exemptions for notified State instrumentalities (s.17(2))

D. International Cyber Security Cooperation

1. Treaties and Instruments

1. **Budapest Convention on Cybercrime (Council of Europe, 2001):** harmonises cybercrime offences, procedural powers (preservation, production orders, search and seizure of computer data) and international cooperation (24/7 network). The Second Additional Protocol (2022) addresses direct cooperation with service providers and disclosure of electronic evidence. **India is not a party**, mainly because it was not involved in drafting and because of concerns about sovereignty (especially Article 32(b) on transborder access to data).
2. **UN Convention against Cybercrime:** negotiated under the UN, adopted by the General Assembly in **December 2024**, and opened for signature in Hanoi in **October 2025**; it covers criminalisation, procedural measures and international cooperation, and has been criticised by human rights groups for possible misuse.
3. **UN norms of responsible State behaviour in cyberspace:** reports of the UN Group of Governmental Experts (2013, 2015, 2021) and the Open-Ended Working Group, affirming that international law, including the UN Charter, applies to cyberspace, and setting voluntary norms (e.g. not attacking critical infrastructure, protecting CERTs).
4. **Tallinn Manual (2013; 2.0 in 2017):** a non-binding academic study of how international law applies to cyber operations.

2. Mechanisms of Cooperation

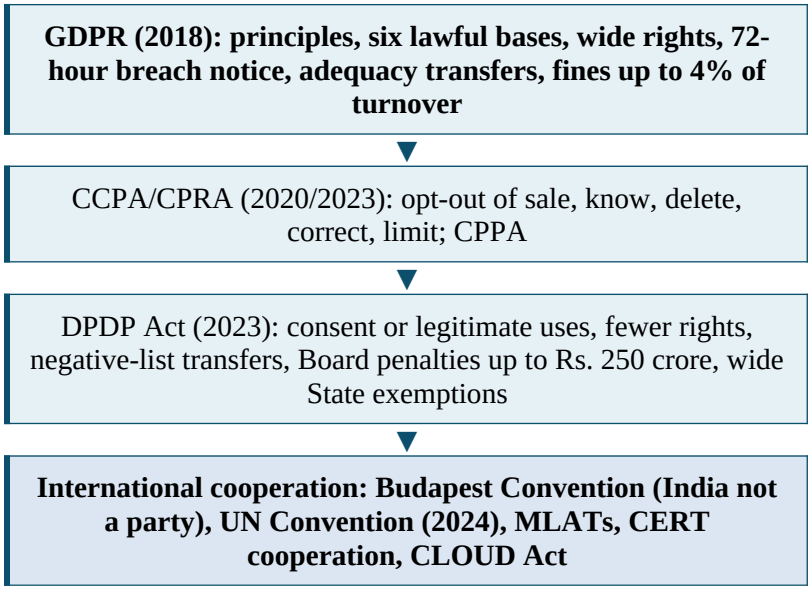
1. **Mutual Legal Assistance Treaties (MLATs):** India has MLATs in criminal matters with many countries; requests are processed through the Ministry of Home Affairs.
2. **Letters rogatory** under s.112 BNSS (s.166A CrPC).
3. **INTERPOL** and police-to-police cooperation.
4. **CERT-to-CERT cooperation:** CERT-In has signed memoranda of understanding with counterpart agencies of many countries.
5. **Direct requests to service providers:** e.g. under the US **CLOUD Act (2018)**, US providers can disclose data in response to requests from countries with executive agreements.
6. **Bilateral and multilateral dialogues:** India has cyber dialogues with several countries and participates in groupings such as the Quad, G20 and BRICS on cyber security.

3. Challenges

1. Slow MLAT processes (often months or years).
2. Conflicts of laws on data access and privacy.

3. Differing views on internet governance (multistakeholder v State-centric).
4. Attribution of State-sponsored attacks.
5. Lack of a binding international framework on cyber warfare.

Flowchart: Comparing data protection regimes and cooperation



Examples

- An Indian e-commerce company sells goods to customers in Germany. It must comply with the GDPR for their data, in addition to the DPDP Act.
- An Indian police agency needs emails stored by a US provider to investigate a fraud. It must usually use the MLAT process, which may take a long time.

Case Laws

Case Law: Google Spain SL v Agencia Española de Protección de Datos (AEPD) and Mario Costeja González

Year: 2014 **Court:** Court of Justice of the European Union (Grand Chamber)

Facts / Background: Mario Costeja González, a Spanish citizen, complained that a Google search of his name showed links to 1998 newspaper notices about the auction of his property for recovery of social security debts, which had long been resolved. He asked the newspaper to remove the notices and Google to remove the links. The Spanish data protection authority upheld the complaint against Google, which challenged the decision.

Legal Issue: Does a search engine "process" personal data and act as a controller, and can an individual require it to remove links to information about him that is inadequate, irrelevant or no longer relevant?

Court's Decision / What the Court Held: The Court held that a search engine operator processes personal data and is a controller under the Data Protection Directive. The Directive applied to Google because it had a subsidiary in Spain promoting and selling advertising. An individual has the right to request the search engine to remove links to

information that is inadequate, irrelevant or no longer relevant, or excessive in relation to the purposes of the processing, even if the original publication was lawful, subject to a balance with the public interest in access to the information (e.g. where the person plays a role in public life).

Relevance to the Topic: The leading case on the **"right to be forgotten"**, later codified in Art 17 of the GDPR. (Indian High Courts have recognised a limited right to be forgotten in some cases, and the DPDP Act provides a right to erasure under s.12.)

Case Law: Data Protection Commissioner v Facebook Ireland Ltd and Maximillian Schrems (Schrems II)

Year: 2020 **Court:** Court of Justice of the European Union (Grand Chamber)

Facts / Background: Maximillian Schrems, an Austrian privacy activist, complained that Facebook's transfer of his personal data from Ireland to the United States exposed it to US government surveillance programmes, without adequate protection. The question concerned the validity of the EU–US Privacy Shield and Standard Contractual Clauses.

Legal Issue: Do EU–US data transfer mechanisms ensure a level of protection essentially equivalent to that guaranteed in the EU?

Court's Decision / What the Court Held: The Court invalidated the EU–US Privacy Shield adequacy decision, because US surveillance laws allowed access to the data of EU persons that was not limited to what was strictly necessary and proportionate, and EU persons lacked effective judicial remedies. Standard Contractual Clauses remain valid, but exporters must assess, case by case, whether the law of the destination country provides adequate protection, and suspend transfers if it does not.

Relevance to the Topic: A leading case on **cross-border data transfers** and the relationship between surveillance and data protection. (A new EU–US Data Privacy Framework was adopted in 2023.)

Important Points

- GDPR (applies from 25 May 2018): extra-territorial; Art 5 principles; six lawful bases; wide rights (erasure, portability, Art 22); 72-hour breach notice; adequacy and SCCs; fines up to €20 million or 4%.
- CCPA (2020) and CPRA (2023): opt-out model; rights to know, delete, correct, opt out, limit, non-discrimination; California Privacy Protection Agency.
- DPDP Act compared: no legitimate interests basis, no portability, no automated-decision right, negative-list transfers, no compensation, wide State exemptions.
- Cases: *Google Spain* (2014): right to be forgotten; *Schrems II* (2020): Privacy Shield invalid.
- Cooperation: Budapest Convention (India not a party), UN Convention against Cybercrime (2024), UN GGE norms, Tallinn Manual, MLATs, CERT cooperation, CLOUD Act.

Remember: India has not joined the Budapest Convention on Cybercrime.

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- The Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, as amended in 2022, 2023, 2025 and 2026.
- The Information Technology (Procedure and Safeguards for Blocking for Access of Information by Public) Rules, 2009.
- The Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011.
- CERT-In Directions under section 70B(6) of the IT Act, dated 28 April 2022.
- The Digital Personal Data Protection Act, 2023 and the Digital Personal Data Protection Rules, 2025.
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- The Protection of Children from Sexual Offences Act, 2012.
- UNCITRAL Model Law on Electronic Commerce, 1996 and Model Law on Electronic Signatures, 2001.
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- Gerald R. Ferrera et al., Cyberlaw: Text and Cases (Cengage Learning).
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- Harish Chander and Gagandeep Kaur, Cyber Laws and IT Protection (PHI Learning).

Landmark Decisions Prescribed for Study (where explained)

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- K.S. Puttaswamy (Retd.) v Union of India, (2017) 10 SCC 1: Topic 5.1.
- Anvar P.V. v P.K. Basheer, (2014) 10 SCC 473: Topic 3.5.
- Arjun Panditrao Khotkar v Kailash Kushanrao Gorantyal, (2020) 7 SCC 1: Topic 3.5.
- Anuradha Bhasin v Union of India, (2020) 3 SCC 637: Topic 4.4.
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- Google India Pvt. Ltd. v Visaka Industries, (2020) 4 SCC 162: Topic 4.1.
- Christian Louboutin SAS v Nakul Bajaj, 2018 SCC OnLine Del 12215: Topic 4.1.
- Banyan Tree Holding (P) Ltd. v A. Murali Krishna Reddy, 2009 SCC OnLine Del 3780: Topic 1.6.
- Yahoo! Inc. v Akash Arora, 1999 SCC OnLine Del 133: Topic 1.6.

Note on Currency of the Law

- Cyber law changes quickly. These notes state the law as at November 2026, including the DPDP Rules, 2025 (notified in November 2025 with phased commencement up to May 2027), the October 2025 amendment to Rule 3(1)(d) of the IT Rules, and the February 2026 amendment on synthetically generated information. Students should check for later amendments and judgments, especially the pending Supreme Court appeal in the fact check unit case and the appeal in X Corp v Union of India.